Where Nehari's Theorem Survives

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Preface

This book organizes a body of research notes around one question: *where does Nehari's theorem survive in infinitely many variables, and why?* In one variable every bounded Hankel form has a bounded symbol. On $\mathbb{T}^\infty$ this fails, and the multiplicative Hilbert matrix sat for years exactly on the fault line. The chapters record what is now known on either side of it.

The book assumes the standard theory: Hardy spaces of Dirichlet series and the Bohr lift, Helson's inequality, the Cole–Gamelin estimate, weighted Bergman spaces of Dirichlet series in the sense of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt, and Perfekt's recent Fejér–Riesz inequality. Known results are quoted, not reproved, and attributed where we know their source. For the remaining results we have not completed a literature search, so "new" here means only that we did not find them; it is not a priority claim. Failed approaches appear only as brief remarks, where they explain why a result has the shape it has.

How the proofs were checked. The results and most proofs were developed with the assistance of large language models. Each result carries one label. means that the proof was checked line by line by the model that wrote it. means that, in addition, a separate model run read the proof as a referee and its objections were resolved. marks uncertified computations. A *probabilistic certificate* is a computation whose conclusion holds except on an event of explicitly bounded probability, under stated assumptions on the random-number generator and on floating-point arithmetic. It is not a proof. None of these is formal verification, and none replaces reading by a specialist: *no proof in this book has been independently verified.* The proofs most in need of such a reading are the envelope argument of section 8.1, the Schatten characterization Theorem 5.6, and the Hellinger–Bessel chain of chapters 1 and 2 on which most of the book depends.

The parts are arranged as an argument. Part I proves the stability theorem for point evaluations and the Hellinger–Bessel inequality, and applies them to Carleson measures. Part II uses them, with sign averages, to prove Nehari-type theorems for structured Helson matrices. Part III shows where Nehari's theorem fails, in Schatten classes, and why the known constructions stop where they do. Part IV returns to the borderline object, the multiplicative Hilbert matrix, and to the constants and coefficient inequalities that remain open there.

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Overview

The problem

A Helson matrix $M(\alpha) = (\alpha(mn))_{m,n\geq 1}$ is a Hankel form in infinitely many variables: under the Bohr lift it becomes $(F, G) \mapsto \int FG\,\overline{g}\,\mathrm{d}m$ on $H^2(\mathbb{T}^\infty)$, for any g whose analytic coefficients are α. On the circle Nehari's theorem says that every bounded Hankel form has such a g in L^∞, of the same norm. On $\mathbb{T}^\infty$ this fails: the best constant on $\mathbb{T}^d$ grows exponentially in d [38], and there are bounded Helson matrices with no bounded symbol at all. The one-variable proof goes through inner–outer factorization, $H^1 = H^2 \cdot H^2$, and that is exactly what is lost.

This book asks which Helson matrices still have bounded symbols, and why. Duality reduces the question to a Carleson embedding: a positive matrix $M(\alpha)$ with $\alpha(n) = \int z^{\kappa(n)}\,\mathrm{d}\mu$ has a bounded symbol as soon as $\int |F|\,\mathrm{d}\mu \lesssim \|F\|_{H^1}$, whereas boundedness of the matrix is the same estimate in H^2. So the question becomes: when does an H^2 Carleson measure on $\mathbb{D}^\infty$ embed H^1? In one variable the answer is always, by factorization; here we need a replacement.

The test case is Perfekt's theorem [39]: the multiplicative Hilbert matrix $(1/(\sqrt{mn}\log mn))$, which is the moment matrix of Lebesgue measure on the curve $\gamma(s) = (p^{-s})_p$, has a bounded symbol. His proof localizes point evaluations with Möbius maps and Hoeffding's inequality, and works for H^1.

Separation: the Hellinger–Bessel inequality

Part I replaces factorization by a one-line inequality and a Gram estimate. For a point $r \in \mathbb{D}^\infty \cap \ell^2$ with reproducing kernel K_r, every $F \in H^p$ and every $p > 0$ satisfy

$$|F(r)|^{p/2} \leq \int |F|^{p/2} |K_r|\,\mathrm{d}m$$

(Theorem 1.7). Equivalently, a function that is efficient at r must have $|F|^p$ close, in Hellinger distance, to the harmonic measure $\omega_r = |K_r|^2 dm/\|K_r\|^2$. Since $|F|^{p/2}$ lies in L^2, the evaluations at many points can then be added up by Bessel's inequality for the vectors $|K_r|$. This gives the *Hellinger–Bessel inequality* (Theorem 2.1): for every positive measure μ and every $p > 0$,

$$\int |F|^p\,\mathrm{d}\mu \leq \|\mathcal{G}_\mu\|\,\|F\|_p^p, \qquad \mathcal{G}(r,s) = \|K_r\|\|K_s\|\,H(\omega_r, \omega_s).$$

The condition does not see p, and it holds in the Hardy, Bergman and Gaussian geometries alike. Everything quantitative then rests on computing Hellinger affinities. In each coordinate they are hypergeometric, and across coordinates they multiply,

so $H(\omega_r, \omega_s) \le e^{-E(r,s)/4}$ with $E = \sum_p \rho_p^2$ the sum of squared pseudo-hyperbolic distances. Along the prime curve E is, up to a bounded error, the hyperbolic distance in the half-plane (Theorem 1.6).

The consequences in Part I:

- the Fejér–Riesz inequality $\int_0^1 |f(1/2+\sigma)|^p \, \mathrm{d}\sigma \le (5+2\sqrt{2})\|f\|^p_{\mathscr{H}^p}$ for every $p > 0$ (Theorem 2.5), and, keeping the phases of one prime, the exact local constant $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ on $(0, 2)$ (Theorem 2.7);
- Carleson estimates for measures along curves in all three geometries and on prime sets of any density, with constant below 99 for the prime curve (Theorem 2.9);
- a complete description of Carleson measures on families separated in energy: for these, H^p Carleson for one p is the same as for all p, and the same as the one-point condition (Theorem 3.1).

The method has limits. The Gram condition is not necessary: angular averages in one coordinate are Carleson for every p while $\mathcal{G}_\mu$ is unbounded (chapter 2). And near the critical line its constant is exactly $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.24$, against π for the corresponding H^2 estimate.

Symmetry: sign averages

The second mechanism is independent. For $g \in H^p(\mathbb{T})$ and $|x| < 1$,

$$\frac{|g(x)|^p + |g(-x)|^p}{2} \le \frac{\|g\|_p^p}{1 - |x|^4},$$

sharply, and this tensorizes (Proposition 3.10). Averaging signs in a set of coordinates therefore costs $\prod(1 - |x_p|^4)^{-1}$, which is finite for $x \in \ell^4$, whereas a single evaluation needs $x \in \ell^2$. Part II combines the two mechanisms. Keep the primes of a set R of density θ on the curve and average the signs of the rest. The resulting family of positive Helson matrices interpolates between the multiplicative Hilbert matrix (R all primes) and its full sign average (R empty). For every member, boundedness, the H^1 Carleson condition and the existence of a bounded symbol are equivalent to $\nu((1/2, 1/2 + t]) = O(t^\theta)$, with a constant depending only on R (Theorem 4.2). On the curve, the Nehari constant is below 200; for the full sign average it is at most $\zeta(2)$.

The same family has a complete compactness theory. There is a Hartman-type theorem: compactness is the little-o condition, and the symbol then lies in the closure of $C(\mathbb{T}^\infty) + N$ (Theorem 4.6). There is an essential Nehari theorem: the essential norm is comparable to the essential size of the symbol, and equals $\pi w(1/2)$ for measures with a density (Theorem 4.8). Finally, the Schatten classes are described exactly, for every $p \ge 1$: $M \in S_p$ if and only if the dyadic profile $(e^{j\theta}\nu(I_j))_j$ lies in ℓ^p (Theorem 5.6). In particular, positive curve matrices occur in every Schatten class and all have bounded symbols. Off the curve the spectrum collapses: root-exponential decay on sparse prime sets, and $1/(n \log n)$ for the full sign average. The constructions and compactness arguments carry over to Bergman and Fock spaces, where Kulikov's contractive embedding, and its Gaussian limit, replace Carleman's inequality (chapter 6).

Failure: the threshold 4

Part III turns to Schatten-class forms with no bounded symbol. Helson's theorem rules them out in S_2. Brevig and Perfekt found them in S_p for $p > 5.74$, and Bampouras has recently shown that they exist for every $p > 4$ [4]. Chapter 7 gives an independent, constructive proof of the same bound. It uses Gaussian blocks $1 + cZ_1 \cdots Z_n$ against perturbed squares, with parameters fixed by large deviations, and it is Legendre dual to Bampouras's integration over scales.

The point of Part III is the threshold itself. *These methods stop at* 4*; whether the true threshold is* 4 *is open.* For the constant-plus-product family, no test function and no choice of coefficients beats S_4, for any n (Theorem 8.1). For admissible blocks, $\|\varphi\|_{L^4} \le \|H_\varphi\|_{S_4}$ on every torus, with equality in the Gaussian model, so scale integration can never go below 4 (Corollary 7.11). Affine, even quadratic and zero-free symbols are stuck in the same way. Every family of bounded degree in a bounded number of variables is excluded on an interval above 2 (Theorem 8.9). All of this reduces to one concrete question: is $|\int f\overline{\varphi}\,\mathrm{d}\mu| \le \|f\|_{L^1(\mu)}\|\varphi\|_{L^4(\mu)}$ for Gaussian polynomials (Problem 7)? A counterexample would go below 4. Bampouras's Paley–Wiener threshold, also 4, turns out to have a different source: square-root cancellation in L^1 combined with parabolic scaling, not a Gaussian moment (section 8.5).

The borderline: is the constant π?

Part IV returns to the multiplicative Hilbert matrix. In one variable four inequalities share the constant π: Hilbert's, Fejér and Riesz's, Hardy's, and the bound for $\sum a_k/(k+1)$ on H^1, all because every H^1 function is a product of two H^2 functions (Proposition 9.1). On $\mathbb{T}^\infty$ the first survives with the same constant and factorization fails, and chapters 9 and 10 measure what that costs. The norm C of the functional $\sum a_n/(\sqrt{n}\log n)$ on $\mathscr{H}^1_0$ is finite by Perfekt's theorem, satisfies $\pi \le C \le 5.515$ (Proposition 10.4), and no counterexample to $C = \pi$ uses only primes below 10^{216}. Near-extremal vectors for the Hilbert matrix vary slowly under multiplication, so no fixed finite-prime factor can raise an extremal square above π (Theorem 9.3 and Corollary 9.4); the problem is equivalent to a Hankel completion on the ordered rationals (Theorem 9.5); and the six per cent excess seen in two to four primes is an Ortega-Cerdà–Seip effect that disappears in the limit (Proposition 9.8). Is $C = \pi$? Very probably not. For quadratic combinations of Euler kernels at many scales near the critical line, the H^1 norm has an exact Gaussian limit (Theorem 10.6), so C is at least a constant of a log-correlated Gaussian field, and we conjecture equality (Conjecture 1). A probabilistic certificate puts that constant at 3.1450 or more, with failure probability at most $2 \cdot 10^{-30}$ (section 10.2); a deterministic proof is open. The companion question with absolute values on the coefficients is proved on every piece of $\mathscr{H}^1$ that a degree constraint visible to the Euler product cuts out (Theorem 10.13), and contrasted with the signed critical exponent $(\log n)^{-\theta}$ on prime sets of density θ (Corollary 10.15).

Reading guide

Chapters 1 and 2 are the foundation and should be read first. After them the parts are largely independent. Part II uses Part I throughout. Part III uses only the Gaussian model, set up in section 7.1. Part IV uses chapter 2 and the rectangular Schur test of section 9.2. Each chapter opens with the question it answers and closes with what remains open. Chapter A collects the open problems.

Part 1

Point evaluations on the infinite torus

CHAPTER 1

Harmonic measure, energy and stability

How much of the torus does a point evaluation use, and when can two point evaluations share it? Section 1.2 answers this for one point: the mass of an extremal function is exactly the harmonic measure of the point. Section 1.3 answers it for two points: harmonic measures overlap by an amount exponentially small in an energy, with an exact formula in one variable and a quantitative form of Kakutani's dichotomy. Section 1.5 identifies that energy, along the Dirichlet half-plane, with hyperbolic distance. Section 1.6 proves the converse: a function that is nearly extremal at a point distributes its mass like that point's harmonic measure, in every H^p. Chapter 2 turns this into the Hellinger–Bessel inequality.

Three geometries run in parallel: the Hardy space of the torus, Bergman spaces of the polydisc, and the Fock space. The Gaussian case is the flat model, where every overlap is an exact exponential. Statements are made on finitely many coordinates, with constants independent of their number, and pass to $\mathbb{T}^\infty$ by approximation.

What is new, to our knowledge, is this point of view: one change-of-variables rule for three geometries reduces every statement about one or two point evaluations to a computation with harmonic measures. In this form the stability theorem, its two-point and many-point consequences, and the identification of energy with hyperbolic distance are new. In one variable the stability theorem is a quantitative form of the factorization $F = Ih^{2/p}$, but its proof uses no factorization, which is what lets it pass to infinitely many variables.

1.1. Three geometries

Each geometry consists of a product probability measure λ, the closure X^p of the analytic polynomials in $L^p(\lambda)$, a reproducing kernel K_r of X^2, and a coordinatewise family of maps T_r with $T_r(0) = r$.

	λ per coordinate	$K_r(z)$ per coordinate	$T_r(w)$ per coordinate
Hardy	arc length dm on $\mathbb{T}$	$(1-\overline{r}z)^{-1}$	$(r+w)/(1+\overline{r}w)$
Bergman	$m_\beta = (\beta-1)(1-\lvert z\rvert^2)^{\beta-2}\,\mathrm{d}A$	$(1-\overline{r}z)^{-\beta}$	$(r+w)/(1+\overline{r}w)$
Gauss	$\gamma = \pi^{-1}e^{-\lvert z\rvert^2}\,\mathrm{d}A$ on $\mathbb{C}$	$e^{\overline{r}z}$	$r+w$

Here $\beta > 1$, and dA is area measure on $\mathbb{D}$, normalized to have mass one in the Bergman row. Write $k_r = K_r/\|K_r\|_2$ for the normalized kernel. Then

$$\|K_r\|_2^2 = \prod_p (1-|r_p|^2)^{-1}, \quad \prod_p (1-|r_p|^2)^{-\beta}, \quad \exp\Big(\sum_p |r_p|^2\Big)$$

in the three cases. All three have the same change-of-variables rule.

LEMMA 1.1. *In each geometry, for every r,*

$$\lambda \circ T_r = \|K_r\|_2^2 \, |K_r \circ T_r|^{-2} \, \lambda,$$

and $K_r \circ T_r$ is holomorphic and zero-free on a neighbourhood of the closed domain.

PROOF. In one coordinate:

- *Hardy.* $K_r(T_r w) = (1+\overline{r}w)/(1-|r|^2)$, and the Jacobian of T_r on $\mathbb{T}$ is $(1-|r|^2)/|1+\overline{r}w|^2$.
- *Bergman.* The density of $m_\beta \circ T_r$ is $(1-|r|^2)^\beta |1+\overline{r}w|^{-2\beta}$, and $K_r(T_r w) = \big((1+\overline{r}w)/(1-|r|^2)\big)^\beta$.
- *Gauss.* $\gamma(r+dw) = e^{-|r|^2 - 2\Re(\overline{r}w)}\gamma(dw)$, and $K_r(r+w) = e^{|r|^2+\overline{r}w}$.

In each case the identity is checked by multiplying out. Products over coordinates preserve it. □

1.2. One point

A *representative* of evaluation at r is a function ψ with $\int F\overline{\psi}\,\mathrm{d}\lambda = F(r)$ for every analytic polynomial F. Every representative has $\|\psi\|_\infty$ at least the norm of evaluation on X^1. Two representatives sit at opposite extremes.

- *The positive representative* $|k_r|^2 = |K_r|^2/\|K_r\|_2^2$. In the Hardy case this is the Poisson kernel P_r. It has L^1 norm one, but it is unbounded, with $\sup P_r = \prod_p (1+|r_p|)/(1-|r_p|)$, which is infinite unless $\sum |r_p| < \infty$.
- *The flat representative* $\Phi_r = \|K_r\|_2^2 \, K_r/\overline{K_r}$, of constant modulus $\|K_r\|_2^2$, introduced in the Hardy case by Perfekt [39]. That it represents evaluation is the case $D = \Omega$ of Lemma 1.5 below.

In the Hardy case the norm of evaluation is due to Cole and Gamelin [19], who identified $\mathbb{D}^\infty \cap \ell^2$ as the set of points where evaluation on $H^1(\mathbb{T}^\infty)$ is bounded.

PROPOSITION 1.2. *In each geometry, evaluation at r has norm exactly $\|K_r\|_2^2$ on X^1. The flat representative is extremal:*

$$\min\{\|\psi\|_\infty : \psi \text{ represents evaluation at } r\} = \|K_r\|_2^2 = \|\Phi_r\|_\infty.$$

The function $F_r = K_r^2/\|K_r\|_2^2$ attains the norm, with $F_r(r) = \|K_r\|_2^2$, $\|F_r\|_1 = 1$, and

$$|F_r|\,\mathrm{d}\lambda = \omega_r := |k_r|^2\,\mathrm{d}\lambda = (T_r)_*\lambda.$$

PROOF. The flat representative gives $|F(r)| \le \|K_r\|_2^2 \|F\|_1$. For F_r we have $\|F_r\|_1 = \|K_r\|_2^2/\|K_r\|_2^2 = 1$ and $F_r(r) = \|K_r\|_2^4/\|K_r\|_2^2$. So the norm is $\|K_r\|_2^2$, and no representative has smaller sup norm. The identity $|k_r|^2\lambda = (T_r)_*\lambda$ is Lemma 1.1, read backwards. □

We call ω_r the *harmonic measure* of r. In the Hardy case it is the usual product of Poisson measures, and in the Gaussian case it is the standard Gaussian translated to r. Proposition 1.2 says that the mass of the extremal function at r is distributed exactly like ω_r. So the question of where near-extremal functions live becomes a question about harmonic measures.

1.3. Two points

Let $\rho(a,b) = |a-b|/|1-\overline{a}b|$ be the pseudo-hyperbolic distance on $\mathbb{D}$, and define the *energy* between two points:

$$E(r,s) = \sum_p \rho(r_p,s_p)^2 \quad \text{(Hardy, Bergman)}, \qquad E(r,s) = \sum_p |r_p - s_p|^2 \quad \text{(Gauss)}.$$

Two numbers measure how much two points share. In X^2 it is the overlap $|\langle k_r, k_s\rangle|$ of the normalized kernels. In X^1 it is the Hellinger affinity

$$H(r,s) = \int \sqrt{\mathrm{d}\omega_r\,\mathrm{d}\omega_s} = \int |k_r||k_s|\,\mathrm{d}\lambda$$

of the harmonic measures.

PROPOSITION 1.3. *The two overlaps compare as follows.*

(1) Exact X^2 overlaps.

$$|\langle k_r,k_s\rangle|^2 = \prod_p (1-\rho_p^2), \quad \prod_p (1-\rho_p^2)^\beta, \quad e^{-E(r,s)}$$

in the three cases, where $\rho_p = \rho(r_p,s_p)$.

(2) Comparison. *In every geometry, $H(r,s) \geq |\langle k_r,k_s\rangle|$.*

(3) Exact X^1 overlaps. *In the Gauss case $H(r,s) = e^{-E(r,s)/4}$. In the Hardy case $H(r,s) = \prod_p h(\rho_p)$ with*

$$h(\rho) = (1-\rho^2)^{1/2}\,{}_2F_1\big(\tfrac{1}{2},\tfrac{1}{2};1;\rho^2\big) = \frac{2}{\pi}(1-\rho^2)^{1/2}\mathbf{K}(\rho) \leq 1 - \frac{1}{4}\rho^2,$$

where $\mathbf{K}$ is the complete elliptic integral of the first kind. Hence $H(r,s) \leq e^{-E(r,s)/4}$. In the Bergman case $H(r,s) = \prod_p h_\beta(\rho_p)$, where

$$h_\beta(\rho) = (1-\rho^2)^{\beta/2}{}_2F_1(\frac{\beta}{2},\frac{\beta}{2};\beta;\rho^2) \leq (1-\frac{1}{2}\rho^2)^{\beta/2} \leq e^{-\beta\rho^2/4}.$$

PROOF. (1) In one coordinate, $|\langle k_a,k_b\rangle|^2 = (1-|a|^2)(1-|b|^2)/|1-\overline{a}b|^2 = 1-\rho(a,b)^2$, and the Bergman kernel is the βth power of the Hardy kernel. In the Gauss case, $|\langle k_r,k_s\rangle|^2 = e^{2\Re\langle s,r\rangle - |r|^2-|s|^2} = e^{-|r-s|^2}$.

(2) $\int |k_r||k_s|\,\mathrm{d}\lambda \geq |\int k_r\overline{k_s}\,\mathrm{d}\lambda|$.

(3) In the Gauss case ω_r is the complex Gaussian centred at r with identity covariance. The Hellinger affinity of two such measures is $e^{-|r-s|^2/4}$. In the Hardy case, Möbius invariance of the pseudo-hyperbolic distance and of the construction $\omega_a = (T_a)_* m$ reduce one coordinate to $a=0$, $b=c$ with $|c|=\rho$, and

$$h(\rho) = \int \sqrt{P_c}\,\mathrm{d}m = (1-\rho^2)^{1/2}\int |1-\overline{c}z|^{-1}\,\mathrm{d}m = (1-\rho^2)^{1/2}\sum_k c_k^2\rho^{2k},$$

with $c_k = \binom{2k}{k}4^{-k}$, the coefficients of $(1-x)^{-1/2}$. Write $x=\rho^2$. The inequality $h \leq 1-x/4$ is equivalent to $\sum c_k^2 x^k \leq (1-x/4)\sum c_k x^k$, which holds coefficient by coefficient:

$$c_k^2 \leq c_k - \frac{1}{4}c_{k-1} = c_k\,\frac{3k-2}{2(2k-1)} \qquad (k \geq 1),$$

since c_k decreases, $c_1 = 1/2$ gives equality at $k = 1$, and for $k \geq 2$ we have $c_k \leq 3/8 < 2/3 \leq (3k-2)/(4k-2)$. Finally $1 - x/4 \leq e^{-x/4}$. In the Bergman case the same reduction gives $h_\beta(\rho) = (1-\rho^2)^{\beta/2}\|(1-\overline{c}z)^{-\beta/2}\|^2_{A^2_\beta}$. Put $a = \beta/2$, $x = \rho^2$ and $u = x/(2-x)$. Gauss's quadratic transformation writes the hypergeometric factor as

$$(1-x/2)^{-a}\,{}_2F_1\Big(\frac{a}{2}, \frac{a+1}{2}; a + \frac{1}{2}; u^2\Big).$$

Its coefficient of u^{2k} is at most $(a)_k/k!$, coefficient by coefficient, since

$$\frac{(a/2+j)((a+1)/2+j)}{a+1/2+j} \leq a + j.$$

The last hypergeometric function is therefore at most $(1-u^2)^{-a}$. Since $1 - u^2 = (1-x)/(1-x/2)^2$, this gives $h_\beta(\rho) \leq (1-x/2)^a \leq e^{-ax/2}$. □

So in every geometry the overlap of harmonic measures decays exponentially in the energy, and it is never smaller than the X^2 overlap. The Gauss case is the exact model. In the Hardy and Bergman cases the upper rates are $1/4$ and $\beta/4$, respectively; both are optimal at small energy, and near $\rho = 1$ the affinities are far smaller. For constants near the boundary the relevant Hardy bound is $h \leq (1-\rho^2)^{1/4}$ (Lemma 2.3).

COROLLARY 1.4 (Kakutani's dichotomy, quantitatively). *In the Hardy geometry on infinitely many coordinates, ω_r and ω_s are mutually singular if and only if $E(r,s) = \infty$. In particular ω_r is absolutely continuous with respect to Haar measure exactly when $r \in \ell^2$, and any two points of $\mathbb{D}^\infty \cap \ell^2$ have equivalent harmonic measures, with*

$$2\big(1 - H(r,s)\big) \leq \|\omega_r - \omega_s\| \leq 2\sqrt{1 - H(r,s)^2}.$$

PROOF. By Kakutani's theorem [29], two product measures with equivalent factors are equivalent or mutually singular according as the product of the factor affinities is positive or zero. By Proposition 1.3 that product is positive exactly when $\sum \rho_p^2 < \infty$. Take $s = 0$ for the second statement. The total variation bounds are the standard Hellinger inequalities. □

This explains two things. First, it explains the domain $\mathbb{D}^\infty \cap \ell^2$ of Cole and Gamelin: it is where harmonic measure lives on the torus at all. Second, it shows that between two points of $\mathbb{D}^\infty \cap \ell^2$ mutual singularity is only ever approximate. Any argument that separates point evaluations must therefore be quantitative in the energy. The rest of the chapter makes it so.

1.4. Localization

The flat representative spreads its mass over the whole space. The key observation, due to Perfekt [39] in the Hardy case, is that it can be cut down to the image under T_r of any rotation-invariant set, at the price of the measure of that set. Lemma 1.1 makes the proof the same in all three geometries.

LEMMA 1.5 (Localization). *Let D have positive measure and be invariant under the common rotations $w \mapsto e^{i\vartheta} w$. Then*

$$\psi_{r,D} = \frac{\|K_r\|_2^2}{\lambda(D)} \frac{K_r}{\overline{K_r}} \mathbf{1}_{T_r(D)}$$

represents evaluation at r, and $|\psi_{r,D}| = \|K_r\|_2^2/\lambda(D)$ on its support $T_r(D)$.

PROOF. Substitute $z = T_r w$ and use Lemma 1.1. Since $\overline{K_r}/K_r \cdot |K_r|^{-2} = K_r^{-2}$,

$$\int F\,\overline{\psi_{r,D}}\,\mathrm{d}\lambda = \frac{\|K_r\|_2^4}{\lambda(D)} \int_D G\,\mathrm{d}\lambda, \qquad G = \frac{F \circ T_r}{(K_r \circ T_r)^2}.$$

The function G is holomorphic near the closed domain (in the Gauss case it is entire, with growth at most $e^{c|w|}$). Since D and λ are invariant under common rotations, averaging $G(e^{i\vartheta}w)$ over ϑ shows $\int_D G\,\mathrm{d}\lambda = G(0)\lambda(D) = F(r)\lambda(D)/\|K_r\|_2^4$. □

The cost of localization is the factor $1/\lambda(D)$. As long as D has measure at least $1/2$, the localized representative is at most twice the flat one, and it lives on $T_r(D)$. That set carries at least half of the harmonic measure $\omega_r = (T_r)_*\lambda$.

1.5. The hyperbolic plane on the prime curve

Lift the right half-plane $\{\Re s > 1/2\}$ to the torus by $s \mapsto \gamma(s) = (p^{-s})_p$. Write d for the hyperbolic distance in the half-plane $\{\Re w > 0\}$, $w = s - 1/2$, with metric $|\mathrm{d}w|/\Re w$.

THEOREM 1.6. *Uniformly for $0 < \Re w, \Re w' \le 1$ and $|\Im w - \Im w'| \le 1$,*

$$E\big(\gamma(\tfrac{1}{2} + w), \gamma(\tfrac{1}{2} + w')\big) = 2 \log\cosh \frac{d(w, w')}{2} + O(1).$$

PROOF. Put $a_p = p^{-1/2-w}$, $b_p = p^{-1/2-w'}$. Then $\rho_p^2 = |a_p - b_p|^2/|1 - \overline{b_p} a_p|^2$, and $|a_p - b_p|^2(1 + |a_p b_p|)^{-2} \le \rho_p^2 \le |a_p - b_p|^2(1 - |a_p b_p|)^{-2}$. The difference contributes $O(\sum_p p^{-2}) = O(1)$. Expanding,

$$\sum_p |a_p - b_p|^2 = \mathcal{P}(1 + 2\Re w) + \mathcal{P}(1 + 2\Re w') - 2\Re\,\mathcal{P}(1 + w + \overline{w'}),$$

where $\mathcal{P}(1 + z) = \sum_p p^{-1-z} = \log\zeta(1 + z) + O(1) = \log(1/z) + O(1)$ uniformly for $0 < \Re z \le 2$ and $|\Im z| \le 1$, since ζ has only the simple pole at 1 there. The sum is therefore $\log\big(|w + \overline{w'}|^2/(4\Re w \Re w')\big) + O(1)$. For the right half-plane, $\cosh^2(d/2) = |w + \overline{w'}|^2/(4\Re w \Re w')$. □

So near the critical line, the geometry that governs harmonic measures on $\mathbb{T}^\infty$ is the hyperbolic plane, and $H(\omega, \omega') \lesssim \cosh(d/2)^{-1/2} \approx e^{-d/4}$. Hyperbolic area grows like e^d, while the number of points within distance d along a curve going to the boundary grows only like d. That difference explains why curves carry H^p-Carleson measures for every p (chapter 2), while vertical segments, which are horocycles with $e^{d/2}$ growth, fail at $p = 1$ (chapter 3).

1.6. Stability of point evaluation

For $F \in X^p$ define the efficiency at r, $\lambda_r^{(p)}(F) = |F(r)|^p/(\|K_r\|_2^2\|F\|_p^p) \in [0,1]$.

THEOREM 1.7 (Stability). *In each geometry, for $0 < p < \infty$, nonzero $F \in X^p$ and every point r,*

$$H\Big(\frac{|F|^p\lambda}{\|F\|_p^p},\, \omega_r\Big) \geq \lambda_r^{(p)}(F)^{1/2}, \qquad \textit{equivalently} \qquad \int |F|^{p/2}\kappa_r \,\mathrm{d}\lambda \geq |F(r)|^{p/2},$$

where $\kappa_r = |K_r|$.

PROOF. By Lemma 1.1, $K_r \circ T_r$ is holomorphic and zero-free near the closed domain, so $G = (F\circ T_r)\left(\|K_r\|_2^2/(K_r\circ T_r)^2\right)^{1/p}$ is holomorphic there, and $|G|^{p/2} = |F\circ T_r|^{p/2}\|K_r\|_2|K_r\circ T_r|^{-1}$. Substituting $z = T_r w$ and using the cocycle $\lambda\circ T_r = \|K_r\|_2^2|K_r\circ T_r|^{-2}\lambda$,

$$\int |F|^{p/2}\kappa_r \,\mathrm{d}\lambda = \int |F\circ T_r|^{p/2}\,\frac{\|K_r\|_2^2}{|K_r\circ T_r|}\,\mathrm{d}\lambda = \|K_r\|_2 \int |G|^{p/2}\,\mathrm{d}\lambda.$$

Since $\log|G|$ is plurisubharmonic, $|G|^{p/2}$ is subharmonic on complex lines through 0, and λ is invariant under common rotations. So $\int|G|^{p/2}\,\mathrm{d}\lambda \geq |G(0)|^{p/2} = |F(r)|^{p/2}\|K_r\|_2^{-1}$, using $K_r(r) = \|K_r\|_2^2$. Hence $\int|F|^{p/2}\kappa_r\,\mathrm{d}\lambda \geq |F(r)|^{p/2}$. The affinity form follows on dividing by $\|K_r\|_2\|F\|_p^{p/2}$. □

Taking $H \leq 1$ recovers the point estimate $|F(r)|^p \leq \|K_r\|_2^2\|F\|_p^p$. The theorem is that estimate with a defect term. For Banach-valued F the same proof applies with $\log\|F\|_X$ in place of $\log|F|$. In one variable it is a stability form of the factorization $F = Ih^{2/p}$, but its proof uses no factorization: the square root of $|F|^p$ plays the role of the H^2 vector.

COROLLARY 1.8 (Two-point necessity). *In the Hardy and Gauss geometries, if $\lambda_r^{(p)}(F), \lambda_s^{(p)}(F) \geq 1-\eta$ with $\eta < 1/2$, then $E(r,s) \leq 4\log\big(1/(1-2\eta)\big) = 8\eta + O(\eta^2)$. In the Bergman geometry the same holds with E replaced by βE.*

PROOF. The Hellinger angle $\arccos H$ is a metric (the angle between square-root densities in L^2). With $\mu = |F|^p\lambda/\|F\|_p^p$, $\arccos H(\omega_r,\omega_s) \leq 2\arccos\sqrt{1-\eta}$, so $H(\omega_r,\omega_s) \geq 1-2\eta$. Combine with $H \leq e^{-E/4}$ (Proposition 1.3). □

The bound is sharp up to a factor 2. The function K_t^2 at the coordinatewise hyperbolic midpoint t of r and s has efficiency $1 - E/4 + O(E^2)$ at both points. Along the prime curve, a Dirichlet series is $(1-\eta)$-efficient only on a set of scales of logarithmic diameter at most $7 + 4\log(1/(1-2\eta))$. So separation in energy is not only sufficient for adding up evaluations (chapter 2) but necessary.

For more than two points the same Hilbert-space picture does better than iterating the triangle inequality.

COROLLARY 1.9 (Many points). *In each geometry let $F \in X^p$ be nonzero, let $r_1, \dots, r_N$ be points, $\lambda_i = \lambda_{r_i}^{(p)}(F)$ and $H_{ij} = H(\omega_{r_i}, \omega_{r_j})$. Then for all $a_1, \dots, a_N \geq$*

0,

$$\Big(\sum_i a_i\sqrt{\lambda_i}\Big)^2 \le \sum_{i,j} a_i a_j H_{ij}.$$

In particular, if every $\lambda_i \ge 1-\eta$, the average of H_{ij} over the pairs $i<j$ is at least $1 - N\eta/(N-1)$.

PROOF. The square roots v of $|F|^p\lambda/\|F\|_p^p$ and w_i of ω_{r_i} are unit vectors in $L^2(\lambda)$ with $\langle v, w_i\rangle \ge \sqrt{\lambda_i}$ by Theorem 1.7. So $\sum_i a_i\sqrt{\lambda_i} \le \langle v, \sum_i a_i w_i\rangle \le \|\sum_i a_i w_i\|_2$. For the second statement take equal weights. □

Iterating the two-point estimate gives only $1-2\eta$ for each pair; for $N>2$ the average bound is better, and combined with $H \le e^{-E/4}$ it forces some pair to have energy at most $-4\log(1 - N\eta/(N-1))$.

Equality and near-extremizers

COROLLARY 1.10. *In each geometry, let $0<p<\infty$ and $F \in X^p$ with $F(r)\neq 0$.*

(1) $\lambda_r^{(p)}(F) = 1$ if and only if $F = cK_r^{2/p}$ for a constant c.

(2) With total variation $\|\cdot\|_{\mathrm{TV}}$ the total mass of the difference,

$$\Big\| \frac{|F|^p\lambda}{\|F\|_p^p} - \omega_r \Big\|_{\mathrm{TV}} \le 2\sqrt{1-\lambda_r^{(p)}(F)}.$$

PROOF. (1) Equality forces equality in $\int |G|^{p/2}\,\mathrm{d}\lambda \ge |G(0)|^{p/2}$ in the proof above. Disintegrating λ over complex lines through 0, the one-variable slices $g_w(\zeta) = G(\zeta w)$ satisfy $\int_{\mathbb{T}} |g_w|^{p/2}\,\mathrm{d}m = |g_w(0)|^{p/2} \neq 0$ for almost every w. A zero of g_w in $\mathbb{D}$ would make the inequality strict, so g_w is zero-free and $g_w^{p/2}$ is analytic with $\int|g_w^{p/2}| = |\int g_w^{p/2}|$; hence $g_w^{p/2}$ has constant argument on $\mathbb{T}$ and is constant. So G is constant, which is $F = cK_r^{2/p}$. (2) For probability measures, $\|\mu-\nu\|_{\mathrm{TV}} \le 2\sqrt{1-H(\mu,\nu)^2}$, and $H \ge \lambda^{1/2}$ by Theorem 1.7. □

So an $(1-\eta)$-efficient function at r distributes its mass like ω_r up to $2\sqrt{\eta}$ in total variation. The converse fails. In the Hardy geometry, let I be a Möbius map in one coordinate with $|I(r)| = (1-\eta)^{1/p}$ and put $F = IK_r^{2/p}$. Then $|F|^p\,\mathrm{d}m/\|F\|_p^p = \omega_r$ exactly, while $\lambda_r^{(p)}(F) = 1-\eta$. Modulus information cannot distinguish near-extremizers from inner perturbations of the extremizer; this is the phase blindness that limits the method in chapter 10.

The proofs above use T_r in every coordinate, which is harmless when r has finitely many nonzero coordinates. The equality statement holds on the whole domain of bounded point evaluations, in each geometry.

PROPOSITION 1.11 (Equality in infinitely many variables). *In each geometry, let $0<p<\infty$, let $r \in \ell^2$, with $r \in \mathbb{D}^\infty$ in the Hardy and Bergman cases, and let $F \in X^p$ be nonzero. Then $\lambda_r^{(p)}(F) = 1$ if and only if $F = cK_r^{2/p}$ for a constant c.*

PROOF. For s with finitely many nonzero coordinates put

$$V_s F = (F\circ T_s)\Big(\frac{\|K_s\|_2}{K_s\circ T_s}\Big)^{2/p},$$

with the branch positive at 0. By Lemma 1.1, V_s is an isometry of X^p onto itself, with $V_s F(0) = F(s)\|K_s\|_2^{-2/p}$. Let $r^{(N)}$ keep the first N coordinates of r. On a polynomial in the first N variables, $V_{r^{(M)}}P = V_{r^{(N)}}P \cdot B_{N,M}$ for $M > N$, where

$$B_{N,M} = \prod_{N<j\le M} k_{-r_j}(z_j)^{2/p},$$

because the formulas in the proof of Lemma 1.1 give $\|K_s\|_2/(K_s \circ T_s) = k_{-s}$ in each geometry. The logarithms of the factors are independent, with mean $O(|r_j|^2)$ and L^2 norm $O(|r_j|)$, so $\log B_{N,M} \to 0$ in L^2 uniformly in M, because $\sum_{j>N}|r_j|^2 \to 0$. Also $|B_{N,M}|^p$ is the squared modulus of a unit vector of X^2 close to 1, hence uniformly integrable; so $B_{N,M} \to 1$ in L^p uniformly in M. By density, $V_{r^{(N)}}$ converges strongly to an isometry V_r of X^p onto itself, with inverse obtained in the same way, and $V_r F(0) = F(r)\|K_r\|_2^{-2/p}$.

So $g = V_r F$ satisfies $|g(0)|^p = \|g\|_p^p$. Setting the coordinates beyond M to zero does not increase the X^p quasi-norm, by subharmonicity in those coordinates and rotation invariance of λ, and fixes $g(0)$; so each truncation g_M attains equality as well. In finitely many variables a nonconstant h with $h(0) \neq 0$ has a local power $h^{p/2}$ near 0, and Parseval on a small torus gives $\int|h(\rho z)|^p \,\mathrm{d}m > |h(0)|^p$. The means of $|h|^p$ over tori increase in each radius, so $\int|h|^p \,\mathrm{d}\lambda > |h(0)|^p$ in each geometry, which contradicts equality. Hence every g_M is the constant $g(0)$, and so is g. Then $F = V_r^{-1} g(0)$, a multiple of $K_r^{2/p}$. □

REMARK (Density of evaluations). Point evaluations at the curve points $\gamma(\sigma)$, $\sigma > 1$, already span a weak-$*$ dense subspace of $(H^1(\mathbb{T}^\infty))^*$, by uniqueness of Dirichlet coefficients. So superpositions of point evaluations, the functionals controlled in chapter 2, are weak-$*$ dense in the dual.

Open

QUESTION 1. Is the factor 2 in Corollary 1.8 necessary? What is the sharp many-point version of Corollary 1.9?

CHAPTER 2

The Hellinger–Bessel inequality

When can point evaluations be added up in H^p, for all p at once? The answer has a single source. By the stability theorem of chapter 1, the square root of $|F|^p$ is at least as aligned with the square root of the harmonic measure ω_r as the efficiency of F at r requires. Bessel's inequality then bounds the sum over many points by the Gram norm of the square roots $\sqrt{\omega_r}$. The results below are consequences of this one inequality; they hold for every $p > 0$, and the constants are explicit and small. To our knowledge the inequality itself, its sharpness on prime curves (Corollary 2.10) and the analysis of what it loses (Proposition 2.14) are new.

Throughout, the setting is one of the three geometries of section 1.1. We write $\kappa_r = |K_r|$ for the modulus of the reproducing kernel, so that $\kappa_r^2 = \|K_r\|_2^2 \, \mathrm{d}\omega_r/d\lambda$, and X^p for the analytic closure in $L^p(\lambda)$.

2.1. The inequality

THEOREM 2.1 (Hellinger–Bessel). *Let μ be a positive Borel measure on the set of admissible points, with the ℓ^2 topology. Let $\mathcal{G}_\mu$ be the integral operator on $L^2(\mu)$ with kernel*

$$\mathcal{G}(r,s) = \langle \kappa_r, \kappa_s \rangle = \|K_r\|_2 \|K_s\|_2 \, H(\omega_r, \omega_s).$$

Then for every $0 < p < \infty$ and $F \in X^p$,

$$\int |F(r)|^p \, \mathrm{d}\mu(r) \le \|\mathcal{G}_\mu\| \, \|F\|_p^p.$$

Moreover, for every positive weight w on $\operatorname{supp}\mu$,

$$\|\mathcal{G}_\mu\| \le \operatorname*{ess\,sup}_{r \in \operatorname{supp}\mu} \frac{1}{w(r)} \int \mathcal{G}(r,s)\, w(s) \, \mathrm{d}\mu(s).$$

With $w(r) = \|K_r\|_2$ the right side becomes $\sup_r \int H(\omega_r, \omega_s) \|K_s\|_2^2 \, \mathrm{d}\mu(s)$.

PROOF. Put $u = |F|^{p/2}$, so $\|u\|_2^2 = \|F\|_p^p$. By the stability theorem (Theorem 1.7), in the form $\int |F|^{p/2} \kappa_r \, \mathrm{d}\lambda \ge |F(r)|^{p/2}$, we have $|F(r)|^p \le \langle u, \kappa_r \rangle^2$. The map $r \mapsto \kappa_r \in L^2(\lambda)$ is continuous, because $\big\| \kappa_r/\|K_r\|_2 - \kappa_s/\|K_s\|_2 \big\|_2^2 = 2 - 2H(\omega_r, \omega_s)$ and $H \ge |\langle k_r, k_s \rangle|$. Let $V : L^2(\mu) \to L^2(\lambda)$, $Vc = \int c(r) \kappa_r \, \mathrm{d}\mu(r)$. Then $V^* u(r) = \langle u, \kappa_r \rangle$ and $V^* V = \mathcal{G}_\mu$. Hence

$$\int |F(r)|^p \, \mathrm{d}\mu \le \|V^* u\|^2 \le \|V\|^2 \|u\|^2 = \|\mathcal{G}_\mu\| \, \|F\|_p^p.$$

The weighted Schur test needs the row bound only μ-almost everywhere, since the kernel is non-negative. □

In X^2 the corresponding statement is Bessel's inequality for the kernels themselves, with $\langle K_r, K_s\rangle$ in place of $\mathcal{G}(r,s)$. Since $H(\omega_r,\omega_s) \ge |\langle k_r, k_s\rangle|$ (Proposition 1.3), the X^p condition is the stronger one. It is a single, p-independent condition that is sufficient for every p simultaneously.

COROLLARY 2.2 (Signed and complex measures). *If μ is a complex measure with $S = \operatorname{ess\,sup}_{|\mu|} \int H(\omega_r,\omega_s)\|K_s\|_2^2\, d|\mu|(s) < \infty$, then $h \mapsto \int h\,\mathrm{d}\mu$ is bounded on X^1 with norm at most S, and the Helson matrix with moments $\alpha(n) = \int r^{\kappa(n)}\,\mathrm{d}\mu(r)$ has a bounded symbol of norm at most S.*

PROOF. $|\int h\,\mathrm{d}\mu| \le \int |h|\, d|\mu|$. Apply Theorem 2.1 with $p = 1$, then Hahn–Banach. □

2.2. The Gram kernel along the curve

Let $\gamma(s) = (p^{-s})_p$ and $r(\sigma) = \gamma(1/2 + \sigma)$.

LEMMA 2.3. *The one-coordinate Hardy affinity satisfies $h(\rho) \le (1-\rho^2)^{1/4}$.*

PROOF. With $x = \rho^2$ and $c_k = (1/2)_k/k!$, $h = (1-x)^{1/2}\sum c_k^2 x^k$, and $(1-x)^{-1/4} = \sum (1/4)_k x^k/k!$. The claim $c_k^2 \le (1/4)_k/k!$ follows by induction, since $((k+1/2)/(k+1))^2 \le (k+1/4)/(k+1)$ is equivalent to $k/4 \ge 0$. □

LEMMA 2.4 (Gram kernel). *In the Hardy geometry,*

$$\mathcal{G}(r(\sigma), r(\tau)) \le \zeta(1+2\sigma)^{1/4}\zeta(1+2\tau)^{1/4}\zeta(1+\sigma+\tau)^{1/2}.$$

PROOF. In one coordinate with $a = p^{-1/2-\sigma}$, $b = p^{-1/2-\tau}$, $1-\rho(a,b)^2 = (1-a^2)(1-b^2)/(1-ab)^2$. So by Lemma 2.3 the factor $(1-a^2)^{-1/2}(1-b^2)^{-1/2}h(\rho)$ is at most $(1-a^2)^{-1/4}(1-b^2)^{-1/4}(1-ab)^{-1/2}$. Take the product over p. □

Near the boundary the kernel is homogeneous of degree -1. With $\zeta(1+x) \approx 1/x$,

$$\mathcal{G} \approx \frac{(\sigma\tau)^{-1/4}}{\sqrt{2}\,(\sigma+\tau)^{1/2}},$$

and in the variable $t = \log(\sigma/\tau)$, after the unitary $f \mapsto \sqrt{\sigma} f$, it becomes the convolution kernel $1/(2\sqrt{\cosh(t/2)})$. Its integral,

$$\int_{\mathbb{R}} \frac{\mathrm{d}t}{2\sqrt{\cosh(t/2)}} = B\big(\tfrac{1}{4},\tfrac{1}{2}\big) = \frac{\Gamma(1/4)^2}{\sqrt{2\pi}} \approx 5.244,$$

is the best constant the inequality can give near the boundary. The same chain with $|\langle K_r, K_s\rangle|$ in place of $\mathcal{G}$, which is the H^2 statement, gives the Carleman kernel with norm exactly π. The factor $5.244/\pi \approx 1.67$ is the price of discarding the phases of F (chapter 10).

2.3. Fejér–Riesz inequalities in every $\mathscr{H}^p$

Perfekt [39] proved the case $p = 1$ of the next theorem, with a constant of order $3\cdot 10^5$, by localizing the flat representative of chapter 1 and controlling the overlaps through separation and concentration estimates. The Hellinger–Bessel inequality replaces that machinery by a single Bessel inequality and gives every p at once. Interpolation could not give this, since it fails on $H^p(\mathbb{T}^\infty)$; for $p \neq 1$ these inequalities seem not to have been recorded before.

THEOREM 2.5. *For every $0 < p < \infty$ and $f \in \mathscr{H}^p$,*

$$\int_0^1 |f(1/2+\sigma)|^p \,\mathrm{d}\sigma \le (5+2\sqrt{2})\,\|f\|_{\mathscr{H}^p}^p.$$

PROOF. Apply Theorem 2.1 to Lebesgue measure on $(0,1)$, pushed forward by r. It suffices to bound the operator norm of the kernel $\overline{\mathcal{G}}(\sigma,\tau) = (1+1/(2\sigma))^{1/4}(1+1/(2\tau))^{1/4}(1+1/(\sigma+\tau))^{1/2}$, which dominates $\mathcal{G}$ by Lemma 2.4 and $\zeta(1+x) \le 1+1/x$. Split $(0,1) = I_0 \cup I_1$ with $I_0 = (0,1/10)$.

- *On I_0^2:* $\overline{\mathcal{G}} \le 6/5 \cdot 1/\sqrt{2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$. The latter kernel is unitarily a convolution by $1/(2\sqrt{\cosh(t/2)}) \le (\sqrt{2}/2)e^{-|t|/4}$, of norm at most $4\sqrt{2}$. So the block has norm $< 24\sqrt{2}/5 < 7$.
- *Off the diagonal block:* with $q(s) = 1 + (2s)^{-1/2}$ we have $\overline{\mathcal{G}}(s,t)^2 \le q(s)q(t)(1+1/(s+t))$. For $s \in I_0$, $t \in I_1$ use $(s+t)^{-1} \le t^{-1}$. The Hilbert–Schmidt norm squared is at most $\int_{I_0} q \cdot \int_{I_1} q(1+t^{-1}) = (1/10 + 1/\sqrt{5})(9/10 + \log 10 + \sqrt{20} - 1/\sqrt{5}) < 4$.
- *On I_1^2:* use $(s+t)^{-1} \le (2\sqrt{st})^{-1}$. The Hilbert–Schmidt norm squared is at most $A_1^2 + (1/2)C_1^2 < 8$, with $A_1 = \int_{I_1} q < 1.87$ and $C_1 = \int_{I_1} q(s)s^{-1/2}\,\mathrm{d}s < 3$.

The norm of the block operator is at most the top eigenvalue of the symmetric matrix with rows $(7,2)$ and $(2,3)$, which is $5+2\sqrt{2}$. Positivity of the kernels gives $\|\mathcal{G}\| \le \|\overline{\mathcal{G}}\|$. □

For $p = 1$ this is Perfekt's inequality with an explicit constant; the consequence for the multiplicative Hilbert matrix, $C < 9$, is Corollary 10.1.

One exact prime, and the exact local constant

The Gram bound loses the phases of all coordinates. In one coordinate they can be kept, by factorizing slices.

THEOREM 2.6. *Write $z = (z_2, w)$ and $a_\sigma = 2^{-1/2-\sigma}$, and let*

$$\mathcal{K}(\sigma,\tau) = \zeta(1+2\sigma)^{1/4}\zeta(1+2\tau)^{1/4}\zeta(1+\sigma+\tau)^{1/2}\Big(\frac{(1-a_\sigma^2)(1-a_\tau^2)}{(1-a_\sigma a_\tau)^2}\Big)^{1/4}.$$

For $0 < \delta \le \infty$, $0 < p < \infty$ and $F \in H^p(\mathbb{T}^\infty)$, $\int_0^\delta |F(r(\sigma))|^p \,\mathrm{d}\sigma \le \|\mathcal{K}_\delta\|\,\|F\|_p^p$, where $\mathcal{K}_\delta$ is the integral operator with kernel $\mathcal{K}$ on $L^2(0,\delta)$.

PROOF. For a polynomial F, factor each slice $F(\cdot,w) = I_w G_w^{2/p}$ with I_w inner and $G_w \in H^2(\mathbb{T})$ outer, and put $U(w) = G_w$, so $U \in L^2(\mathbb{T}^{\{p\ge 3\}}; H^2(\mathbb{T}))$ with $\|U\|^2 =$

$\|F\|_p^p$. Stability in the variables w (Theorem 1.7) and $|F(a,w)|^{p/2} \le |G_w(a)| = \|K_a\|\,|\langle G_w, k_a\rangle|$ give $|F(r(\sigma))|^{p/2} \le \|K_{r(\sigma)}\|\langle U, v_\sigma\rangle$ with the unit vector $v_\sigma(w) = \theta_\sigma(w) P_{r'(\sigma)}(w)^{1/2} k_{a_\sigma}$, θ_σ a phase. Then $|\langle v_\sigma, v_\tau\rangle| \le |\langle k_{a_\sigma}, k_{a_\tau}\rangle|\, H(\omega_{r'(\sigma)}, \omega_{r'(\tau)})$: the prime 2 contributes $(1-\rho_2^2)^{1/2}$ instead of the affinity $h(\rho_2) \le (1-\rho_2^2)^{1/4}$. With Lemma 2.3 for $p \ge 3$ this gives $\|K_{r(\sigma)}\|\|K_{r(\tau)}\||\langle v_\sigma, v_\tau\rangle| \le \mathcal{K}(\sigma,\tau)$, and the Bessel argument of Theorem 2.1 concludes. □

The factorization is not analytic in the remaining variables, so the argument cannot be repeated in a second coordinate.

THEOREM 2.7 (The local constant). $\|\mathcal{K}_2\| = \Gamma(1/4)^2/\sqrt{2\pi}$. *Hence, for every* $0 < p < \infty$ *and* $f \in \mathscr{H}^p$,

$$\int_0^2 |f(1/2+\sigma)|^p \,\mathrm{d}\sigma \le \frac{\Gamma(1/4)^2}{\sqrt{2\pi}}\, \|f\|_{\mathscr{H}^p}^p.$$

PROOF. *Lower bound.* As $\sigma, \tau \to 0$, $\mathcal{K}$ is asymptotic to

$$L_0(\sigma,\tau) = 2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2},$$

homogeneous of degree -1, with Mellin multiplier $2^{-1/2}B(1/4,1/4) = \Gamma(1/4)^2/\sqrt{2\pi}$ on the critical line; test with $\sigma^{-1/2+it}$ cut off to (e^{-2T}, e^{-T}).

Upper bound. Schur's test with $w(\tau) = \tau^{-1/2}e^{3\tau/10}$. For $\sigma \le 1/100$ it is analytic: $\zeta(1+x) \le x^{-1}(1 + 7x/12 + x^2/12)$ gives $\mathcal{K} \le L_0$ times explicit factors $1 + O(\sigma+\tau)$, and the saving $\int_2^\infty L_0(\sigma,\tau)\tau^{-1/2}\,\mathrm{d}\tau \asymp \sigma^{1/4}$ beats the loss $O(\sigma^{1/4})$ from these factors, the margin being 0.0926 at $\sigma = 1/100$. On $[1/100, 2]$ the row sums were bounded in ball arithmetic (`arb`) on 268 cells of ratio at most 1.02, with smallest margin 0.177. □

This replaces $5 + 2\sqrt{2}$ on $(0,1)$ by the exact near-boundary constant of the method on $(0,2)$. Numerically $\|\mathcal{K}_\delta\|$ stays equal to the same value up to some $\delta_* \in (2,3)$ and then grows.

2.4. Measures on curves

A set R of primes has *density* $\theta \in [0,1]$ *with constant* B if

$$\Big|\sum_{p\in R} p^{-1-t} - \theta \log\frac{1}{t}\Big| \le B \qquad (0 < t \le 2). \tag{2.1}$$

All primes have density 1. Unions of residue classes have rational densities. Sets with $\sum_{p\in R} 1/p < \infty$ have density 0. Let $r_R(\sigma) = (p^{-1/2-\sigma})_{p\in R}$.

LEMMA 2.8 (Energy and kernel growth). *Let* R *have density* $\theta \in (0,1]$ *with constant* B, *and* $u, v \in (0,1]$. *In each geometry,*

$$E\big(r_R(u), r_R(v)\big) \ge \theta|\log u - \log v| - \theta\log 4 - 4B.$$

The kernel norm $\|K_{r_R(\sigma)}\|_2^2$ *is at most* $e^{1+B}(2\sigma)^{-\theta}$, $e^{\beta(1+B)}(2\sigma)^{-\beta\theta}$ *and* $e^B(2\sigma)^{-\theta}$ *in the Hardy, Bergman and Gauss cases. We write* κ *for the exponent:* θ, $\beta\theta$ *and* θ *respectively.*

PROOF. Since $\rho(a,b) \ge |a-b|$ for $a,b \in [0,1)$, every energy is at least

$$\sum_{p\in R} (p^{-1/2-u} - p^{-1/2-v})^2.$$

Expanding gives three values of $\sum_{p\in R} p^{-1-t}$, with $t \in \{2u, 2v, u+v\} \subset (0,2]$. By (2.1) the sum is at least $\theta \log((u+v)^2/(4uv)) - 4B$, and $(u+v)^2/(4uv) \ge e^{|\log u - \log v|}/4$. For the kernels, use $-\log(1-x) \le x + x^2$ for $x \le 1/2$, $\sum_p p^{-2} < 1$, and (2.1) at $t = 2\sigma$. □

THEOREM 2.9 (Curves in three geometries). *Let R have density $\theta \in (0,1]$ with constant B, and let κ be the kernel exponent. There is $C = C(\theta, B, \text{geometry})$ such that, for every $0 < p < \infty$ and every F in the coordinates of R:*

(1) (block maximum) for every choice of $\tau_j \in [e^{-j-1}, e^{-j}]$,

$$\sum_{j\ge 0} e^{-j\kappa} |F(r_R(\tau_j))|^p \le C \|F\|_p^p;$$

(2) (measures) for every positive ν on $(0,1]$ with $[\nu]_\kappa = \sup_t \nu((0,t])/t^\kappa < \infty$,

$$\int |F(r_R(\sigma))|^p \, \mathrm{d}\nu(\sigma) \le C [\nu]_\kappa \|F\|_p^p.$$

For all primes in the Hardy geometry, (2) holds with $C < 99$.

PROOF. In each geometry $H(\omega_r, \omega_s) \le e^{-cE(r,s)}$, with $c = 1/4$ (Hardy and Gauss, Proposition 1.3) or $c = \beta/4$ (Bergman, Proposition 1.3).

(1) Apply Theorem 2.1 to $\mu = \sum_j e^{-j\kappa} \delta_{r_R(\tau_j)}$ with the weight $\|K\|$. Since $\|K_{r_R(\tau_k)}\|_2^2 \le C_K e^{(k+1)\kappa}$, the row sums are at most

$$C_K e^{\kappa} \sum_k \min\bigl(1, e^{-c(\theta(|j-k|-1) - \theta\log 4 - 4B)}\bigr),$$

which is bounded uniformly in j by Lemma 2.8.

(2) Apply Theorem 2.1 to ν itself with the weight $\|K\|$, and split $(0,1]$ into the blocks $I_k = (e^{-k-1}, e^{-k}]$. Then $\int_{I_k} \|K\|^2 \, \mathrm{d}\nu \le C_K e^{(k+1)\kappa} [\nu]_\kappa e^{-k\kappa}$, while for $\sigma \in I_j$ and $\tau \in I_k$ the affinity is at most $\min(1, e^{-c(\theta(|j-k|-1) - C_2)})$. Sum over k.

For all primes in the Hardy geometry, Perfekt's estimate $\sum_p (p^{-1/2-u} - p^{-1/2-v})^2 > |\log u - \log v| - 7$ and $\zeta(1+2\sigma) \le 3/(2\sigma)$ give the explicit bound $(3/2)e\bigl(17 + 2e^{-1/4}/(1-e^{-1/4})\bigr) < 99$. □

In the block maximum the points can be chosen to maximize $|F|$ in each block, which is what atomic measures need (chapter 4). Averaging over the position in the block recovers $\int_0^1 |F(r_R(\sigma))|^p \sigma^{\kappa-1} \, \mathrm{d}\sigma \lesssim \|F\|_p^p$. For prime sets this is the Fejér–Riesz inequality with weight $\sigma^{\theta-1}$; for the Bergman geometry with all primes it is $\int_0^1 |f(1/2+\sigma)|^p \sigma^{\beta-1} \, \mathrm{d}\sigma \lesssim \|f\|_{\mathscr{A}^p_\beta}^p$.

On the curves themselves the condition of Theorem 2.9 is also necessary, so there the Hellinger–Schur bound loses nothing.

COROLLARY 2.10 (Carleson measures on curves). *Let R have density $\theta \in (0,1]$ with constant B, let κ be the kernel exponent, and let ν be a positive measure on $(0,1]$. If $\int |F(r_R(\sigma))|^p \,\mathrm{d}\nu(\sigma) \le M\|F\|_p^p$ for some $0 < p < \infty$, then $[\nu]_\kappa \le CM$. Hence the following are equivalent: the embedding holds for one p; it holds for every p; $[\nu]_\kappa < \infty$; and the Gram operator $\mathcal{G}_\nu$ is bounded.*

PROOF. Test with $F_t = K_{r_R(t)}^{2/p}\|K_{r_R(t)}\|_2^{-2/p}$, of norm one. For $\sigma \le t$ the kernel estimates behind Lemma 2.8 give $|F_t(r_R(\sigma))|^p = |K_{r_R(t)}(r_R(\sigma))|^2/\|K_{r_R(t)}\|_2^2 \ge ct^{-\kappa}$, so $\nu((0,t]) \le CMt^\kappa$. Part (2) of the proof of Theorem 2.9 bounds $\|\mathcal{G}_\nu\|$ by $C[\nu]_\kappa$, and Theorem 2.1 closes the circle. □

2.5. Further consequences

COROLLARY 2.11 (Energy-separated families). *Let (r^j) be points with*

$$S = \sup_j \sum_k H(\omega_{r^j}, \omega_{r^k}) < \infty,$$

and let $c_j \ge 0$. The following are equivalent:

(1) $\sum c_j |F(r^j)|^p \lesssim \|F\|_p^p$ for one, equivalently every, $p > 0$;
(2) $\sum c_j |F(r^j)|^2 \lesssim \|F\|_2^2$;
(3) $c_j \lesssim \|K_{r^j}\|_2^{-2}$.

PROOF. The one-point condition and Theorem 2.1 with the weight $\|K\|$ give every p. Testing $p = 2$ with K_{r^j} gives the one-point condition. The summability of the affinities holds, for instance, when there are boundedly many near neighbours and $\sum_{\text{far } k} e^{-E_{jk}/4} < \infty$. □

Summability is used only to bound the affinity matrix on ℓ^2, and that bound is all the argument needs.

PROPOSITION 2.12 (Bounded affinity matrices). *(1) Let (r^j) be points whose affinity matrix $\mathbf{H} = (H(\omega_{r^j}, \omega_{r^k}))_{j,k}$ is bounded on ℓ^2, and let $c_j \ge 0$. Then $\sum c_j |F(r^j)|^p \lesssim \|F\|_p^p$ for one $p > 0$ if and only if $c_j \lesssim \|K_{r^j}\|_2^{-2}$, and then the Gram operator is bounded and the embedding holds for every $p > 0$.*

(2) In the Hardy and Gauss geometries there are energy-separated families with $\mathbf{H}$ bounded on ℓ^2 but $\sum_k H(\omega_{r^j}, \omega_{r^k}) = \infty$ for every j.

PROOF. (1) Testing with $K_{r^j}^{2/p}/\|K_{r^j}\|_2^{2/p}$, as in Corollary 2.10, gives $c_j\|K_{r^j}\|_2^2 \le C_p$. For the measure $\mu = \sum_j c_j\delta_{r^j}$, the map $f \mapsto (\sqrt{c_j} f(r^j))_j$ identifies $L^2(\mu)$ with ℓ^2 and turns $\mathcal{G}_\mu$ into $D\mathbf{H}D$, with D diagonal and $D_{jj} = (c_j\|K_{r^j}\|_2^2)^{1/2}$. So $\|\mathcal{G}_\mu\| \le C_p\|\mathbf{H}\|$, and Theorem 2.1 gives every p.

(2) Let T be the tree in which every vertex has degree 3, with adjacency operator A, and let $c = 3/5$. The weighted Schur test with weight $2^{-|v|/2}$, where $|v|$ is the distance to a fixed root, gives $\|A\| \le 2\sqrt{2}$. For $v \ne w$ exactly one neighbour of v is closer to w, so $h_w(v) = c^{d(v,w)}$ satisfies $((1+2c^2)I - cA)h_w = (1-c^2)e_w$, and $h_w \in \ell^2(T)$ since $2c^2 < 1$. As $(1+2c^2)I - cA \ge (1-\sqrt{2}c)^2 I$, the matrix

$(c^{d(v,w)})$ equals $(1-c^2)((1+2c^2)I - cA)^{-1}$ and is bounded, while its row sums $1 + 3c\sum_{n\ge 0}(2c)^n$ diverge.

Index the coordinates by the edges of T. To the vertex v assign the point r^v with coordinate t on each edge of the path from the root to v and 0 elsewhere. In the Hardy geometry choose t with $h(t) = c$; it exists because h is continuous, $h(0) = 1$, and $h(t) \le (1-t^2)^{1/4}$ (Lemma 2.3). In the Gauss geometry choose t with $e^{-t^2/4} = c$. Shared edges contribute affinity 1 and the $d(v,w)$ edges of the symmetric difference contribute c each, so $H(\omega_{r^v}, \omega_{r^w}) = c^{d(v,w)}$ by Proposition 1.3, and $E(r^v, r^w) = t^2 d(v,w)$. □

With $c_v = \|K_{r^v}\|_2^{-2}$ the Gram operator of the tree family is $\mathbf{H}$ itself, so it is bounded and every embedding holds. The Hellinger–Schur rows with the weight $\|K\|$ are the divergent row sums of $\mathbf{H}$, so a different weight is needed to show boundedness.

Corollary 2.13 (Independent general frequencies). *Let $b_j > 0$ be linearly independent over $\mathbb{Q}$, let $0 < q_j < 1$ with $\sum_j [-\log(1-q_j^2) - q_j^2] < \infty$, put $r_j(\sigma) = q_j e^{-b_j\sigma}$, and suppose that $A(\sigma) = \sum_j q_j^2 e^{-2b_j\sigma}$ satisfies $|A(\sigma) - \theta\log(1/\sigma)| \le M$ on $(0,1]$. Then for every $p > 0$,*

$$\int_0^1 |F(r(\sigma))|^p \, \sigma^{\theta-1}\,\mathrm{d}\sigma \le C\|F\|_p^p.$$

Proof. The proof of Theorem 2.9 uses only the analogue of (2.1). For Hardy spaces of general Dirichlet series $\sum a_n e^{-\lambda_n s}$ whose frequencies are generated by the b_j, this is the corresponding Fejér–Riesz inequality, by the identification of Defant and Schoolmann [20] with $H^p(\mathbb{T}^\infty)$. The literal condition "density of the frequencies" is not enough: without the remainder hypothesis the inequality can fail. □

Remark. Two further consequences need no new argument.

- *Vector-valued H^p.* For X-valued F, the function $\|F\|_X^{p/2}$ is plurisubharmonic. So stability, and with it every result above, holds for $H^p(\mathbb{T}^\infty; X)$ and every Banach space X.
- *Translation-average Bergman spaces.* Averaging Theorem 2.5 over horizontal translations gives $\int_0^1 |f(1/2+t)|^p \mu([0,t])\,\mathrm{d}t \le (5+2\sqrt{2})\|f\|^p_{\mathscr{A}^p_\mu}$ for the spaces of Gómez-Cabello, Lefèvre and Queffélec.

2.6. What the Gram bound sees

The inequality replaces each evaluation by the positive function $\kappa_r = |K_r|$, and so forgets the phases of the kernels. The cleanest way to see what this costs is to compare, in one coordinate, the exact Carleson constant $C(\mu) = \sup \int |F|^p \,\mathrm{d}\mu / \|F\|_p^p$ with $\|\mathcal{G}_\mu\|$ for measures that interpolate between a point and a circle.

Proposition 2.14. *In the Hardy geometry, in one coordinate and for $0 < r < 1$, let μ_k be the average of the point masses at the k points ωr, $\omega^k = 1$, and let m_r be normalized arc length on $\{|z| = r\}$. For every $0 < p < \infty$:*

(1) *$C(\mu_k) = 1/(1-r^{2k})$ and $C(\mu_k) \le \|\mathcal{G}_{\mu_k}\| \le kC(\mu_k)$. For fixed k the ratio tends to 1 as $r \to 1$ and as $r \to 0$.*
(2) *$C(m_r) = 1$, while $\|\mathcal{G}_{m_r}\| \ge c\log^2\bigl(1/(1-r)\bigr)$.*
(3) *For $\mu = (6/\pi^2)\sum_j j^{-2} m_{1-e^{-j}}$, $C(\mu) \le 1$ while $\mathcal{G}_\mu$ is unbounded.*
(4) *Separation does not help. Every probability measure ν on $\{|z| = r\}$ has $\|\mathcal{G}_\nu\| \ge c\log^2\bigl(1/(1-r)\bigr)$. For every $\delta < 1$ there are radii $r_j \to 1$ and grids ν_j, each the normalized counting measure on n_j equally spaced points of $\{|z| = r_j\}$, such that any two points of the union are at pseudo-hyperbolic distance at least δ, and $\mu = \sum_j j^{-3/2}\nu_j$ satisfies $C(\mu) < \infty$ for every p while $\mathcal{G}_\mu$ is unbounded.*

Proof. (1) The constant is Lemma 3.9, sharp. The Gram matrix is circulant with positive entries $(1-r^2)^{-1}h(\rho(r,\omega r))/k$, so its norm is the row sum $(1-r^2)^{-1}k^{-1}\sum_\omega h(\rho(r,\omega r))$. This lies between $(1-r^2)^{-1}/k$ and $(1-r^2)^{-1}$, and $k(1-r^2) \ge 1-r^{2k} \ge 1-r^2$. As $r \to 1$, $\rho(r,\omega r) \to 1$ for $\omega \ne 1$, so the row sum tends to 1 and the ratio to $\lim(1-r^{2k})/(k(1-r^2)) = 1$. As $r \to 0$, both sides tend to 1. (2) Subharmonicity gives $\int|F|^p\,\mathrm{d}m_r \le \|F\|_p^p$, with equality for constants. With V as in the proof of Theorem 2.1, $\|\mathcal{G}\| = \|V^*\|^2 \ge \int\langle 1,\kappa_z\rangle^2\,\mathrm{d}m_r(z)$, and $\langle 1,\kappa_z\rangle = \int_{\mathbb{T}}|1-\overline{z}\zeta|^{-1}\,\mathrm{d}m(\zeta) \asymp \log(1/(1-r))$. (3) Sum (2) over the circles. (4) Since (4) The bound in (2) used only that $\langle 1,\kappa_z\rangle$ depends on $|z|$ alone, so it holds for every probability measure on the circle. Let $\lambda,\eta > 0$, put $a_j = e^{-\lambda j}$, $r_j = 1-a_j$ and $n_j = \lceil \eta/a_j\rceil$, and keep $j \ge j_0$ with $\eta/a_{j_0} \ge 2$. Since $|F|^p$ is subharmonic, averaging the Poisson kernel over the grid gives $\int|F|^p\,\mathrm{d}\nu_j \le (1+r_j^{n_j})(1-r_j^{n_j})^{-1}\|F\|_p^p$, and $r_j^{n_j} \le e^{-\eta}$; the weights $j^{-3/2}$ are summable. For points on different circles, $\rho(z,w) \ge \rho(|z|,|w|) \ge (a_j-a_k)/(a_j+a_k) \ge \tanh(\lambda/2)$ when $j < k$. On one circle, two distinct points differ by an angle θ with $|\sin(\theta/2)| \ge 2/n_j \ge a_j/\eta$, and $1-r_j^2 \le 2a_j$, so

$$\rho^2 = \frac{4r_j^2\sin^2(\theta/2)}{(1-r_j^2)^2 + 4r_j^2\sin^2(\theta/2)} \ge \frac{r_j^2/\eta^2}{1+r_j^2/\eta^2}.$$

Both lower bounds exceed δ once λ is large and η small. Finally $\|\mathcal{G}_{\nu_j}\| \gtrsim \lambda^2 j^2$, so the compression of $\mathcal{G}_\mu$ to the jth grid has norm $\gtrsim \lambda^2 j^{1/2}$. □

So the Gram bound loses even when each kernel's support, of angular width about $1-r$, contains at most one point of the measure. The loss is in the tails: the exact formula of Proposition 1.3 gives $h(\rho) \sim (2/\pi)(1-\rho^2)^{1/2}\log\bigl(4(1-\rho^2)^{-1/2}\bigr)$ as $\rho \to 1$, so on a grid the affinity to the mth neighbour is about $m^{-1}\log m$, and the rows of the (circulant) affinity matrix sum to about $\log^2(1/(1-r))$. Thus the affinity matrix is unbounded on ℓ^2, which is exactly the hypothesis of Proposition 2.12 that fails. The loss comes from directions in which the kernels rotate: on a circle, $K_{\omega r}$ and K_r differ by a phase, and the cancellation that makes $\int|F|^p\,\mathrm{d}m_r \le \|F\|_p^p$ trivial is invisible to $|K|$. Along the prime curve the kernels at different heights have no relative rotation, every coordinate being real and positive, and Theorem 2.9 is efficient there. Rotational averages are handled instead by the symmetry mechanism of section 3.4, which uses analyticity in the rotated variable.

Open

QUESTION 2. On prime curves, and for families whose affinity matrix is bounded on ℓ^2, the Hellinger–Schur condition is necessary (Corollary 2.10 and Proposition 2.12); for merely separated families it is not (part (4) of Proposition 2.14). Is there a geometric condition on the family, between the two, under which it is necessary? What exactly does the passage from K_r to $|K_r|$ lose?

QUESTION 3. What is the sharp constant in Theorem 2.5? Gram methods give exactly $\Gamma(1/4)^2/\sqrt{2\pi}$ near the boundary (Theorem 2.7), and exactness in finitely many primes does not lower this. For $p = 2$ the truth is at most π, the norm of the multiplicative Hilbert matrix.

CHAPTER 3

Carleson measures near the critical line

Question. Which measures in $\mathbb{D}^\infty \cap \ell^2$, or in the half-plane $\mathbb{C}_{1/2}$, are Carleson measures for H^p, and when does the answer depend on p?

The Hellinger–Bessel inequality (chapter 2) gives a sufficient condition that does not see p. This chapter tests how far it reaches. It settles discrete families separated in energy completely, for every p at once; it handles twisted points, and hence complex measures on the half-plane, at no extra cost; and on vertical lines it gives the upper rate $\varepsilon^{-1/2}$ for every p. Averaging over roots of unity is a second, independent mechanism, and it too works for every p. The chapter ends with the exact norms of decimation operators. Apart from the embedding theorems quoted in Proposition 3.4, the results of the chapter are, to our knowledge, new.

3.1. Families separated in energy

For $r, s \in \mathbb{D}^\infty \cap \ell^2$ let $E(r,s) = \sum_p |a_p(r,s)|^2$, where a_p is the coordinatewise Möbius transition point, as in chapter 1.

THEOREM 3.1 (Energy-separated families). *Let $(r^j)_{j\in J}$ be a countable family in $\mathbb{D}^\infty \cap \ell^2$ with*

$$\Sigma := \sup_j \sum_k e^{-E(r^j,r^k)/4} < \infty,$$

and let $c_j \geq 0$. For each $0 < p < \infty$ the following are equivalent:

(1) $\sum_j c_j |F(r^j)|^p \leq C_p \|F\|_p^p$ for all $F \in H^p(\mathbb{T}^\infty)$;

(2) $c_j \leq C \|K_{r^j}\|_2^{-2}$ for all j.

One may take $C = C_p$ in (1) $\Rightarrow$ (2) and $C_p = \Sigma C$ in (2) $\Rightarrow$ (1). In particular the H^p Carleson measures of the form $\sum_j c_j \delta_{r^j}$ are the same for every p.

PROOF. (1) $\Rightarrow$ (2): test with $F = K_{r^j}^{2/p}$, which has $\|F\|_p^p = \|K_{r^j}\|_2^2$ and $|F(r^j)|^p = \|K_{r^j}\|_2^4$, and keep the jth term. (2) $\Rightarrow$ (1): apply Theorem 2.1 to $\mu = \sum_j c_j \delta_{r^j}$ with weight $\|K\|_2$. The row sums are $\sum_k c_k \|K_{r^k}\|_2^2 H(\omega_{r^j}, \omega_{r^k}) \leq C \sum_k e^{-E(r^j,r^k)/4}$ by Proposition 1.3. □

REMARK (Only in infinitely many variables). In $\mathbb{D}^d$ every energy is less than d, so $\Sigma \geq e^{-d/4} \# J$: a family satisfying the hypothesis is finite. In one variable, uniformly separated sequences are the analogue, but there the condition is a product of distances, and here it is a sum of squares over coordinates. In many coordinates far pairs are *very* far, and a random point of the torus sees essentially one localized kernel at a time.

3.2. Twists and the half-plane

For a Dirichlet series f and a completely multiplicative $\chi : \mathbb{N} \to \mathbb{T}$ write $f_\chi = \sum a_n \chi(n) n^{-s}$. Vertical translation is the case $\chi(n) = n^{-iy}$. The point is that twisting the evaluation points costs nothing: kernel norms depend only on moduli, and energies only increase, since $|a_p(r,s)| \geq \big||r_p| - |s_p|\big|$.

THEOREM 3.2 (Twisted block maximum). *Let R have density $\theta \in (0,1]$ with constant B. For every $0 < p < \infty$ and every $f \in \mathscr{H}^p$ supported on the R-integers,*

$$\sum_{j \geq 0} e^{-j\theta} \max_{\tau \in \overline{I_j}} \sup_\chi |f_\chi(1/2+\tau)|^p \leq C \|f\|^p_{\mathscr{H}^p},$$

$$\int_0^1 \sup_\chi |f_\chi(1/2+\sigma)|^p \sigma^{\theta-1} \, \mathrm{d}\sigma \leq C \|f\|^p_{\mathscr{H}^p},$$

with C as in Theorem 2.9.

PROOF. For a polynomial the supremum over χ is a maximum over a finite torus. Choose maximizing (τ_j, χ_j) and the points $r^j = (\chi_j(p) p^{-1/2-\tau_j})_p$. The proof of Theorem 2.9 (1) uses the points only through $\|K_{r^j}\|_2$ and lower bounds on the energies by moduli. The second estimate follows from the first, since $\int_{I_j} \sigma^{\theta-1} \, \mathrm{d}\sigma \leq e^{-j\theta}/\theta$. □

COROLLARY 3.3 (Complex measures on the half-plane). *Let μ be a complex measure on $\mathbb{C}_{1/2}$ with*

$$[\mu] = \sup_{0<t\leq 1} \frac{|\mu|(\{1/2 < \Re s \leq 1/2+t\})}{t} < \infty.$$

Then $\int |f|^p \, d|\mu| \leq (C[\mu] + \zeta(3)|\mu|(\mathbb{C}_{1/2}))\|f\|^p_{\mathscr{H}^p}$ for every $0 < p < \infty$, and the Helson matrix with moments $\alpha(n) = \int n^{-s} \, \mathrm{d}\mu(s)$ has a bounded symbol.

PROOF. On the strip $\Re s - 1/2 \in I_j$, $|f(s)| \leq \max_\tau \sup_\chi |f_\chi(1/2+\tau)|$. Apply Theorem 3.2; for $\Re s > 3/2$ use $|f(s)|^p \leq \zeta(2\Re s)\|f\|_p^p$. With $p = 1$, Hahn–Banach gives the symbol. □

The strip condition is sufficient but not necessary.

PROPOSITION 3.4 (Necessity on the half-plane).
(1) An $\mathscr{H}^p$ Carleson measure satisfies the box condition $|\mu|(\{\Re s - 1/2 \leq \delta,\ |\Im s - y| \leq \delta\}) \lesssim \delta$. For $2 \leq p < \infty$ and μ supported in a bounded box above the line, the box condition is also sufficient: this is the local embedding theorem of Chen, Fang, Guo, Hou, Shao and Zhou [17], which for $p = 2$ goes back to the Montgomery–Vaughan inequality, combined with Carleson's theorem and the box test of Olsen and Saksman [36].

(2) The strip condition is not necessary for $\mathscr{H}^2$: with λ_ε Lebesgue measure on $\{1/2 + \varepsilon + it : 0 \leq t \leq 1\}$ and $\varepsilon_k = e^{-k^2}$, the measure $\sum_k 2^{-k} \lambda_{\varepsilon_k}$ is $\mathscr{H}^2$-Carleson.

(3) There is an explicit $\mathscr{H}^2$- but not $\mathscr{H}^1$-Carleson measure of the form $\sum_k 2^{-k} \lambda_{\varepsilon_k}$.

Proof. (1) Test with $\zeta(s+1/2+\delta-iy)^{2/p}$ truncated to finitely many primes, using $|\zeta(1+w)|\geq 1/|w|-3$, and let the truncation grow. (2) The local embedding for $\mathscr{H}^2$ and contractivity of horizontal translation make each λ_ε uniformly Carleson. (3) The norms $N_1(\varepsilon)=\sup\int|f|\,\mathrm{d}\lambda_\varepsilon/\|f\|_{\mathscr{H}^1}$ are finite but unbounded, for otherwise $\int_0^1|f(1/2+it)|\,\mathrm{d}t\lesssim\|f\|_{\mathscr{H}^1}$, which Harper's theorem [22] rules out. Choose ε_k with $N_1(\varepsilon_k)\geq 8^k$. □

Perfekt and Pushnitski [40] observed that the $\mathscr{H}^1$ and $\mathscr{H}^2$ Carleson classes differ, and asked whether they coincide for measures on real points. The measure in part (3) is an explicit witness, but it lives on complex points. For real points the question is Problem 3, and by chapter 4 and Theorem 3.1 every family we can handle has the same Carleson class for all p.

3.3. Vertical lines

The most prominent open problem about Carleson measures for Dirichlet series concerns a single segment of the critical line.

Problem 1 (The embedding problem; Saksman and Seip [43, Problem 2.1]). *For which $0<p<\infty$ is there a constant C_p with $\int_0^1|f(1/2+it)|^p\,\mathrm{d}t\leq C_p\|f\|_{\mathscr{H}^p}^p$ for every Dirichlet polynomial f?*

The answer is yes for $p=2$ and hence for $p\in 2\mathbb{N}$, by applying the case $p=2$ to $f^{p/2}$. It is no for every $0<p<2$: Harper's low-moment estimates [22] show that for $f=\sum_{n\leq x}n^{-s}$ the pth power of the $\mathscr{H}^p$ norm is smaller than $x^{p/2}$ by a power of $\log\log x$, while $\int_0^1|f(1/2+it)|^p\,\mathrm{d}t\asymp x^{p/2}$. Chen, Fang, Guo, Hou, Shao and Zhou [17] have announced an affirmative answer for every $p>2$, by reducing the dual problem to a log-correlated Gaussian field and a critical branching random walk; if their proof stands, the answer is exactly $p\geq 2$. We study the problem at a fixed distance from the line. For $0<\varepsilon\leq 1/2$ let

$$N_p(\varepsilon)=\sup_{f\neq 0}\frac{\int_0^1|f(1/2+\varepsilon+it)|^p\,\mathrm{d}t}{\|f\|_{\mathscr{H}^p}^p}.$$

For $p\in 2\mathbb{N}$ these are bounded, and for $p=1$ they are unbounded (Proposition 3.4). The following theorem gives a rate valid for every p; it is our contribution to Problem 1.

Theorem 3.5. *There is an absolute C with $N_p(\varepsilon)\leq C\varepsilon^{-1/2}$ for all $0<p<\infty$ and $0<\varepsilon\leq 1/2$.*

Proof. Let $r_t=\gamma(1/2+\varepsilon+it)$. Then $\|K_{r_t}\|_2^2=\zeta(1+2\varepsilon)\leq\varepsilon^{-1}$ for all t. By Theorem 1.6 and Proposition 1.3, with $\cosh^2(d/2)=1+(t-u)^2/(4\varepsilon^2)$ for two points at the same height,

$$H(\omega_{r_t},\omega_{r_u})\leq C_0\bigl(1+(t-u)^2/(4\varepsilon^2)\bigr)^{-1/4}\qquad(|t-u|\leq 1).$$

Apply Theorem 2.1 to $\zeta(1+2\varepsilon)^{-1}\,\mathrm{d}t$ on $[0,1]$; the Schur bound is $C_0\sup_t\int_0^1(1+(t-u)^2/(4\varepsilon^2))^{-1/4}\,\mathrm{d}u\leq C_0(4\varepsilon+4\sqrt{2\varepsilon})$, and multiplying back by $\zeta(1+2\varepsilon)$ gives $C\varepsilon^{-1/2}$. □

The exponent $1/2$ is what one point evaluation per unit hyperbolic length gives: the hyperbolic length of $[0,1]$ at height ε is $1/\varepsilon$, and the affinity decays like $e^{-d/4}$, so the Gram operator sees $\varepsilon^{-1}\cdot\varepsilon^{1/2}$. What the true rate is for $p<2$, or even whether it is a power, is open; Harper's work on low moments suggests $(\log 1/\varepsilon)^{1/4}$ at $p=1$ (B5).

Translating vertically, the theorem answers the remaining question about the strip condition: it is not necessary, for any p.

COROLLARY 3.6. *The measure* $\mathrm{d}\mu=\varepsilon^{-1/4}e^{-|t|}\,\mathrm{d}\varepsilon\,\mathrm{d}t$ *on* $\{1/2+\varepsilon+it:0<\varepsilon<1/4,\ t\in\mathbb{R}\}$ *is an* $\mathscr{H}^p$ *Carleson measure for every* $0<p<\infty$, *but* $\mu(\{1/2<\Re s\le 1/2+\delta\})=(8/3)\delta^{3/4}$, *so it violates the strip condition.*

PROOF. Vertical translation preserves the $\mathscr{H}^p$ norm, so Theorem 3.5 bounds $\int_k^{k+1}|f(1/2+\varepsilon+it)|^p\,\mathrm{d}t$ by $C\varepsilon^{-1/2}\|f\|_{\mathscr{H}^p}^p$ for every integer k. Summing against $e^{-|t|}$ and integrating $\varepsilon^{-3/4}$ over $(0,1/4)$ gives the embedding. □

The measure of Corollary 3.6 also satisfies the Hellinger–Schur condition of Theorem 2.1, though with a weight other than $\|K\|$. Let $r=\gamma(1/2+a+iy)$ and $s=\gamma(1/2+\varepsilon+it)$, with $\gamma(z)=(q^{-z})_q$. Since $|r_qs_q|\le 1/q$, we have $\rho(r_q,s_q)^2\ge(1-2/q)|r_q-s_q|^2$, and $\Re\sum_q q^{-z}=\log|\zeta(z)|+O(1)$ for $\Re z>1$. Hence

$$E(r,s)\ge\log\zeta(1+2a)+\log\zeta(1+2\varepsilon)-2\log|\zeta(1+a+\varepsilon+i(t-y))|-C.$$

With $H\le e^{-E/4}$ (Proposition 1.3), $\zeta(1+x)\le 1+1/x$ for $x>0$, and $|\zeta(1+w)|\le C/|w|$ for $|w|\le 3$ and $\le C\log(2+|w|)$ otherwise, the Gram kernel satisfies $\mathcal{G}(r,s)\le C(a\varepsilon)^{-1/4}k(t-y)$, where $k(\tau)=|\tau|^{-1/2}$ for $|\tau|\le 2$ and $k(\tau)=\log^{1/2}(2+|\tau|)$ otherwise. Schur's test with the weight $\varepsilon^{-1/4}e^{|t|/2}$ bounds every row by

$$C\int_0^{1/4}\varepsilon^{-3/4}\,\mathrm{d}\varepsilon\cdot\sup_y e^{-|y|/2}\int_{\mathbb{R}}e^{-|t|/2}k(t-y)\,\mathrm{d}t<\infty.$$

So this example is consistent with the Hellinger–Schur condition being necessary below $p=2$.

For $p=2$ the limit of $N_2(\varepsilon)$ is the constant of the local embedding on the line itself. The author's thesis proposed a value for it.

PROBLEM 2 (Søvik [45, Section 4.4]). *Is the least constant* C_0 *in* $\int_0^1|f(1/2+it)|^2\,\mathrm{d}t\le C_0\|f\|_{\mathscr{H}^2}^2$ *equal to* 2*?*

The answer is no, and the constant can be located quite precisely.

PROPOSITION 3.7 (The constant at $p=2$). *Let* C_0 *be the least constant with* $\int_0^1|P(1/2+it)|^2\,\mathrm{d}t\le C_0\|P\|_{\mathscr{H}^2}^2$ *for all Dirichlet polynomials* P. *Then* $N_2(\varepsilon)\le C_0$ *for every* ε, $N_2(\varepsilon)\to C_0$ *as* $\varepsilon\to 0$, *and*

$$2\pi\le C_0<2\pi+1/8.$$

PROOF. Horizontal translation is a contraction on $\mathscr{H}^2$, so $N_2(\varepsilon)\le C_0$; conversely, for a fixed polynomial P the integral over $1/2+\varepsilon+[0,1]i$ tends to the one over the line, so $\liminf N_2(\varepsilon)\ge C_0$.

For the lower bound, write $\zeta(1+w) = 1/w + h(w)$ with h bounded on $\{|w| \le 2\}$, and test $N_2(\varepsilon)$ with $f(s) = \zeta(1/2 + \eta + s - i/2)$, for which $\|f\|^2_{\mathscr{H}^2} = \zeta(1+2\eta)$. With $\delta = \eta + \varepsilon$,

$$\int_0^1 |f(1/2+\varepsilon+it)|^2 \,\mathrm{d}t = \int_{-1/2}^{1/2} \frac{\mathrm{d}v}{\delta^2+v^2} + O\big(\log(1/\delta)\big) = \frac{\pi}{\delta} + O\big(\log(1/\delta)\big),$$

and $\zeta(1+2\eta) = 1/(2\eta) + O(1)$. Taking $\eta = \sqrt{\varepsilon}$, the quotient tends to 2π as $\varepsilon \to 0$.

For the upper bound, let $T_\varepsilon a(t) = \sum_n a_n n^{-1/2-\varepsilon-it}$ on $[0,1]$, so that $N_2(\varepsilon) = \|T_\varepsilon\|^2$. Then $T_\varepsilon T_\varepsilon^*$ has kernel $\zeta(1+2\varepsilon+i(t-u)) = k(t-u) + \psi(1+2\varepsilon+i(t-u))$, where $k(x) = 1/(2\varepsilon+ix) = \int_0^\infty e^{-(2\varepsilon+ix)y}\,\mathrm{d}y$ and $\psi(s) = \zeta(s) - 1/(s-1)$. Convolution with k on $L^2(\mathbb{R})$ is the Fourier multiplier $2\pi e^{2\varepsilon\xi}\mathbf{1}_{\xi<0}$, which lies below $2\pi\mathbf{1}_{\xi<0}$. Compressing to $[0,1]$, $T_\varepsilon T_\varepsilon^* \le S + \Psi_\varepsilon$, where S is the compression of $2\pi\mathbf{1}_{\xi<0}$ and Ψ_ε has kernel $\psi(1+2\varepsilon+i(t-u))$. In terms of the finite Hilbert transform $Hf(t) = \mathrm{PV}\int_0^1 f(u)(t-u)^{-1}\,\mathrm{d}u$, $S = \pi I - iH$.

The operator S can be diagonalized. The substitution $t = e^x/(1+e^x)$ gives a unitary map $Uf(x) = f(t)(\mathrm{d}t/\,\mathrm{d}x)^{1/2}$ from $L^2(0,1)$ onto $L^2(\mathbb{R})$. Since $t-u = \sinh((x-y)/2)/(2\cosh(x/2)\cosh(y/2))$, it carries H to convolution with $\mathrm{PV}\,1/(2\sinh(x/2))$, the Fourier multiplier $-i\pi\tanh(\pi\xi)$. So S becomes multiplication by $\pi(1-\tanh(\pi\xi))$, and $0 \le S \le 2\pi$. The constant function goes to $1/(2\cosh(x/2))$, whose Fourier transform is $\pi\,\mathrm{sech}(\pi\xi)$. Substituting $s = \pi(1-\tanh(\pi\xi))$,

$$\langle \phi(S)1, 1\rangle = \frac{1}{2\pi}\int_0^{2\pi} \phi(s)\,\mathrm{d}s$$

for bounded Borel ϕ: the spectral measure of 1 is uniform on $[0, 2\pi]$.

Let $J = 1 \otimes 1$ and $c > 0$. For $\lambda > 2\pi$ the rank-one operator $c(\lambda - S)^{-1/2}J(\lambda - S)^{-1/2}$ has norm $c\langle(\lambda-S)^{-1}1,1\rangle = (c/2\pi)\log(\lambda/(\lambda-2\pi))$. When this is less than 1, $\lambda - S - cJ > 0$. Hence

$$S + cJ \le \lambda_c = \frac{2\pi}{1-e^{-2\pi/c}}.$$

Take $c = \psi(1) = \gamma$, Euler's constant. Then $\lambda_\gamma < 2\pi + 2\cdot 10^{-4}$ and $T_\varepsilon T_\varepsilon^* \le \lambda_\gamma + R_\varepsilon$, where $R_\varepsilon = \Psi_\varepsilon - \gamma J$ has kernel $\psi(1+2\varepsilon+i(t-u)) - \psi(1)$. By the Euler–Maclaurin formula,

$$\psi(s) = \frac{1}{2} + \frac{s}{12} - \frac{s(s+1)}{2}\int_1^\infty \widetilde{B}_2(x)x^{-s-2}\,\mathrm{d}x,$$

where $\widetilde{B}_2$ is the periodic Bernoulli function, with $|\widetilde{B}_2| \le 1/6$. Differentiating, and using $\int_1^\infty x^{-3}\,\mathrm{d}x = 1/2$ and $\int_1^\infty x^{-3}\log x\,\mathrm{d}x = 1/4$, for $s = 1+iv$ with $|v| \le 1$

$$|\psi'(s)| \le \frac{1}{12} + \frac{\sqrt{13}}{24} + \frac{\sqrt{10}}{48} < \frac{3}{10},$$

and the same bound holds up to $O(\varepsilon)$ for $1 \le \Re s \le 1+2\varepsilon$. So the kernel of R_ε is at most $(3/10 + O(\varepsilon))|t-u| + O(\varepsilon)$. Its Hilbert–Schmidt norm is at most $(3/10)(\int_0^1\int_0^1 (t-u)^2\,\mathrm{d}t\,\mathrm{d}u)^{1/2} + O(\varepsilon) = 3/(10\sqrt{6}) + O(\varepsilon) < 0.1225 + O(\varepsilon)$. Hence $N_2(\varepsilon) < 2\pi + 0.0002 + 0.1225 + O(\varepsilon)$, and letting $\varepsilon \to 0$ gives the upper bound. □

This answers Problem 2 in the negative; already a five-term Dirichlet polynomial gives a quotient above 2. The upper bound improves the constant $5\pi/2$ that follows from Brevig's weighted inequality $\pi^{-1}\int_{\mathbb{R}}|f(1/2+it)|^2(1+t^2)^{-1}\,\mathrm{d}t \le 2\|f\|^2_{\mathscr{H}^2}$.

The proof shows more. The edge 2π is the top of the spectrum of S, and the constant function sits at the edge with a logarithmically divergent resolvent. So any positive multiple of J pushes an eigenvalue above 2π, by the exponentially small amount $2\pi/(e^{2\pi/c}-1)$. The true remainder Ψ is not a multiple of J, so this heuristic does not decide whether $C_0 > 2\pi$. A computation does.

PROPOSITION 3.8 (Above the edge). *Let $A\colon \mathscr{H}^2 \to L^2(0,1)$ be the bounded extension of $P \mapsto P(1/2+it)$.*

(1) $AA^* = S + \Psi$, where Ψ has the continuous kernel $\psi(1+i(t-u))$.
(2) If $C_0 > 2\pi$, then C_0 is an isolated eigenvalue of AA^* of finite multiplicity, and there is $f \in \mathscr{H}^2$ with $\|Af\|^2 = C_0\|f\|^2_{\mathscr{H}^2}$.
(3) $2\pi + 2.8\cdot 10^{-7} < C_0 < 2\pi + 10^{-3}$.

PROOF. (1) With D_ε the diagonal operator $n^{-\varepsilon}$, $T_\varepsilon = AD_\varepsilon$ and $D_\varepsilon^2 \to I$ strongly, so $T_\varepsilon T_\varepsilon^* \to AA^*$ strongly. In the proof of Proposition 3.7, the compressed multipliers converge strongly to S, and $\Psi_\varepsilon \to \Psi$ uniformly.

(2) Since Ψ is compact and $S \le 2\pi$, the essential spectrum of AA^* lies in $[0, 2\pi]$. Hence the spectrum above 2π consists of isolated eigenvalues of finite multiplicity, and $C_0 = \|AA^*\|$ is one of them. If $AA^*g = C_0 g$ with $g \neq 0$, then $f = A^*g$ is nonzero and $\|Af\|^2 = \langle AA^*g, AA^*g\rangle = C_0\|f\|^2$.

(3) For the lower bound, take $g = \sum_{n<192} a_n P_n(2t-1)$ with rational a_n, real for even n and imaginary for odd n, found by an eigenvalue computation. By (1), $\langle AA^*g, g\rangle = \pi\|g\|^2 + \langle -iHg, g\rangle + \langle \Psi g, g\rangle$. The Hilbert form is exact: $\int_0^1 P_n(2t-1)\,(HP_m(2\cdot -1))(t)\,\mathrm{d}t$ equals $2/((n-m)(n+m+1))$ when $n+m$ is odd, and 0 otherwise. On $|w| \le 1$, $\psi(1+w)$ is replaced by the degree-96 Taylor polynomial of an Euler–Maclaurin approximant with cutoff 64. The error is below $2\cdot 10^{-18}$ by the remainder formula and Cauchy's estimate, and the moments of g are exact. Exact rational arithmetic then gives $\langle AA^*g, g\rangle > (2\pi + 2.8\cdot 10^{-7})\|g\|^2$.

For the upper bound, repeat the proof of Proposition 3.7 with the rank-two model $c + ia(t-u)$ in place of γJ, where c and a are rationals close to γ and $-\gamma_1$ and $\psi(1+w) = \gamma - \gamma_1 w + \cdots$. Its positive part is $\mu\, q \otimes q$ with q in the span of 1 and $t - 1/2$. Under U, $t - 1/2$ goes to a multiple of ξ times the image of 1, so $\langle(\lambda - S)^{-1}q, q\rangle$ is an explicit integral over $[0, 2\pi]$ with logarithmic and dilogarithmic terms. At $\lambda = 2\pi + 1/340000$, $\mu\langle(\lambda - S)^{-1}q, q\rangle < 0.999$, in exact arithmetic. The remainder kernel has Hilbert–Schmidt norm below $9.7\cdot 10^{-4}$, by exact integration of the Taylor polynomial above. □

3.4. Averages over roots of unity

Between a single evaluation and the full circle average there is a ladder: average over the kth roots of unity.

LEMMA 3.9. *For $k \geq 2$, $0 < p < \infty$, $x \in \mathbb{D}$ and $g \in H^p(\mathbb{T})$,*

$$\frac{1}{k} \sum_{\omega^k=1} |g(\omega x)|^p \leq \frac{\|g\|_{H^p}^p}{1-|x|^{2k}},$$

with equality for $g = (1-\overline{x}^k z^k)^{-2/p}$. For $k = 2$ this is $(|g(x)|^p + |g(-x)|^p)/2 \leq \|g\|_p^p/(1-|x|^4)$.

PROOF. Write $g = Ih^{2/p}$ with I inner and h outer, so $|g(\omega x)|^p \leq |h(\omega x)|^2$ and $\|h\|_2^2 = \|g\|_p^p$. The Gram matrix of the Szegő kernels at the points ωx is circulant over the group of kth roots of unity, with eigenvalues $\sum_{n \equiv -m \ (k)} k|x|^{2n}$, $0 \leq m < k$. The largest, at $m = 0$, is $k/(1-|x|^{2k})$, with eigenvector the constant vector, that is $k^{-1}\sum_\omega k_{\omega x} = (1-\overline{x}^k z^k)^{-1}$. So $k^{-1}\sum_\omega |h(\omega x)|^2 \leq (1-|x|^{2k})^{-1}\|h\|_2^2$, with equality for that h and $I = 1$. □

As $k \to \infty$ the constant tends to 1, which is the trivial bound for the full circle average. For a single point ($k = 1$) the constant is $1/(1-|x|^2)$, the Cole–Gamelin bound.

PROPOSITION 3.10 (Root-of-unity averages in every H^p). *Let $0 < p < \infty$, let Q be a set of coordinates with orders $k_q \geq 2$ and points $x_q \in \mathbb{D}$, and fix the remaining coordinates at w. With independent ω_q uniform on the k_qth roots of unity,*

$$\mathbb{E}_\omega |F((\omega_q x_q)_{q\in Q}, w)|^p \leq \prod_{q\in Q} \frac{1}{1-|x_q|^{2k_q}} \int_{\mathbb{T}^Q} |F(\zeta, w)|^p \, \mathrm{d}m(\zeta).$$

For $x = \gamma(s)$ with $\Re s \geq 1/2$ and all $k_q \geq k$, the constant is at most $\zeta(k)$; for signs it is $\zeta(4s) \leq \zeta(2)$.

PROOF. Induct on the coordinates, applying Lemma 3.9 in one and Fubini in the rest; $\prod_q (1-q^{-2k_q\Re s})^{-1} \leq \prod_q (1-q^{-k})^{-1}$. □

So averaging over kth roots in each coordinate replaces the domain ℓ^2 of point evaluations by ℓ^{2k}: the averaged evaluation is bounded on every H^p at points where no single evaluation is, and the higher the order, the more of $\mathbb{D}^\infty$ it reaches. The analytic average is decimation in disguise: $\mathbb{E}_\omega F(\omega x) = (D_{k,Q}F)(x^k)$ in the notation of section 3.5; the proposition controls instead the average of $|F|^p$, which decimation does not.

Let R have density θ, Q its complement, orders $k_q \geq 2$ on Q, and ν a finite positive measure on $(1/2, \infty)$. Let $\mu_{\nu,R}$ be the law of $\big((\omega_q q^{-s})_{q\in Q}, (p^{-s})_{p\in R}\big)$ under $d\omega \times d\nu(s)$. Its moments are $\alpha(n) = \mathbf{1}[k_q \mid v_q(n) \ \forall q \in Q] \int n^{-s} \, \mathrm{d}\nu(s) \geq 0$. For signs, $k_q = 2$, these are the measures of chapter 4.

COROLLARY 3.11 (One Carleson class for all p). *For every $0 < p < \infty$, $\mu_{\nu,R}$ is a Carleson measure for $H^p(\mathbb{T}^\infty)$ if and only if $[\nu]_\theta = \sup_t \nu((1/2, 1/2+t])/t^\theta < \infty$, and then, with $k = \min_q k_q$,*

$$\int |F|^p \, \mathrm{d}\mu_{\nu,R} \leq \zeta(k)\big(C[\nu]_\theta + \zeta(3)\nu((1/2,\infty))\big)\|F\|_p^p.$$

In particular the Helson matrix $(\alpha(mn))$ has a bounded symbol of norm at most the constant for $p = 1$.

PROOF. Sufficiency: Proposition 3.10 in the Q-coordinates, then Theorem 2.9 (2) on each slice $F(\zeta, \cdot) \in H^p(\mathbb{T}^R)$, then Fubini. Necessity: functions of the R-coordinates only do not see the averaging, and testing with $K^{2/p}$ for kernels at $\gamma_R(1/2+\delta)$ reduces to the H^2 test with K, which gives $[\nu]_\theta \le 2e^B C_2$, C_2 the H^2 Carleson constant, as in section 4.5. The symbol comes from Hahn–Banach. □

Compactness needs a Montel lemma in infinitely many variables.

LEMMA 3.12. *Let $0 < p < \infty$. Every bounded sequence in $H^p(\mathbb{T}^\infty)$ has a subsequence that converges uniformly on every compact subset of $\mathbb{D}^\infty \cap \ell^2$, compact in the ℓ^2 topology.*

PROOF. Point evaluation satisfies $|F(r)| \le \|F\|_p \prod_j (1-|r_j|^2)^{-1/p}$, which is at most $\|F\|_p \exp(M^2/(p(1-\rho^2)))$ when $\|r\|_2 \le M$ and $\|r\|_\infty \le \rho < 1$. A compact set K has $\sup_K \|r\|_2 < \infty$ and $\sup_K \|r\|_\infty < 1$, and the same bounds hold, with room to spare, on a fixed ℓ^2-neighbourhood of K and of its truncations $r^{(D)} = (r_1, \dots, r_D, 0, \dots)$. So the sequence is uniformly bounded there, and Cauchy's estimate along lines bounds its derivatives uniformly. Since $\sup_K \|r - r^{(D)}\|_2 \to 0$, we get $|F_n(r) - F_n(r^{(D)})| \to 0$ uniformly in n and $r \in K$. The coefficients of a bounded sequence are bounded, so a diagonal subsequence converges coefficientwise; on each $\mathbb{D}^D$ the finite-dimensional Montel theorem then gives uniform convergence on compact sets. Combining the two statements, the subsequence is uniformly Cauchy on K. □

PROPOSITION 3.13 (Compact embeddings). *In the setting of Corollary 3.11, suppose $[\nu]_\theta < \infty$.*

(1) If the embedding $H^p \to L^p(\mu_{\nu,R})$ is compact for some $0 < p < \infty$, then $\nu((1/2, 1/2+t]) = o(t^\theta)$.
(2) If $\nu((1/2, 1/2+t]) = o(t^\theta)$, the embedding is compact for every $0 < p < \infty$.

PROOF. (1) Let K_δ be the kernel at $\gamma_R(1/2+\delta)$ and $f_\delta = K_\delta^{2/p} / \|K_\delta\|_2^{2/p}$, of norm one. At every point of the support, $f_\delta \to 0$ as $\delta \to 0$, because $K_\delta(z)$ stays bounded there while $\|K_\delta\|_2 \to \infty$. If the embedding is compact, every sequence $\delta_n \to 0$ has a subsequence along which f_{δ_n} converges in $L^p(\mu)$, necessarily to 0, the pointwise limit. But $\int |f_\delta|^p d\mu \ge \zeta_R(1+2\delta)\nu((1/2, 1/2+\delta]) \gtrsim \delta^{-\theta}\nu((1/2, 1/2+\delta])$, as in section 4.5. (2) Split ν at $1/2+\delta$. By Corollary 3.11, the part near $1/2$ has embedding norm at most a constant times $\sup_{t \le \delta} \nu((1/2, 1/2+t])t^{-\theta} + \nu((1/2, 1/2+\delta])$, which tends to 0 with δ. The far part is the image of a compact subset K_δ of $\mathbb{D}^\infty \cap \ell^2$: its points have coordinates of modulus at most $q^{-1/2-\delta}$, whose squares have uniformly summable tails, and the root phases range over a compact product. Given a bounded sequence in H^p, Lemma 3.12 provides a subsequence converging uniformly on K_δ, hence in L^p of the finite far measure. Since the near parts are uniformly small, the images form a Cauchy sequence; for $p < 1$ use the p-subadditivity of $\int |\cdot|^p$. □

For signs the Helson matrix is positive and chapter 4 develops its full theory. For some $k_q \geq 3$ it is not: the measure lives on complex points, and already the one-variable Hankel matrix of $\mathbf{1}[3 \mid n]r^n$ has the indefinite block $\left(\begin{smallmatrix} 0 & r^3 \\ r^3 & 0 \end{smallmatrix}\right)$. Boundedness of the matrix and the H^2 Carleson condition then need not coincide, and the corollary gives the bounded symbol directly from the H^1 embedding.

3.5. Decimation

For $k \geq 2$ and a set Q of coordinates let $D_{k,Q}$ replace each exponent ν_q, $q \in Q$, by $k\nu_q$ in the coefficient index: $\widehat{D_{k,Q}F}(\nu) = \widehat{F}(\tilde{\nu})$. For Q all primes, $D_k \sum a_n n^{-s} = \sum a_{n^k} n^{-s}$.

PROPOSITION 3.14. *$\|D_{k,Q}\|_{H^p \to H^p} = 1$ for $1 \leq p \leq \infty$. For $0 < p < 1$ and $|Q| = d$, $\|D_{k,Q}\| = k^{d(1/p-1)}$; so $D_{k,Q}$ is unbounded for infinite Q. If $\alpha(n^k) = \beta(n)$ and $\alpha = 0$ off kth powers, the functionals Λ_α and Λ_β on H^1 have equal norms, hence equal optimal symbol norms.*

PROOF. $(D_{k,Q}F) \circ \pi_k$ is the average of F over kth roots of unity in the Q-coordinates, where π_k raises those coordinates to the kth power and preserves Haar measure. For $p \geq 1$ this is a conditional expectation. For $p < 1$ the upper bound is p-subadditivity; the lower bound uses outer functions of modulus 1 on an arc whose k rotations are disjoint and δ elsewhere, tensored d times. For the functionals, $\Lambda_\alpha = \Lambda_\beta \circ D_k$ and $D_k J_k = I$ with the isometry $J_k G = G \circ \pi_k$. □

Decimation identifies analytic sign averages exactly,

$$\mathbb{E}_\varepsilon F((\varepsilon_q q^{-s})_Q, (p^{-s})_R) = (D_{2,Q}F)((q^{-2s})_Q, (p^{-s})_R),$$

but it bounds $|\mathbb{E}_\varepsilon F|$, not $\mathbb{E}_\varepsilon |F|$ (already $F = z_q$ has $D_2 F = 0$), so it does not give Proposition 3.10.

Open

(1) The exact value of C_0. By Proposition 3.8 it exceeds 2π by between $2.8 \cdot 10^{-7}$ and 10^{-3}.
(2) The rate of $N_p(\varepsilon)$ for $p < 2$; is it $(\log 1/\varepsilon)^{1/4}$ at $p = 1$ (B5)?
(3) Do the H^1 and H^2 Carleson classes coincide for measures on $(-1,1)^\infty$ (Perfekt–Pushnitski)? Every family treated here and in chapter 4 says yes.
(4) A version of Theorem 3.1 for continuous families with many near neighbours, such as uncountably many sign patterns at one scale, beyond the product structure of Proposition 3.10.

Part 2

Structured Helson matrices

CHAPTER 4

Positive Helson matrices on the half-line

In one variable a non-negative Hankel matrix is a moment matrix. Widom characterized the bounded ones by a Carleson condition on the moment measure [48], and Nehari's theorem then supplies a bounded symbol for free. In infinitely many variables Nehari's theorem fails, but the known counterexamples, built from tensor powers of small symbols such as $z_1 + z_2$, have indefinite Hankel forms [38]. This chapter asks whether positivity rescues the theorem.

Two mechanisms produce bounded symbols for positive Helson matrices, and they behave very differently.

- *Separation.* On the curve $\gamma(s) = (2^{-s}, 3^{-s}, 5^{-s}, \dots)$, points at different scales have nearly mutually singular harmonic measures. Perfekt's localization argument [39] turns this into an H^1 Carleson embedding. It requires the moment measure to satisfy $\nu((1/2, 1/2+t]) \lesssim t$, exactly the condition for boundedness.
- *Symmetry.* Averaging a point over independent signs in each coordinate kills every pair (m, n) except those with mn a square, in H^1 as well as in H^2. This needs no condition at all near $s = 1/2$.

The main result of the chapter, Theorem 4.2, runs both mechanisms at once. Keep a set R of primes on the curve and average the signs of the others. If R has density θ among the primes, the condition becomes $\nu((1/2, 1/2+t]) \lesssim t^\theta$. Within the whole family, boundedness, the H^1 embedding and the existence of a bounded symbol are equivalent, with a constant depending only on R. The curve ($\theta = 1$) and the full sign average ($\theta = 0$) are the two endpoints. The same interpolation describes the compact members (Theorem 4.6), which is a Hartman-type theorem whose condition moves from $o(t)$ at one endpoint to no condition at the other. To our knowledge the interpolating family, the main theorem with its uniform constant, the sign-average mechanism and the Hartman-type theorem are new.

Section 4.1 sets up the dictionary between symbols, functionals and Carleson measures. Section 4.2 defines the family and states the results. Sections 4.3 and 4.4 prove the two mechanisms separately, and section 4.5 combines them. Section 4.6 treats compactness, section 4.7 collects consequences, and section 4.8 discusses what remains of the general question.

4.1. Positive Helson matrices and Carleson measures

Throughout, $M(\alpha) = (\alpha(mn))_{m,n\ge 1}$ is a Helson matrix, and $z^{\kappa(n)}$ is the Bohr lift of n^{-s}. For a Dirichlet polynomial $h(s) = \sum_k h_k k^{-s}$ with Bohr lift H, put

$$\Lambda_\alpha(h) = \sum_k h_k\, \alpha(k).$$

For finitely supported $a, b \in \ell^2$ with Dirichlet polynomials f, g, the form of $M(\alpha)$ is $\sum_{m,n} \alpha(mn) a_m b_n = \Lambda_\alpha(fg)$.

DEFINITION. A *bounded symbol* for $M(\alpha)$ is a function $g \in L^\infty(\mathbb{T}^\infty)$ with $\int_{\mathbb{T}^\infty} z^{\kappa(n)} g\, \mathrm{d}m = \alpha(n)$ for every $n \ge 1$.

For real α this agrees, after complex conjugation, with the convention of [40], where a symbol satisfies $\langle A, z^{\kappa(n)}\rangle = \alpha(n)$. The bilinear form is more convenient here.

LEMMA 4.1. *$M(\alpha)$ has a bounded symbol if and only if Λ_α extends to a bounded functional on $\mathscr{H}^1$. In that case*

$$\|M(\alpha)\| \le \inf\{\|g\|_\infty : g \text{ a bounded symbol}\} = \|\Lambda_\alpha\|_{(\mathscr{H}^1)^*}.$$

PROOF. If g is a bounded symbol, then $\Lambda_\alpha(h) = \int H g\, \mathrm{d}m$ on polynomials, so $|\Lambda_\alpha(h)| \le \|g\|_\infty \|h\|_{\mathscr{H}^1}$. Conversely, a bounded functional on $\mathscr{H}^1$, which the Bohr lift identifies with a closed subspace of $L^1(\mathbb{T}^\infty)$, extends by the Hahn–Banach theorem to L^1 with the same norm, and is represented by some $g \in L^\infty$ whose moments are $\Lambda_\alpha(n^{-s}) = \alpha(n)$. Finally, $\|fg\|_{\mathscr{H}^1} \le \|f\|_{\mathscr{H}^2}\|g\|_{\mathscr{H}^2}$ gives $\|M(\alpha)\| \le \|\Lambda_\alpha\|$. □

Positivity

Perfekt and Pushnitski [40] show that, under a mild growth condition, $M(\alpha) \ge 0$ exactly when α is a moment sequence $\alpha(n) = \int t^{\kappa(n)}\, \mathrm{d}\mu(t)$ of a positive measure μ on $\mathbb{R}^\infty$. Then $M(\alpha) = \mathcal{N}^*\mathcal{N}$ with $\mathcal{N} : \ell^2 \to L^2(\mu)$, $(a_n) \mapsto \sum_n a_n t^{\kappa(n)}$, so $M(\alpha)$ is bounded exactly when μ is a Carleson measure for $H^2(\mathbb{T}^\infty)$:

$$\int |F|^2\, \mathrm{d}\mu \le C\|F\|_{H^2}^2.$$

On the other side, if μ is a Carleson measure for $H^1(\mathbb{T}^\infty)$, then $|\Lambda_\alpha(h)| = |\int H\, \mathrm{d}\mu| \le C\|h\|_{\mathscr{H}^1}$, and Lemma 4.1 gives a bounded symbol. So for non-negative Helson matrices, Nehari's theorem follows from the inclusion

$$\{H^2\text{-Carleson measures}\} \subseteq \{H^1\text{-Carleson measures}\} \tag{4.1}$$

for the moment measures in question. The reverse inclusion always holds: apply an H^1 embedding to F^2, using $\|F^2\|_{H^1} = \|F\|_{H^2}^2$. In one variable (4.1) is trivial, because every H^1 function is a product of two H^2 functions with $\|F\|_1 = \|G\|_2\|H\|_2$. On $\mathbb{T}^\infty$ that factorization fails, and (4.1) is false for general measures: arc length on a vertical segment of the critical line is a counterexample, by Harper's work [22, 39], and chapter 3 gives one in the interior of $\mathbb{D}^\infty$. Whether this can happen for measures on real points is the following problem.

PROBLEM 3 (Positive Nehari; Perfekt and Pushnitski [40, Section 4]). *Does every bounded non-negative Helson matrix have a bounded symbol? Equivalently, does* (4.1) *hold for every positive measure on* $(-1,1)^\infty$ *whose moments define a bounded Helson matrix?*

4.2. Partially averaged curve measures

The real half-line $s > 1/2$ lifts to the curve

$$\gamma(s) = (p^{-s})_p \in (0,1)^\infty \cap \ell^2,$$

and $F(\gamma(s)) = f(s)$ for every Dirichlet series f with Bohr lift F. Split the primes into two sets: R, the primes kept on the curve, and Q, the primes whose signs are averaged. Let $\varepsilon = (\varepsilon_p)_{p\in Q}$ be independent uniformly distributed signs, and write $\varepsilon\gamma(s)$ for the point with coordinates $\varepsilon_p p^{-s}$ for $p \in Q$ and p^{-s} for $p \in R$.

For a finite positive Borel measure ν on $(1/2,\infty)$, let $\mu_{\nu,R}$ be the image of $\nu \otimes \mathbb{P}_\varepsilon$ under $(s,\varepsilon) \mapsto \varepsilon\gamma(s)$. Its moments are

$$\alpha_{\nu,R}(n) = \int \mathbb{E}_\varepsilon\big[\varepsilon^{\kappa(n)}\big] n^{-s}\,\mathrm{d}\nu(s) = \mathbf{1}\big[n_Q \text{ is a square}\big] \int n^{-s}\,\mathrm{d}\nu(s),$$

where n_Q is the part of n composed of primes in Q. So $M(\alpha_{\nu,R}) \geq 0$, and $\int |F|\,\mathrm{d}\mu_{\nu,R} = \int \mathbb{E}_\varepsilon |F(\varepsilon\gamma(s))|\,\mathrm{d}\nu(s)$.

The size of R is measured by its density among the primes. Say that R has *density* $\theta \in [0,1]$ *with constant* B if

$$\Big|\sum_{p\in R} p^{-1-t} - \theta \log\frac{1}{t}\Big| \leq B, \qquad 0 < t \leq 2. \tag{4.2}$$

All primes have density 1, since $\sum_p p^{-1-t} = \log(1/t) + O(1)$. Finite sets, and more generally sets with $\sum_{p\in R} 1/p < \infty$, have density 0. The primes in a union of a residue classes modulo q coprime to q have density $a/\varphi(q)$, by the prime number theorem in arithmetic progressions, and the odd-indexed primes have density $1/2$. Put

$$[\nu]_\theta = \sup_{0<t\leq 1} \frac{\nu((1/2, 1/2+t])}{t^\theta},$$

which for $\theta = 0$ is just $\nu((1/2, 3/2])$.

THEOREM 4.2. *Let R have density $\theta \in [0,1]$ with constant B, and let ν be a finite positive Borel measure on $(1/2,\infty)$. The following are equivalent.*

(1) $M(\alpha_{\nu,R})$ is bounded on ℓ^2.
(2) $[\nu]_\theta < \infty$.
(3) $\mu_{\nu,R}$ is a Carleson measure for $H^1(\mathbb{T}^\infty)$.
(4) $M(\alpha_{\nu,R})$ has a bounded symbol.

Moreover there is $C = C(\theta, B)$ such that

$$\|M(\alpha_{\nu,R})\| \leq \inf\|g\|_\infty \leq C\,\|M(\alpha_{\nu,R})\|,$$

the infimum being over all bounded symbols. If $\theta = 0$, all four conditions hold for every finite ν.

The two endpoints are the motivating cases.

COROLLARY 4.3 (The curve). *For R the set of all primes, $\alpha_\nu(n) = \int n^{-s}\,\mathrm{d}\nu(s)$, and $[\nu] = [\nu]_1$: the matrix $M(\alpha_\nu)$ is bounded if and only if $[\nu] < \infty$, if and only if $\int |f|\,\mathrm{d}\nu \lesssim \|f\|_{\mathscr{H}^1}$, if and only if it has a bounded symbol. One may take $\inf\|g\|_\infty \le (198 + \zeta(3))\|M(\alpha_\nu)\| < 200\,\|M(\alpha_\nu)\|$.*

The equivalence of the first two conditions is due to Perfekt and Pushnitski [40, Theorem 5.1]. The multiplicative Hilbert matrix [15] comes from Lebesgue measure, which is not finite but differs from a finite measure by a Hilbert–Schmidt term (Corollary 4.11). The constant does not depend on ν; by contrast, the best constant in Nehari's theorem on $\mathbb{T}^d$ is at least $(\pi^2/8)^{d/4}$ [38].

COROLLARY 4.4 (Full sign averages). *For $R = \emptyset$ and every finite positive ν, the Helson matrix with $\alpha(n) = \mathbf{1}_\square(n) \int n^{-s}\,\mathrm{d}\nu(s)$, where $\mathbf{1}_\square$ is the indicator of the squares, is bounded and has a bounded symbol with*

$$\inf\|g\|_\infty \le \zeta(2)\,\nu((1/2,\infty)) \le \zeta(2)\,\|M(\alpha)\|.$$

So the full sign averages satisfy Nehari's theorem with constant at most $\zeta(2) \approx 1.645$, uniformly in ν. That is far better than the constant for the curve. It is also the only family in this book with an explicit small constant.

4.3. Mechanism I: separation

The difficulty on the curve is a logarithm. Point evaluation at $\gamma(1/2+\sigma)$ has norm $\zeta(1+2\sigma) \asymp 1/\sigma$ on $\mathscr{H}^1$, by the Cole–Gamelin estimate [19]. If $[\nu] < \infty$, then ν puts mass $O(e^{-j})$ on the scale $1/2 + (e^{-j-1}, e^{-j}]$, so each scale contributes $O(\|f\|_{\mathscr{H}^1})$ to $\int |f|\,\mathrm{d}\nu$, and there are $\log(1/\delta)$ scales above height δ. One function cannot be near-extremal at many scales at once: the harmonic measures at different scales have exponentially small Hellinger affinity (chapter 1), and the Hellinger–Bessel inequality turns this into a Carleson estimate (chapter 2). For the Hardy geometry the kernel exponent equals the density, $\kappa = \theta$, and Theorem 2.9 with $p = 1$ reads as follows.

LEMMA 4.5 (Block-maximal evaluation on prime sets). *Let R have density $\theta \in (0,1]$ with constant B. There is $C_1 = C_1(\theta, B)$ such that every $f \in \mathscr{H}^1$ whose coefficients are supported on integers composed of primes in R satisfies*

$$\sum_{j\ge 0} e^{-j\theta} \max_{e^{-j-1} \le t \le e^{-j}} |f(1/2+t)| \le C_1 \|f\|_{\mathscr{H}^1},$$

and $\int_{(1/2,3/2]} |f|\,\mathrm{d}\nu \le C_1 [\nu]_\theta \|f\|_{\mathscr{H}^1}$ for every positive ν. For R the set of all primes the second bound holds with $C_1 = 99$.

PROOF. Both are Theorem 2.9 for $p = 1$, applied to the Bohr lift; the maximum is attained since f is continuous on each closed block, and Fatou's lemma passes from polynomials to $\mathscr{H}^1$. □

For all primes, Perfekt's localization argument [39, Section 2], with Hoeffding's inequality [26] controlling the overlaps, gives the lemma with $C_1 = 3eL$, $L = 10^5$. The Gram route of chapter 2 needs no disjointness of the localized sets, which is why its constant is so much smaller.

Taking R to be all primes and ν Lebesgue measure on $(1/2,3/2]$ recovers Perfekt's inequality $\int_0^1 |f(1/2+\sigma)|\,\mathrm{d}\sigma \lesssim \|f\|_{\mathscr{H}^1}$. The block maximum is the strengthening a general measure needs: ν may be atomic, and an average says nothing about any particular point.

4.4. Mechanism II: symmetry

The second mechanism is the case $k_q = 2$ of the root-of-unity averages of section 3.4. For sign averages Lemma 3.9 reads $(|g(x)| + |g(-x)|)/2 \le \|g\|_{H^1}/(1-|x|^4)$, with equality for $g = (1-\overline{x}^2 z^2)^{-2}$, and Proposition 3.10 gives, for an analytic polynomial F, a set Q of coordinates with points $x_p \in \mathbb{D}$, and the remaining coordinates fixed at w,

$$\mathbb{E}_\varepsilon \big|F\big((\varepsilon_p x_p)_{p\in Q}, w\big)\big| \le \prod_{p\in Q} \frac{1}{1-|x_p|^4} \int_{\mathbb{T}^Q} \big|F(\zeta, w)\big|\,\mathrm{d}m(\zeta). \tag{4.3}$$

For $x = \gamma(s)$ the constant is at most $\zeta(4s) \le \zeta(2)$. Two features matter here. The constant does not blow up as $s \to 1/2$, and $\sum |x_p|^4 < \infty$ is weaker than $x \in \ell^2$: the sign-averaged evaluation is bounded on H^1 at points where no single point evaluation is.

4.5. Proof of the main theorem

(1) $\Rightarrow$ (2). Assume $\theta > 0$; for $\theta = 0$ there is nothing to prove. Test with $a_n = n^{-1/2-\delta}$, $0 < \delta \le 1$, on the integers composed of primes in R. For such m, n the Q-part of mn is 1, so $\alpha_{\nu,R}(mn) = \int (mn)^{-s}\,\mathrm{d}\nu(s)$. By Tonelli's theorem, with $\zeta_R(w) = \prod_{p\in R}(1-p^{-w})^{-1}$,

$$\sum_{m,n} \alpha_{\nu,R}(mn) a_m a_n = \int \zeta_R(1/2+\delta+s)^2\,\mathrm{d}\nu(s) \ge \zeta_R(1+2\delta)^2\,\nu\big((1/2, 1/2+\delta]\big).$$

The left side is at most $\|M(\alpha_{\nu,R})\|\zeta_R(1+2\delta)$, and by (4.2) $\zeta_R(1+2\delta) \ge e^{-B}(2\delta)^{-\theta}$. Hence $[\nu]_\theta \le 2e^B \|M(\alpha_{\nu,R})\|$. We also record $\nu((1/2,\infty)) = \alpha_{\nu,R}(1) \le \|M(\alpha_{\nu,R})\|$.

(2) $\Rightarrow$ (3). Let F be an analytic polynomial. For fixed s, (4.3) with the R-coordinates fixed at $\gamma_R(s)$ gives

$$\mathbb{E}_\varepsilon |F(\varepsilon\gamma(s))| \le \zeta(2) \int_{\mathbb{T}^Q} |F_\zeta(\gamma_R(s))|\,\mathrm{d}m(\zeta),$$

where $F_\zeta = F(\zeta,\cdot)$ is a polynomial in the R-coordinates. It is the Bohr lift of a Dirichlet polynomial f_ζ supported on integers composed of primes in R, with $f_\zeta(s) = F_\zeta(\gamma_R(s))$. Integrate in ν and use Tonelli:

$$\int |F|\,\mathrm{d}\mu_{\nu,R} \le \zeta(2) \int_{\mathbb{T}^Q} \Big(\int |f_\zeta|\,\mathrm{d}\nu \Big)\,\mathrm{d}m(\zeta).$$

If $\theta > 0$, split $(1/2,3/2]$ into the blocks $1/2 + I_j$ with $I_j = (e^{-j-1}, e^{-j}]$. Since $\nu(1/2+I_j) \le [\nu]_\theta e^{-j\theta}$, Lemma 4.5 gives $\int_{(1/2,3/2]} |f_\zeta|\,\mathrm{d}\nu \le C_1[\nu]_\theta \|F_\zeta\|_{L^1(\mathbb{T}^R)}$. If $\theta = 0$, the point estimate gives $|f_\zeta(s)| \le \zeta_R(2\Re s)\|F_\zeta\|_1 \le e^{1+B}\|F_\zeta\|_1$ for every

$s > 1/2$. On $(3/2,\infty)$, $|f_\zeta| \le \zeta(3)\|F_\zeta\|_1$. Since $\int_{\mathbb{T}^Q}\|F_\zeta\|_{L^1(\mathbb{T}^R)}\,\mathrm{d}m(\zeta) = \|F\|_{L^1}$, we obtain

$$\int |F|\,\mathrm{d}\mu_{\nu,R} \le \zeta(2)\big(C_1[\nu]_\theta + \zeta(3)\,\nu((1/2,\infty))\big)\|F\|_{H^1} \tag{4.4}$$

for $\theta > 0$, and $\int|F|\,\mathrm{d}\mu_{\nu,R} \le \zeta(2)e^{1+B}\nu((1/2,\infty))\|F\|_{H^1}$ for $\theta = 0$. Fatou's lemma extends both to $H^1(\mathbb{T}^\infty)$.

(3) $\Rightarrow$ (4). For polynomials, $\Lambda_{\alpha_{\nu,R}}(h) = \int H\,\mathrm{d}\mu_{\nu,R}$. Apply Lemma 4.1.

(4) $\Rightarrow$ (1). Lemma 4.1.

The constant. Combine Lemma 4.1, (4.4) and the bounds from the first step. Corollary 4.3 follows with $\zeta(2)$ omitted, since $Q = \emptyset$, and with $C_1 = 99$, $[\nu] \le 2\|M\|$. Corollary 4.4 follows from (4.3) directly, since then $R = \emptyset$ and $\int|F|\,\mathrm{d}\mu_{\nu,\emptyset} \le \int \zeta(4s)\,\mathrm{d}\nu(s)\,\|F\|_{H^1}$. □

4.6. Compactness

In one variable, Hartman's theorem [23] says that a Hankel operator is compact exactly when its symbol can be taken in $H^\infty + C$, and Sarason's theorem [44] says that this space is closed. Let

$$N = \Big\{ g \in L^\infty(\mathbb{T}^\infty) : \int z^\kappa g\,\mathrm{d}m = 0 \text{ for every multi-index } \kappa \ge 0 \Big\},$$

so that the bounded symbols of a given $M(\alpha)$ form a coset $g + N$. Call a functional on $\mathscr{H}^1$ a *polynomial functional* if it has the form $h \mapsto \sum_{k\le K} h_k\beta(k)$.

THEOREM 4.6. *Let R have density $\theta \in [0,1]$ with constant B, and let ν be a finite positive Borel measure on $(1/2,\infty)$ with $[\nu]_\theta < \infty$. The following are equivalent.*

(1) $M(\alpha_{\nu,R})$ is compact.
(2) $\nu((1/2,1/2+t]) = o(t^\theta)$ as $t \to 0^+$.
(3) $\Lambda_{\alpha_{\nu,R}}$ lies in the $(\mathscr{H}^1)^*$-closure of the polynomial functionals.
(4) Some, equivalently every, bounded symbol lies in the L^∞-closure of $C(\mathbb{T}^\infty)+N$.

In particular, if $\theta = 0$, then $M(\alpha_{\nu,R})$ is compact for every finite ν.

The condition interpolates in the same way as in Theorem 4.2. At the curve endpoint it is $o(t)$; at the sign-average endpoint it is vacuous, because $\nu((1/2,1/2+t]) \to 0$ for every finite measure. The full sign averages of Corollary 4.4 are therefore all compact, while the multiplicative Hilbert matrix, on the curve, is not.

PROOF. (1) $\Rightarrow$ (2). Assume $\theta > 0$. The unit vectors k_δ proportional to $(n^{-1/2-\delta})$ on the R-integers tend to 0 weakly as $\delta \to 0^+$, because $\zeta_R(1+2\delta) \to \infty$. A compact operator maps them to a norm-null family, and the computation in the proof of (1) $\Rightarrow$ (2) above gives $\nu((1/2,1/2+\delta]) \le e^B(2\delta)^\theta\langle M(\alpha_{\nu,R})k_\delta, k_\delta\rangle$.

(2) $\Rightarrow$ (3). Fix $0 < \varepsilon \le 1$ and write $\nu = \nu_\varepsilon + \nu^\varepsilon$, with ν_ε the restriction to $(1/2,1/2+\varepsilon]$. Then $[\nu_\varepsilon]_\theta \le \eta(\varepsilon) := \sup_{0<u\le\varepsilon}\nu((1/2,1/2+u])/u^\theta$, which tends to 0 by (2), and the estimates in the proof of Theorem 4.2 give $\|\Lambda_{\alpha_{\nu_\varepsilon,R}}\| \le \zeta(2)C_1\eta(\varepsilon)$, or $\le \zeta(2)e^{1+B}\nu((1/2,1/2+\varepsilon])$ when $\theta = 0$. The measure ν^ε lives on $[1/2+\varepsilon,\infty)$, so

$0 \le \alpha_{\nu^\varepsilon,R}(n) \le \nu((1/2,\infty))\, n^{-1/2-\varepsilon}$. By Cauchy–Schwarz and Helson's inequality $\sum_k |h_k|^2/d(k) \le \|h\|^2_{\mathscr{H}^1}$ [24],

$$\Big|\Lambda_{\alpha_{\nu^\varepsilon,R}}(h) - \sum_{k\le K} h_k \alpha_{\nu^\varepsilon,R}(k)\Big| \le \Big(\sum_{k>K} d(k)\alpha_{\nu^\varepsilon,R}(k)^2\Big)^{1/2} \|h\|_{\mathscr{H}^1},$$

which tends to 0 as $K\to\infty$ because $\sum_k d(k)k^{-1-2\varepsilon} = \zeta(1+2\varepsilon)^2$.

(3) $\Rightarrow$ (1). A polynomial functional with coefficients $\beta(k)$, $k\le K$, gives the Helson matrix $(\beta(mn))$, which vanishes unless $m,n\le K$ and so has finite rank. By Lemma 4.1, $\|M(\alpha)-M(\beta)\| \le \|\Lambda_\alpha - \Lambda_\beta\|$.

(3) $\Leftrightarrow$ (4). Let $q : L^\infty \to L^\infty/N \cong (\mathscr{H}^1)^*$ be the quotient map, and let V be the space of symbols $\sum_{k\le K}\beta(k)\overline{z}^{\kappa(k)}$ of polynomial functionals. A trigonometric monomial z^μ lies in V if $\mu \le 0$ and in N otherwise, so $V+N$ contains all trigonometric polynomials and $\overline{q(V)} = \overline{q(C(\mathbb{T}^\infty)+N)}$. For a subspace $W \supseteq N$ one has $q^{-1}(\overline{q(W)}) = \overline{W}$: if $\|q(x-w_k)\|\to 0$ with $w_k\in W$, choose $n_k\in N$ with $\|x-w_k-n_k\|_\infty \to 0$. Take $W = C(\mathbb{T}^\infty)+N$; the statement holds for one element of the coset $g+N$ exactly when it holds for all. □

The closure in (4) can be dropped.

PROPOSITION 4.7. *$C(\mathbb{T}^\infty)+N$ is closed in $L^\infty(\mathbb{T}^\infty)$. Hence the symbols of the compact members of the family can be chosen in $C(\mathbb{T}^\infty)+N$ itself.*

PROOF. For $k\ge 1$ let E_k be the conditional expectation onto the first k coordinates and F_k convolution on $\mathbb{T}^k$ with the product of Fejér kernels of order k, and put $T_k = F_kE_k$. Then $\|T_k\|_{L^\infty\to L^\infty}\le 1$, T_k maps L^∞ into trigonometric polynomials, and $T_kc\to c$ uniformly for $c\in C(\mathbb{T}^\infty)$, by density of finite-coordinate polynomials. Moreover $T_kN\subseteq N$: E_k is self-adjoint and sends z^κ either to itself or to 0, and F_k multiplies Fourier coefficients. So for $c\in C$ and $n\in N$, $\operatorname{dist}(c, C\cap N) \le \|c - T_kn\|_\infty \le \|c-T_kc\|_\infty + \|c-n\|_\infty$, and letting $k\to\infty$ gives $\operatorname{dist}(c,C\cap N) = \operatorname{dist}(c,N)$. Thus $C/(C\cap N)\to L^\infty/N$ is isometric, its image $(C+N)/N$ is complete, hence closed, and $C+N$ is its preimage. This is Rudin's argument for $H^\infty + C$, which needs only the maps T_k, not an algebra structure. □

Essential norms

Compactness is the qualitative statement; the quantitative one compares the essential norm $\|M\|_{\mathrm{e}}$ with the essential size of the symbol,

$$\mathcal{E}_{\mathrm{sym}}(M) = \operatorname{dist}_{L^\infty}\big(g, C(\mathbb{T}^\infty)+N\big),$$

which does not depend on the bounded symbol g chosen. A continuous symbol gives a compact matrix, being a uniform limit of trigonometric polynomials, and N gives the zero matrix, so $\|M\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M)$ always. On the curve the converse holds up to an absolute constant.

THEOREM 4.8 (Essential Nehari on the curve). *Let R be all primes, let ν be finite and positive with $[\nu]<\infty$, and put $\ell_\nu = \limsup_{t\to 0^+}\nu((1/2,1/2+t])/t$. Then*

$$\frac{1}{2}\ell_\nu \le \|M(\alpha_\nu)\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M(\alpha_\nu)) \le 99\,\ell_\nu \le 198\,\|M(\alpha_\nu)\|_{\mathrm{e}}.$$

PROOF. The lower bound is the test in (1) ⇒ (2) of Theorem 4.6, with $\delta\zeta(1+2\delta) \to 1/2$ and the weak nullity of k_δ. For the upper bound, split $\nu = \nu_\varepsilon + \nu^\varepsilon$ as in (2) ⇒ (3). The functional of ν^ε is a norm limit of polynomial functionals, and Lemma 4.5 bounds the functional of ν_ε by $99 \sup_{t \le \varepsilon} \nu((1/2, 1/2+t])/t$. Through the quotient map q, $\mathcal{E}_{\mathrm{sym}}$ is the distance of $q(g)$ to the closure of the polynomial functionals, which is at most this bound. Let $\varepsilon \to 0$. □

COROLLARY 4.9. *If $d\nu = w\,\mathrm{d}s$ near $1/2$, with w right-continuous at $1/2$, then $M(\alpha_\nu)$ has essential spectrum $[0, \pi w(1/2)]$; in particular $\|M(\alpha_\nu)\|_{\mathrm{e}} = \pi w(1/2)$.*

PROOF. Let $\nu_0 = w(1/2)\mathbf{1}_{(1/2,3/2]}\,\mathrm{d}s$. The positive and negative parts of $\nu - \nu_0$ have $o(t)$ mass near $1/2$, so their matrices are compact by Theorem 4.6. Extending ν_0 to $(1/2, \infty)$ changes the matrix, after deleting the first row and column, by a Hilbert–Schmidt term, and leaves $w(1/2)$ times the multiplicative Hilbert matrix, whose essential spectrum is $[0, \pi]$ [41]. □

So for a curve measure with a density the essential norm is read off from a single number, and the symbol cannot be made essentially smaller than $\pi w(1/2)$ by any choice in the coset. For the multiplicative Hilbert matrix this gives $\mathcal{E}_{\mathrm{sym}} \ge \pi$: *no* bounded symbol is within distance less than π of $C(\mathbb{T}^\infty) + N$, a statement about the whole coset that the sharp-constant problem (chapter 10) only asks about at the point 0.

4.7. Consequences

COROLLARY 4.10. *In the setting of Theorem 4.2, the following are equivalent: $\mu_{\nu,R}$ is a Carleson measure for $H^2(\mathbb{T}^\infty)$; it is a Carleson measure for $H^1(\mathbb{T}^\infty)$; and $[\nu]_\theta < \infty$. If they hold, $\mu_{\nu,R}$ is a Carleson measure for $H^{2k}(\mathbb{T}^\infty)$ for every integer $k \ge 1$.*

PROOF. By section 4.1, the H^2 condition is boundedness of $M(\alpha_{\nu,R})$. For the last claim, apply the H^2 condition to F^k. □

So the inclusion (4.1) holds throughout the family, and Problem 3 has a positive answer for every measure of the form $\mu_{\nu,R}$.

The best-known special case is the multiplicative Hilbert matrix, with entries $1/(\sqrt{mn}\log mn)$ for $m, n \ge 2$.

PROBLEM 4 (Saksman and Seip [43, Problem 3.2]). *Does the multiplicative Hilbert matrix have a bounded symbol?*

Perfekt [39] answered Problem 4 in the affirmative. The family of Theorem 4.2 contains the matrix, up to a harmless correction, and so gives a second proof.

COROLLARY 4.11 (Perfekt [39, Corollary 1.2]). *The multiplicative Hilbert matrix has a bounded symbol: there is $g \in L^\infty(\mathbb{T}^\infty)$ with $\int g\,\mathrm{d}m = 1$ and $\int z^{\kappa(n)} g\,\mathrm{d}m = 1/(\sqrt{n}\log n)$ for $n \ge 2$.*

PROOF. Take R to be all primes and $\nu = \mathbf{1}_{(1/2,3/2]}\,\mathrm{d}s$. Then $[\nu] = 1$, $\alpha_\nu(1) = 1$, and

$$\alpha_\nu(n) = \frac{1}{\sqrt{n}\log n} - c_n, \qquad c_n = \frac{n^{-3/2}}{\log n} \qquad (n \ge 2).$$

Corollary 4.3 gives a bounded symbol for α_ν. The functional $h \mapsto \sum_{n\geq 2} h_n c_n$ is bounded on $\mathscr{H}^1$ by Helson's inequality, since $\sum_n d(n)n^{-3} < \infty$, so it too has a bounded symbol by Lemma 4.1. The sum of the two symbols works. □

Perfekt's Tauberian argument also gives convergence of the coefficient series in its natural order, which this proof does not.

The same subtraction applied to the odd-indexed primes ($\theta = 1/2$) gives a bounded symbol for the matrix $\big(\mathbf{1}[(mn)_Q \in \square]/(\sqrt{mn}\log mn)\big)_{m,n\geq 2}$, the multiplicative Hilbert matrix with the signs of the even-indexed primes averaged out.

4.8. The positive-Nehari question

Problem 3 therefore has a positive answer for every measure $\mu_{\nu,R}$, and for discrete measures on energy-separated families with weights bounded by the inverse kernel norm (chapter 3). The separation mechanism controls the coordinates kept on a curve. The symmetry mechanism controls the coordinates whose signs are averaged, and costs only the factor $\zeta(2)$. Rotations $z \mapsto \varepsilon z$ of $\mathbb{T}^\infty$ preserve every H^p norm, so each statement also holds for a fixed rotated copy of the family. Since a positive measure dominates each of its parts, finite sums of such measures are covered as well.

Averaging over kth roots of unity instead of signs, with orders $k_q \geq 2$ varying over the coordinates, gives a larger family of Helson matrices with nonnegative moments $\mathbf{1}[k_q \mid v_q(n)] \int n^{-s}\,\mathrm{d}\nu$. They are no longer positive as matrices once some $k_q \geq 3$, but the same condition $[\nu]_\theta < \infty$ still gives an H^1 embedding and a bounded symbol, with constant $\zeta(\min k_q)$ in place of $\zeta(2)$ (Corollary 3.11). Explicit members of all these families at every borderline are listed in section 5.2.

The first case not covered is a countable superposition. A counterexample would need mass that is neither separated in energy nor symmetric under sign changes in any fixed set of coordinates, for instance many rotated copies of the curve with individually bounded constants $[\nu_\varepsilon]$ whose sum diverges.

QUESTION 4. Let $\mu = \sum_\varepsilon (\varepsilon\gamma)_*\nu_\varepsilon$ be a countable superposition of measures on rotated copies of γ whose moments define a bounded Helson matrix. Is μ a Carleson measure for $H^1(\mathbb{T}^\infty)$?

CHAPTER 5

Spectral, Schatten and rank structure

Question. What do positive Helson matrices look like spectrally, and how far can the Nehari constant grow among matrices of bounded rank?

The answer to the first question is a dictionary. For a curve matrix the dyadic profile $a_j = e^{j\theta}\nu(I_j)$ of the measure determines everything: boundedness is $a \in \ell^\infty$, compactness is $a \in c_0$ (chapter 4), and membership in S_p is $a \in \ell^p$ for every $p > 0$, and likewise for every Lorentz class (Theorems 5.6 and 5.10). Below the curve, on sparse prime sets, the spectrum decays root-exponentially; for the full sign average $s_n \sim 1/(2n \log n)$. The second question has a short answer at the level of S_2, where Helson's inequality is exactly a weak-factorization identity, and a gap between the exponents 0.174 and 1/2.

To our knowledge the following are new: the dictionary between the profile and the Schatten–Lorentz classes for every $p > 0$, with its pointwise corollaries and the separated-atom theorem; the exact asymptotics of the full sign average; the critical spectrum for general prime sets, including the band structure for oscillating densities; and the reading of Helson's inequality as an S_2 weak-factorization identity.

5.1. The companion kernel

Throughout, R is a set of primes, $\mathcal{S}_R$ the multiplicative semigroup it generates, and $\zeta_R(w) = \prod_{p\in R}(1 - p^{-w})^{-1}$. In this chapter a curve measure is a finite positive measure ν on $(0, \infty)$, shifted so that $u = 0$ is the critical line:

$$\alpha_\nu(n) = \int_0^\infty n^{-1/2-u}\, \mathrm{d}\nu(u), \qquad M_{\nu,R} = \big(\alpha_\nu(mn)\big)_{m,n\in\mathcal{S}_R}.$$

For R all primes we write M_ν. Density of R is defined as in (2.1). Put $I_j = (e^{-j-1}, e^{-j}]$ for $j \geq 0$.

LEMMA 5.1. *Let $\mathcal{N} : \ell^2(\mathcal{S}_R) \to L^2(\nu)$, $\mathcal{N}e_n(u) = n^{-1/2-u}$, on finitely supported vectors. Then $M_{\nu,R} = \mathcal{N}^*\mathcal{N}$ as forms, and whenever either side is bounded, $M_{\nu,R}$ and the integral operator on $L^2(\nu)$ with kernel $\zeta_R(1 + u + v)$ have the same nonzero spectrum with multiplicities.*

PROOF. $\langle \mathcal{N}e_n, \mathcal{N}e_m\rangle_{L^2(\nu)} = \int (mn)^{-1/2-u}\, \mathrm{d}\nu(u) = \alpha_\nu(mn)$, and $\mathcal{N}\mathcal{N}^*$ has kernel $\sum_{n\in\mathcal{S}_R} n^{-1-u-v} = \zeta_R(1 + u + v)$. The nonzero spectra of $\mathcal{N}^*\mathcal{N}$ and $\mathcal{N}\mathcal{N}^*$ coincide. □

The kernel $\zeta_R(1 + u + v)$ is a Hankel kernel in the additive variable $u + v$, singular like $(u + v)^{-\theta}$ at the origin. Every result in the first half of this chapter is a statement about this one kernel restricted to ν.

PROPOSITION 5.2 (Traces). *For every finite positive ν and integer $k \geq 1$,*

$$\operatorname{Tr} M_{\nu,R} = \int \zeta_R(1+2u)\,\mathrm{d}\nu(u),$$

$$\operatorname{Tr} M_{\nu,R}^k = \int \prod_{i=1}^{k} \zeta_R(1+u_i+u_{i+1})\,\mathrm{d}\nu(u_1)\cdots d\nu(u_k),$$

with $u_{k+1} = u_1$, both sides possibly infinite. In particular $\|M_{\nu,R}\|_{S_2}^2 = \iint \zeta_R(1+u+v)^2\,\mathrm{d}\nu(u)\,\mathrm{d}\nu(v)$.

PROOF. Restrict ν to $[\varepsilon,\infty)$. Then $\mathcal{N}$ is Hilbert–Schmidt, the formula is the trace of the kth power of the companion, and the compressions to $n \leq N$ increase to it. Monotone convergence as $N \to \infty$ and $\varepsilon \to 0$ gives the general case. □

COROLLARY 5.3. *Let R have density $\theta \in [0,1)$. The multiplicative Hilbert matrix restricted to $\mathcal{S}_R \setminus \{1\}$ is positive and trace class, with*

$$\operatorname{Tr} = \frac{1}{2}\int_0^\infty \big(\zeta_R(1+t)-1\big)\,\mathrm{d}t.$$

More generally, $\big(\Gamma(\kappa)(mn)^{-1/2}(\log mn)^{-\kappa}\big)$ on $\mathcal{S}_R \setminus \{1\}$ is trace class for $\kappa > \theta$, bounded and not compact for $\kappa = \theta > 0$, and unbounded for $\kappa < \theta$.

PROOF. The matrix is $M_{\nu,R}$ for Lebesgue measure ν on $(0,\infty)$ with the index 1 removed, and Proposition 5.2 with $\zeta_R(1+2u)-1$ in place of $\zeta_R(1+2u)$ gives the trace, which is finite because $\zeta_R(1+t) \lesssim t^{-\theta}$ near 0 and $\zeta_R(1+t)-1 \leq 2^{1-t}(\zeta(2)-1)$ for $t \geq 1$. The second family is $\nu = u^{\kappa-1}\,\mathrm{d}u$. Its block profile is $a_j \asymp e^{-j(\kappa-\theta)}$. The part of ν on $(1,\infty)$ has infinite mass, but once the index 1 is removed its companion kernel $\zeta_R(1+u+v)-1$ decays exponentially, so it contributes a trace-class term; for the part on $(0,1]$, Theorem 5.6 below, with Theorems 4.2 and 4.6, gives the three cases. □

At the critical exponent the matrix is bounded but not compact. For the multiplicative Hilbert matrix the essential spectrum is $[0,\pi]$ [41], and the Mellin structure of the companion kernel suggests a universal answer.

PROBLEM 5 (A universal spectral constant). *Let the companion kernel of a positive Helson matrix grow like $c\,x^{-\kappa}$ at the boundary. Is its critical essential spectrum always $[0, c\,\Gamma(\kappa/2)^2/\Gamma(\kappa)]$?*

The answer is yes when the kernel and the weight vary regularly at the boundary.

THEOREM 5.4 (Critical essential spectrum). *Let $\kappa > 0$, let $h \geq 0$ be continuous on $(0,\infty)$ with $x^\kappa h(x) \to c$ as $x \to 0$, and let $\nu = g(u)\,\mathrm{d}u$ be a positive measure on $(0,\infty)$ with $u^{1-\kappa}g(u) \to b$ as $u \to 0$. Suppose that $\iint_{\max(u,v)\geq\delta} h(u+v)^2\,\mathrm{d}\nu(u)\,\mathrm{d}\nu(v) < \infty$ for every $\delta > 0$. Then the integral operator T on $L^2(\nu)$ with kernel $h(u+v)$ is bounded, and*

$$\sigma_{\mathrm{ess}}(T) = \Big[0,\ bc\,\frac{\Gamma(\kappa/2)^2}{\Gamma(\kappa)}\Big].$$

PROOF. Multiplication by $g^{1/2}$ is unitary from $L^2(\nu)$ onto $L^2(0,\infty)$ and carries T to the kernel $K(u,v) = g(u)^{1/2}h(u+v)g(v)^{1/2}$. Compare it with $K_0(u,v) = (uv)^{(\kappa-1)/2}(u+v)^{-\kappa}$, which is homogeneous of degree -1. The Mellin transform $f \mapsto \int_0^\infty f(u)u^{-1/2-i\xi}\,\mathrm{d}u$ is a multiple of a unitary map onto $L^2(\mathbb{R})$ and carries K_0 to multiplication by

$$m(\xi) = \int_0^\infty t^{\kappa/2-1+i\xi}(1+t)^{-\kappa}\,\mathrm{d}t = \frac{|\Gamma(\kappa/2+i\xi)|^2}{\Gamma(\kappa)},$$

a continuous even function, decreasing in $|\xi|$, with maximum $m_0 = \Gamma(\kappa/2)^2/\Gamma(\kappa)$ at $\xi = 0$ and limit 0 at infinity. So K_0 is bounded with $\sigma(K_0) = \sigma_{\mathrm{ess}}(K_0) = [0, m_0]$. Let P_δ be multiplication by the indicator of $(0,\delta)$. The functions $f_n(u) = n^{-1/2}u^{-1/2+i\xi}\mathbf{1}_{(e^{-2n},e^{-n})}(u)$ are, up to $o(1)$ in norm, approximate eigenvectors of K_0 with eigenvalue $m(\xi)$; they tend weakly to zero and are eventually supported in $(0,\delta)$. Hence $\sigma_{\mathrm{ess}}(P_\delta K_0 P_\delta) = [0, m_0]$ for every $\delta > 0$.

Now $K = P_\delta K P_\delta + R_\delta$, where R_δ has kernel supported in $\{\max(u,v) \geq \delta\}$ and is Hilbert–Schmidt by hypothesis. On $(0,\delta)^2$ the hypotheses give $|K - bcK_0| \leq \varepsilon(\delta)K_0$ pointwise, with $\varepsilon(\delta) \to 0$. A kernel dominated by a positive kernel has at most its norm, so $\|P_\delta(K - bcK_0)P_\delta\| \leq \varepsilon(\delta)m_0$. In particular T is bounded. Compact perturbations do not change essential spectra, and a self-adjoint perturbation of norm η moves the essential spectrum by at most η in Hausdorff distance. Hence $\sigma_{\mathrm{ess}}(T)$ lies within $\varepsilon(\delta)m_0$ of $[0, bcm_0]$, for every δ, and the two sets coincide. $\square$

For a prime set R with

$$c_R = \lim_{x\to 0} x^\theta \zeta_R(1+x) \quad \text{existing},$$

for instance when $\sum_{p\in R} p^{-1-t} = \theta\log(1/t) + C + o(1)$, take $\nu = u^{\theta-1}\,\mathrm{d}u$ and $h(x) = \zeta_R(1+x) - 1$, which decays exponentially. By Lemma 5.1 the matrix $\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)$ on $\mathcal{S}_R \setminus \{1\}$ and this operator have the same nonzero spectrum, so

$$\sigma_{\mathrm{ess}}\Big(\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)_{m,n\in\mathcal{S}_R\setminus\{1\}}\Big) = \Big[0,\, c_R\frac{\Gamma(\theta/2)^2}{\Gamma(\theta)}\Big].$$

For R all primes, $c_R = 1$, $\theta = 1$, and this is the interval $[0,\pi]$ of the multiplicative Hilbert matrix found by Perfekt and Pushnitski [41]; the proof above is the standard Mellin argument for kernels homogeneous of degree -1. The theorem applies verbatim to any positive family whose companion kernel and weight vary regularly at the boundary, and so answers Problem 5 in that setting.

Ma, Lu and Zu [33, Theorem 3.3] prove a sharper statement under stronger hypotheses. For kernels $w(x)b(x+y)w(y)$ whose normalized profiles $t^\kappa b(t)$ and $t^{(1-\kappa)/2}w(t)$ are C^2, have limits at 0 and at ∞, and satisfy $|(t\partial_t)^2 f(t)| \lesssim (1 + |\log t|)^{-2}$, they find the same essential spectrum, with a contribution from each end, and by a Mourre estimate they exclude singular continuous spectrum and show that each compact window inside the open bands contains only finitely many eigenvalues. For the Bergman forms with weights d_α on all primes this gives essential and absolutely continuous spectrum $[0, \Gamma(\alpha/2)^2/\Gamma(\alpha)]$, the case $\kappa = \alpha$ of the theorem above, together with finitely many simple eigenvalues above it, which occur exactly

when α exceeds a threshold $\alpha_* \in (1,2)$ [33, Theorem 1.1]. Theorem 5.4 asks only for limits, which is what lets it reach prime sets, where ζ_R need not be smooth in the required sense; it says nothing about the finer spectral type. If only the bounded remainder (2.1) is known, $\psi(L) = \log\big(x^\theta \zeta_R(1+x)\big)$, $L = \log(1/x)$, may oscillate without converging, and then the answer changes.

PROPOSITION 5.5 (Oscillating densities). *Let R have density $\theta \in (0,1]$, and suppose that $\psi(L) - \psi_P(L) \to 0$ as $L \to \infty$ for a continuous P-periodic function ψ_P. Let T_P be the operator on $L^2(\mathbb{R})$ with kernel*

$$\Big(2\cosh\frac{s-t}{2}\Big)^{-\theta} e^{\psi_P(\Lambda(s,t))}, \qquad \Lambda(s,t) = -\log(e^{-s}+e^{-t}).$$

Then the essential spectrum of the critical matrix

$$\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)_{m,n\in\mathcal{S}_R\setminus\{1\}}$$

is $\sigma(T_P)$, the closure of the union over $k \in [0,2\pi)$ of the spectra of the compact fibre operators on $L^2(0,P)$ with kernels $\sum_{n\in\mathbb{Z}} K_P(s,t+nP)e^{ikn}$, where K_P is the kernel of T_P.

PROOF. In the variable $u = e^{-s}$ the companion operator of Lemma 5.1, with $\nu = u^{\theta-1}\,\mathrm{d}u$, becomes the operator on $L^2(\mathrm{d}s)$ with kernel $(2\cosh((s-t)/2))^{-\theta} e^{\tilde\psi(\Lambda(s,t))}$, where $x^\theta(\zeta_R(1+x)-1) = e^{\tilde\psi(\log(1/x))}$ and $\tilde\psi - \psi \to 0$. Since $\Lambda(s+P,t+P) = \Lambda(s,t)+P$, the operator T_P commutes with translation by P, and the Floquet–Bloch decomposition gives its spectrum as stated; the fibres are Hilbert–Schmidt because the kernel decays like $e^{-\theta|s-t|/2}$. As in the proof of Theorem 5.4, the part of the companion operator with $\min(s,t) \le s_0$ is Hilbert–Schmidt, and on $\{s,t > s_0\}$ its kernel differs from that of T_P by at most $\varepsilon(s_0)$ times the kernel of T_P, with $\varepsilon(s_0) \to 0$. So the essential spectrum is that of the compression of T_P to a half-line. The pieces of T_P coupling $(-\infty, s_0)$ with (s_0,∞) are Hilbert–Schmidt, by the exponential decay, so the compressions to the two half-lines together have essential spectrum $\sigma(T_P)$; and Weyl sequences for T_P, truncated Bloch waves translated by multiples of P, can be placed in (s_0,∞). Hence the compression to (s_0,∞) has essential spectrum $\sigma(T_P)$ as well. □

When ψ_P is constant, the fibres are diagonalized by the Mellin transform, the bands of T_P fill $[0, c_R\Gamma(\theta/2)^2/\Gamma(\theta)]$, and Theorem 5.4 is recovered. Genuine oscillations occur for prime sets. If $R_P = \{p : \{\log\log p/P\} < \theta\}$, then by the prime number theorem ψ is asymptotically the P-periodic smoothing of the indicator of $\{y : \{y/P\} < \theta\}$ by the kernel e^{-e^z}, whose kth Fourier coefficient carries the factor $\Gamma(2\pi ik/P)$, and R_P has density θ with bounded remainder. Numerically, for $\theta = 1/2$ and $P = 8$, where ψ oscillates between ± 0.53, the two largest bands of T_P are $[3.137, 8.236]$ and $[0.359, 0.790]$, stable to four digits under refinement of the discretization; for $P = 12$ they are $[7.43, 10.80]$ and $[1.03, 1.46]$. So for densities with bounded remainder the critical essential spectrum is in general not an interval, and Problem 5 has a negative answer; the reduction to T_P is proved, while the gaps are established numerically.

5.2. Schatten classes along the curve

Put

$$a_j(\nu) = e^{j\theta}\nu(I_j), \qquad j \ge 0.$$

Chapter 4 shows that $M_{\nu,R}$ is bounded exactly when a is bounded, and compact exactly when $a_j \to 0$. The trace formula shows that $M_{\nu,R}$ is trace class exactly when $\int_0^1 u^{-\theta}\,\mathrm{d}\nu < \infty$, that is, when $a \in \ell^1$. The intermediate classes interpolate.

THEOREM 5.6 (Schatten classes of curve matrices). *Let R have density $\theta \in (0,1]$ with constant B, and let $1 \le p < \infty$. There are $c, C > 0$ depending only on θ, B and p such that, for every finite positive ν on $(0,\infty)$,*

$$c\sum_{j\ge 0} a_j(\nu)^p \le \|M_{\nu,R}\|_{S_p}^p \le C\Big(\sum_{j\ge 0} a_j(\nu)^p + \nu((1,\infty))^p\Big).$$

In particular $M_{\nu,R} \in S_p$ if and only if $a(\nu) \in \ell^p$.

PROOF. The part of ν on $(1,\infty)$ gives a positive matrix whose trace is at most $\zeta(3)\,\nu((1,\infty))$. So assume that ν lives on $(0,1]$.

Upper bound. Let $\nu_j = \nu|_{I_j}$ and, for those j with $a_j > 0$, let $B_j = M_{\nu_j,R}/a_j$. For a finitely supported complex sequence c put $Tc = \sum_j c_j B_j$, so that $M_{\nu,R} = Ta$. Since $\zeta_R(1+2u) \le e^{1+B}(2u)^{-\theta}$,

$$\|B_j\|_{S_1} = \operatorname{Tr} B_j \le e^{1+B}2^{-\theta}e^{(j+1)\theta}\nu(I_j)/a_j \le e^{2+B},$$

so $\|Tc\|_{S_1} \le e^{2+B}\|c\|_{\ell^1}$. For real c and finitely supported x, positivity gives $|\langle Tc\,x, x\rangle| \le \|c\|_\infty \langle M_{\nu',R}\,x, x\rangle$ with $\nu' = \sum_j \nu_j/a_j$. Every block of ν' has $a_j(\nu') = 1$, so $\nu'((0,t]) \le (e^\theta/(1-e^{-\theta}))\,t^\theta$ and Theorem 2.9 with $p = 2$ bounds $\|M_{\nu',R}\|$ by a constant. Splitting complex c into real and imaginary parts, $\|Tc\| \le C\|c\|_{\ell^\infty}$. Complex interpolation, with $[\ell^1, \ell^\infty]_{1-1/p} = \ell^p$ and $[S_1, B(H)]_{1-1/p} = S_p$, gives $\|Tc\|_{S_p} \le C\|c\|_{\ell^p}$. Apply this to $c = a$.

Lower bound. For $0 < \delta \le 1$ let $k_\delta(n) = n^{-1/2-\delta}\zeta_R(1+2\delta)^{-1/2}$, a unit vector in $\ell^2(\mathcal{S}_R)$. For $u \le \delta$ one has $\zeta_R(1+\delta+u) \ge \zeta_R(1+2\delta)$, so

$$\langle M_{\nu,R}k_\delta, k_\delta\rangle = \frac{\int \zeta_R(1+\delta+u)^2\,\mathrm{d}\nu(u)}{\zeta_R(1+2\delta)} \ge \zeta_R(1+2\delta)\,\nu((0,\delta]) \ge e^{-B}(2\delta)^{-\theta}\nu((0,\delta]).$$

With $\delta_j = e^{-j}$ this is at least $e^{-B}2^{-\theta}a_j$. The density bounds, used as in Lemma 2.8, give

$$\langle k_{\delta_j}, k_{\delta_k}\rangle = \frac{\zeta_R(1+\delta_j+\delta_k)}{\sqrt{\zeta_R(1+2\delta_j)\zeta_R(1+2\delta_k)}} \le e^{2B+1}\Big(\frac{2\sqrt{\delta_j\delta_k}}{\delta_j+\delta_k}\Big)^\theta \le 2e^{2B+1}e^{-\theta|j-k|/2}.$$

Choose L with $\sum_{h\ge1} 2e^{2B+1}e^{-\theta Lh/2} \le 1/2$. For a residue r modulo L, Schur's test shows that the Gram matrix of $(k_{\delta_{r+Li}})_{i\ge0}$ has norm at most $3/2$, so $V : e_i \mapsto k_{\delta_{r+Li}}$ satisfies $\|V\|^2 \le 3/2$. For $p \ge 1$ the diagonal of an S_p operator lies in ℓ^p with norm at most its S_p norm, hence

$$\sum_i \langle M_{\nu,R}k_{\delta_{r+Li}}, k_{\delta_{r+Li}}\rangle^p \le \|V^*M_{\nu,R}V\|_{S_p}^p \le (3/2)^p\|M_{\nu,R}\|_{S_p}^p.$$

Sum over the L residues. □

COROLLARY 5.7. *With $F(t) = \nu((0,t])$ and $0 < \theta \le 1$:*

(1) $M_{\nu,R} \in S_1$ if and only if $\int_0^1 u^{-\theta}\,\mathrm{d}\nu(u) < \infty$;
(2) $M_{\nu,R} \in S_2$ if and only if $\int_0^1 F(t)^2 t^{-1-2\theta}\,\mathrm{d}t < \infty$;
(3) if $\theta = 0$, then $M_{\nu,R}$ is trace class for every finite ν, with $\operatorname{Tr} M_{\nu,R} \le e^{1+B}\nu((0,\infty))$.

PROOF. (1) and (2) restate $a \in \ell^1$ and $a \in \ell^2$; for (2), note $e^{j\theta}F(e^{-j}) = \sum_{k\ge j} e^{-(k-j)\theta}a_k$, a convolution of a with a summable kernel, so $(e^{j\theta}F(e^{-j})) \in \ell^2$ exactly when $a \in \ell^2$. (3) is the trace formula with $\zeta_R(1+2u) \le e^{1+B}$. □

For R all primes, (1) and (2) are due to Perfekt and Pushnitski [40, Theorem 5.1] and the trace formula.

Part (3) sharpens the $\theta = 0$ case of Theorem 4.6, which gave only compactness.

REMARK (Positive matrices in every Schatten class have symbols). Take $\nu = \sum_j j^{-1/p} e^{-j\theta}\delta_{e^{-j}}$. Then $M_{\nu,R}$ lies in S_q exactly for $q > p$, and by Theorem 4.2 it has a bounded symbol. So positive curve matrices occupy every Schatten class, each with a bounded symbol. The counterexamples of chapter 7, which exist in S_p for every $p > 4$, are therefore necessarily not of this form. Inside the positive family the S_p condition is a pure size condition on ν; the S_p failure of Nehari's theorem is a phenomenon of cancellation.

A catalogue

The criteria of chapter 4 and Theorem 5.6 turn every borderline into an explicit matrix. Let R have density $\theta \in (0,1]$.

(1) *Power weights.* $d\nu = u^{\kappa-1}\,\mathrm{d}u$ gives $\alpha(n) = \Gamma(\kappa)n^{-1/2}(\log n)^{-\kappa}$ and $a_j \asymp e^{-j(\kappa-\theta)}$: in every S_p for $\kappa > \theta$, bounded but not compact for $\kappa = \theta$, unbounded for $\kappa < \theta$ (Corollary 5.3). For R all primes and $\kappa = 1$ this is the multiplicative Hilbert matrix.
(2) *Logarithmic weights.* $d\nu = u^{\theta-1}(\log(e/u))^{-\alpha}\,\mathrm{d}u$ on $(0,1]$ has $a_j \asymp (1+j)^{-\alpha}$. It is bounded exactly for $\alpha \ge 0$, compact exactly for $\alpha > 0$, and in S_p, $p \ge 1$, exactly for $\alpha p > 1$.
(3) *Iterated logarithms.* With an extra factor $(\log(e\log(e/u)))^{-\beta}$ at $\alpha = 1$, $a_j \asymp (1+j)^{-1}\log^{-\beta}(e+j)$: compact and in every S_p, $p > 1$, and trace class exactly for $\beta > 1$.
(4) *Sparse atoms.* $\nu = \sum_j c_j\delta_{e^{-Lj}}$ has $a_{Lj} = e^{Lj\theta}c_j$ and $a = 0$ elsewhere. With $c_j = e^{-Lj\theta}j^{-1/q}$ the singular values are exactly of order $n^{-1/q}$ (Theorem 5.9 below), so $M \in S_p$ iff $p > q$, for every $p > 0$.
(5) *Same profile, different spectra.* One atom in a block gives a rank-one matrix; two atoms in the same block, with the same total mass, give rank two (Proposition 5.8). So the profile cannot see individual singular values, and Theorem 5.9 needs its separation hypothesis.

Each example has a bounded symbol whenever it is bounded, with the constant of Theorem 4.2.

5.3. What the block profile determines

The profile $a(\nu)$ determines the operator norm, compactness and the S_p norms for $p \geq 1$, up to constants. It does not determine finer spectral information.

PROPOSITION 5.8. *Let R have density $\theta \in (0,1]$ with constant B.*

(1) $c \sup_j a_j(\nu) \leq \|M_{\nu,R}\| \leq C \sup_j a_j(\nu) + C\nu((1,\infty))$.

(2) If ν consists of N atoms at distinct points of $(0,\infty)$, then $\operatorname{rank} M_{\nu,R} = N$. In particular measures with the same profile can have different ranks, and no statement about individual singular values depends on $a(\nu)$ alone.

PROOF. (1) The lower bound is the test in the proof of Theorem 5.6; the upper bound is Theorem 2.9 with $[\nu]_\theta \lesssim \sup_j a_j$. (2) $M = \mathcal{N}^*\mathcal{N}$ with $\mathcal{N}$ mapping into $L^2(\nu) \cong \mathbb{C}^N$, and the vectors $(n^{-1/2-u_i})_{n \in \mathcal{S}_R}$ are linearly independent by uniqueness of Dirichlet coefficients, so $\mathcal{N}$ is onto. □

Individual singular values are controlled once each atom has its own scale.

THEOREM 5.9 (Separated atoms). *Let R have density $\theta \in (0,1]$ with constant B. There is $L = L(\theta, B)$ such that, if $\nu = \sum_i c_i \delta_{u_i}$ with $u_i \in (0,1]$ and $|\log u_i - \log u_j| \geq L$ for $i \neq j$, then*

$$\frac{1}{2} b_n^* \leq s_n(M_{\nu,R}) \leq \frac{3}{2} b_n^* \qquad (n \geq 1),$$

where $b_i = c_i \zeta_R(1+2u_i) \asymp c_i u_i^{-\theta}$ and b^* is its decreasing rearrangement. In particular $M_{\nu,R} \in S_p$ if and only if $b \in \ell^p$, for every $0 < p < \infty$.

PROOF. By Lemma 5.1 the nonzero spectrum of $M_{\nu,R}$ is that of the matrix $\big(\sqrt{c_i c_j}\, \zeta_R(1+u_i+u_j)\big)_{i,j} = D^{1/2} \widetilde{K} D^{1/2}$, with $D = \operatorname{diag}(b_i)$ and $\widetilde{K}$ the Gram matrix of the normalized kernels k_{u_i}. As in the proof of Theorem 5.6, $\widetilde{K}_{ij} \leq 2e^{2B+1} (2\sqrt{u_i u_j}/(u_i+u_j))^\theta \leq 2^{1+\theta} e^{2B+1} e^{-\theta |\log u_i - \log u_j|/2}$. At most two indices j have $|\log u_i - \log u_j| \in [mL, (m+1)L)$, so for L large Schur's test gives $\|\widetilde{K} - I\| \leq 1/2$. Hence $D/2 \leq D^{1/2} \widetilde{K} D^{1/2} \leq 3D/2$, and the min-max principle compares eigenvalues. □

So the profile cannot determine individual singular values, but it does determine their distribution up to dilations, and with it every Schatten and Lorentz class. For a positive sequence or operator x write $n_x(t) = \#\{n : x_n^* > t\}$, with x^* the decreasing rearrangement of the sequence or the singular values of the operator, and let $\ell^{p,q}$ and $S_{p,q}$ be the Lorentz sequence spaces and the corresponding Schatten–Lorentz classes, $0 < p < \infty$, $0 < q \leq \infty$; $S_{p,p} = S_p$, and $S_{p,\infty}$ is weak S_p.

THEOREM 5.10 (Distribution of the spectrum). *Let R have density $\theta \in (0,1]$ with constant B, and let ν be a finite positive measure on $(0,1]$. There are $C, \varepsilon > 0$ depending only on θ and B such that, for all $\tau > 0$ and $0 < r \leq 1$,*

$$n_a(\tau) \leq C\, n_M(\varepsilon\tau), \tag{5.1}$$

$$n_M(C\tau) \leq C_r\, \tau^{-r} \sum_{a_j > \tau} a_j^r, \tag{5.2}$$

where $M = M_{\nu,R}$, $a = a(\nu)$, and C_r depends also on r. Consequently, for every finite positive ν on $(0,\infty)$, every $0 < p < \infty$ and every $0 < q \le \infty$,

$$\|M_{\nu,R}\|_{S_{p,q}} \asymp \|a(\nu)\|_{\ell^{p,q}} + \nu((1,\infty)),$$

with constants depending only on θ, B, p and q. In particular Theorem 5.6 holds for every $0 < p < \infty$.

The proof rests on two estimates. The first is that one block is almost of finite rank: with $\nu_j = \nu|_{I_j}$ and $M_j = M_{\nu_j,R}$, there is $\eta = \eta(\theta, B) < 1$ with

$$s_{N+1}(M_j) \le C a_j \eta^{2N} \qquad (N \ge 0), \qquad \text{hence} \qquad \operatorname{Tr} M_j^r \le C_r a_j^r. \tag{5.3}$$

Indeed, $M_j = \mathcal{N}_j^* \mathcal{N}_j$ by Lemma 5.1, and $\mathcal{N}_j^*$ is synthesis $f \mapsto \int k(u) f(u)\, \mathrm{d}\nu_j(u)$ by the $\ell^2(\mathcal{S}_R)$-valued function $k(u) = (n^{-1/2-u})_n$, which is analytic in $\Re u > 0$ with $\|k(u)\|^2 = \zeta_R(1 + 2\Re u)$. Write $u = t(c + z)$ with $t = e^{-j}$ and $c = (1 + e^{-1})/2$; then I_j corresponds to $|z| \le \delta = (1 - e^{-1})/2$, and on $|z| = 1/2$ we have $\Re u \ge t e^{-1}/2$, so $\|k\| \le C t^{-\theta/2}$ there. Truncating the Taylor series in z after N terms gives an operator of rank at most N, and by Cauchy's estimate its Hilbert–Schmidt distance to $\mathcal{N}_j^*$ is at most $C(t^{-\theta}\nu(I_j))^{1/2}(2\delta)^N = C\sqrt{a_j}\,\eta^N$ with $\eta = 2\delta$.

The second estimate is the concavity inequality $\operatorname{Tr}(A+B)^r \le \operatorname{Tr} A^r + \operatorname{Tr} B^r$ for positive A, B and $0 < r \le 1$. It follows from $x^r = (\sin \pi r/\pi) \int_0^\infty t^{r-1} x (x+t)^{-1}\, \mathrm{d}t$ and $\operatorname{Tr} A(A+B+t)^{-1} \le \operatorname{Tr} A(A+t)^{-1}$.

Proof of Theorem 5.10. *Upper estimate.* Let $J = \{j : a_j > \tau\}$ and split $M = M' + M''$, with M' coming from the blocks in J. By part (1) of Proposition 5.8, $\|M''\| \le C\tau$, and by (5.3) and concavity, $\operatorname{Tr} M'^r \le C_r \sum_{j \in J} a_j^r$. Weyl's inequality gives $n_M((C+1)\tau) \le n_{M'}(\tau) \le \tau^{-r} \operatorname{Tr} M'^r$.

Lower estimate. Let F be any finite set of N blocks with $a_j > \tau$, and let $\mu = \sum_{j \in F} (\tau/a_j)\nu_j \le \nu$, so that $Q = M_{\mu,R} \le M$ and every nonzero entry of $a(\mu)$ equals τ. Then $\|Q\| \le C\tau$, $\operatorname{Tr} Q \ge cN\tau$ by Proposition 5.2 and $\zeta_R(1+2u) \ge c u^{-\theta}$, and $\operatorname{Tr} Q^r \le C_r N \tau^r$ for $r = 1/2$ by (5.3) and concavity. The eigenvalues of Q that are at most $\varepsilon\tau$ contribute at most $(\varepsilon\tau)^{1/2} \operatorname{Tr} Q^{1/2} \le C\varepsilon^{1/2} N\tau$ to the trace. With ε so small that this is at most $cN\tau/2$, at least $cN/(2C)$ eigenvalues of Q exceed $\varepsilon\tau$, and by the min-max principle so do as many eigenvalues of M. Since F was arbitrary, $n_a(\tau) \le (2C/c)\, n_M(\varepsilon\tau)$.

Lorentz norms. The quasi-norm $\|x\|_{\ell^{p,q}}^q$ is comparable to $\int_0^\infty \tau^{q-1} n_x(\tau)^{q/p}\, \mathrm{d}\tau$ for $q < \infty$, and $\|x\|_{\ell^{p,\infty}} = \sup_\tau \tau n_x(\tau)^{1/p}$, the same for singular values. The lower bound for M follows at once from (5.1). For the upper bound fix $r < \min(p, 1)$ and split the sum in (5.2) into the bands $2^l\tau < a_j \le 2^{l+1}\tau$:

$$\tau^p n_M(C\tau) \le C_r \sum_{l \ge 0} 2^{-l(p-r)} (2^l\tau)^p n_a(2^l\tau).$$

Raise to the power q/p and integrate against $\mathrm{d}\tau/\tau$, which is invariant under the dilations $\tau \mapsto 2^l\tau$; use subadditivity of $x \mapsto x^{q/p}$ if $q \le p$ and Minkowski's inequality in $L^{q/p}$ if $q > p$. The geometric factor makes the series converge. For $q = \infty$ take suprema. Finally, the part of ν on $(1,\infty)$ gives a matrix in every $S_{p,q}$ with norm at most $C\nu((1,\infty))$: write $k(u) = e_1 + \psi(u)$ with $\|\psi(w)\|^2 = \zeta_R(1+2\Re w) - 1 \le C 4^{-\Re w}$, apply the Taylor argument on the unit intervals $[m, m+1)$, $m \ge 1$, to ψ, and

absorb the rank-one part e_1 by $s_{n+1}(X+Y) \le s_n(X)$ for $\operatorname{rank} Y \le 1$. The entry $\alpha_\nu(1) \ge \nu((1,\infty))$ gives the reverse bound. □

In the Banach range of exponents the upper half of the theorem also follows from Olsen's local embedding of $\mathscr{H}^2$ into a weighted Dirichlet space of the half-plane [37], together with Luecking's description of Schatten class Toeplitz operators there [32]. The argument above avoids interpolation, and so covers $p < 1$, $q < 1$ and weak S_1 directly.

Two pointwise consequences answer the question of what lies between the profile and individual singular values.

COROLLARY 5.11. *Let R and ν be as in Theorem 5.10.*

(1) If the profile is doubling, $a_n^* \le D a_{2n}^*$ for all n, then $s_n(M_{\nu,R}) \asymp a_n^*$ for all n, with constants depending only on θ, B and D.

(2) If ν is purely atomic with at most K atoms in each block, and b is the sequence of Theorem 5.9, then for all n

$$c\, b^*_{\lceil Cn \rceil} \le s_n(M_{\nu,R}) \le C\, b_n^*,$$

with c, C depending only on θ, B and K.

PROOF. (1) Iterating the doubling condition gives $a_m^* \le D(n/m)^A a_n^*$ for $m \le n$, with $A = \log_2 D$. Take $\tau = a_n^*$ and $r < \min(1, 1/A)$ in (5.2). Since $n_a(\tau) < n$, $\sum_{a_j > \tau} a_j^r \le \sum_{m<n} (D(n/m)^A \tau)^r \le C\tau^r n$, so $n_M(Ca_n^*) \le Cn$, that is $s_{Cn}(M) \le Ca_n^*$; doubling removes the index change. The lower bound follows in the same way from (5.1) with $\tau = a_n^*/2$.

(2) Let L be as in Theorem 5.9. Colour each atom by the residue of its block index modulo $L+1$ and its position, among at most K, inside the block. Atoms of the same colour are L-separated, so each of the at most $K(L+1)$ coloured parts M_κ satisfies Theorem 5.9. Since $M = \sum_\kappa M_\kappa$ is a sum of positive operators, Weyl's inequality gives $n_M(K(L+1)t) \le \sum_\kappa n_{M_\kappa}(t) \le n_b(2t/3)$, the upper bound. Conversely $M \ge M_\kappa$ for every κ, and among the $K(L+1)n$ largest entries of b some colour has at least n, so $s_n(M) \ge b^*_{K(L+1)n}/2$. □

The index change in (2) cannot be removed, even for $K = 1$: two atoms with the same normalized mass, at $e^{-L}(1 \pm \delta)$ on either side of a block boundary, give almost the same rank-one operator, so the second singular value of the pair is of order δ times the first. Placing such pairs at rapidly separated scales with masses 2^{-j^2} and $\delta_j \to 0$ gives one measure with $s_{2j}/a^*_{2j} \to 0$. Nor does the profile determine the decay scale: with the same blocks and normalized masses e^{-j}, one atom per block gives $s_n \asymp e^{-n}$, while j atoms spread in the jth occupied block give $s_n = e^{-\Theta(\sqrt{n})}$, the root-exponential rate permitted by (5.2). Both measures lie in the same Lorentz classes, as they must.

5.4. Eigenvalue decay

Below the curve the spectrum is far smaller than Theorem 5.6 requires.

LEMMA 5.12. *Let h be completely monotone on $(0,\infty)$ with $h(t) \le Ct^{-\theta}$ on $(0,1]$ for some $\theta < 1$ and $h(t) \le Ce^{-ct}$ on $[1,\infty)$. The integral operator with kernel $h(u+v)$ on $L^2(0,\infty)$ has $s_n \le C'e^{-c'\sqrt{n}}$.*

PROOF. Bernstein's representation gives $|h^{(m)}(t)| \le m!(2/t)^m h(t/2)$. Fix m. Discard the outputs $u < e^{-m}$ and $u > m$, whose Hilbert–Schmidt norms are $O(e^{-cm})$ because $\theta < 1$ and the tail is exponential. Cover $[e^{-m}, m]$ by $O(m)$ intervals $[a,(1+\delta)a]$ with $\delta < 1/2$, and on each replace $h(u+v)$ by its Taylor polynomial of degree $m-1$ in u about a. Each piece has rank m, and the remainders have total Hilbert–Schmidt norm squared at most $C(2\delta)^{2m}\int_0^\infty th(t)^2\,\mathrm{d}t$. A rank-$Cm^2$ operator is thus within Ce^{-cm} in norm. □

THEOREM 5.13. *If R has density $\theta < 1$, the multiplicative Hilbert matrix on $\mathcal{S}_R \setminus \{1\}$ has $s_n \le Ce^{-c\sqrt{n}}$.*

PROOF. By Lemma 5.1, with Lebesgue measure and the index 1 removed, the companion kernel is $h_R(u+v)$ with $h_R(t) = \zeta_R(1+t) - 1$, completely monotone, $O(t^{-\theta})$ at 0 and $O(2^{-t})$ at infinity. Apply Lemma 5.12. □

At the other end of the family of chapter 4, where all signs are averaged, the decay is only polynomial, and its exact asymptotics can be computed.

THEOREM 5.14 (The full sign average). *Let A be the matrix with entries $\mathbf{1}_\square(mn)/(\sqrt{mn}\log mn)$ for $m,n \ge 2$. Then*

$$s_n(A) \sim \frac{1}{2n\log n} \qquad (n \to \infty).$$

In particular $A \in S_p$ exactly for $p > 1$, and A lies in weak S_1 but not in S_1.

PROOF. Write $m = qa^2$, $n = qb^2$ with q squarefree. Then $A = \bigoplus_q A_q$ with $A_q(a,b) = (2qab\log(qab))^{-1}$, where for $q = 1$ only $a, b \ge 2$ occur. For $q \ge 2$ let $L = \log q$, $u_t = (a^{-1-t})_{a\ge 1}$, and let T be exponential with rate L. Then $A_q = (2q)^{-1}\int_0^\infty e^{-Lt}u_t \otimes u_t\,\mathrm{d}t = (2qL)^{-1}\mathbb{E}[u_T \otimes u_T]$. Subtract the rank-one mean, $C_q = (2qL)^{-1}v \otimes v$ with $v = \mathbb{E}u_T$, whose entries are $v(a) = a^{-1}L/(L+\log a)$. The remainder $R_q = A_q - C_q$ is a covariance, hence positive, and since $\|\partial_t u_t\|_2$ is bounded and $\mathbb{E}T^2 = 2/L^2$,

$$\operatorname{Tr} R_q = \frac{\mathbb{E}\|u_T - v\|_2^2}{2qL} \le \frac{\mathbb{E}\|u_T - u_0\|_2^2}{2qL} \le \frac{C}{q\log^3 q}.$$

The eigenvalue of C_q is $c_q = \|v\|_2^2/(2qL) = \zeta(2)/(2q\log q)\,(1 + O(1/\log q))$, and the squarefree integers up to x number $x/\zeta(2) + o(x)$, so $C = \bigoplus_q C_q$ has $n_C(s) \sim 1/(2s\log(1/s))$ as $s \to 0$.

Let R be the sum of the R_q and the block A_1. For $q \le Q = s^{-1/2}$ use $R_q \le A_q \le (2/q)A_2$ and Lemma 5.12: the companion kernel of A_2 is $e^{-(u+v)\log 2/2}\zeta(2+u+v)/4$, completely monotone and exponentially decaying, so $s_j(A_2) \le Ce^{-c\sqrt{j}}$ and $n_{R_q}(s) \le C\log^2(1/s)$; the block A_1, with kernel $(\zeta(2+u+v)-1)/2$, is the same. For $q > Q$

use $n_{R_q}(s) \le \operatorname{Tr} R_q/s$. Altogether

$$n_R(s) \le C s^{-1/2}\log^2(1/s) + \frac{C}{s}\sum_{q>Q}\frac{1}{q\log^3 q} = o\Big(\frac{1}{s\log(1/s)}\Big).$$

Since $C \le A = C+R$, Weyl's inequality gives $n_C(s) \le n_A(s) \le n_C((1-\varepsilon)s)+n_R(\varepsilon s)$ for every $\varepsilon \in (0,1)$, so $n_A(s) \sim 1/(2s\log(1/s))$, which inverts to the claim. □

So along the family $\mu_{\nu,R}$ of chapter 4, with ν Lebesgue measure, the spectrum moves from $[0,\pi]$ (all coordinates on the curve) through root-exponential decay (a sparse set on the curve) to $1/(2n\log n)$ (all signs averaged). The intermediate cases, with R and its complement both infinite, are not computed.

5.5. Rank and Hilbert–Schmidt structure

For a submonoid $S \subseteq \mathbb{N}$ let $d_S(k) = \#\{(m,n) \in S^2 : mn = k\}$. The space $W_2(S)$ is the completion of polynomials under

$$\|F\|_{W_2(S)} = \inf\Big\{\Big(\sum_j \|G_j\|_2^2\|H_j\|_2^2\Big)^{1/2} : F = \sum_j G_j H_j\Big\},$$

with (G_j) and (H_j) orthogonal families in $H^2(S)$; equivalently the quotient of $H^2(S)\otimes_2 H^2(S)$ under multiplication.

THEOREM 5.15 (Helson's inequality is S_2 weak factorization). *For $F = \sum_{k\in S} a_k\chi_k$,*

$$\|F\|_{W_2(S)}^2 = \sum_{k\in S}\frac{|a_k|^2}{d_S(k)}.$$

Hence $\sum_k |a_k|^2/d_S(k) \le C^2\|F\|_1^2$ is equivalent, with the same constant, to the statement that every Hilbert–Schmidt S-Hankel form H_b has a bounded symbol of norm at most $C\|H_b\|_{S_2}$.

PROOF. A tensor $U = (u_{mn})$ maps to F when $\sum_{mn=k} u_{mn} = a_k$ for each k. The fibres are disjoint, and Cauchy–Schwarz in each fibre gives $\sum_{mn=k}|u_{mn}|^2 \ge |a_k|^2/d_S(k)$, with equality for the constant choice $u_{mn} = a_k/d_S(k)$. The dual norm is $\sum_k d_S(k)|b_k|^2 = \|H_b\|_{S_2}^2$, and Hahn–Banach supplies the symbol. □

For $S = \mathbb{N}$ this is Helson's inequality [24] read as $H^1(\mathbb{T}^\infty) \hookrightarrow W_2$ contractively. Classical weak factorization is the W_1 endpoint, with cost $\sum_j \|G_j\|_2\|H_j\|_2$, and it fails on $\mathbb{T}^\infty$ [38]; Nehari's theorem in S_q corresponds to embedding into $W_{q'}$. The divisor weight therefore supplies exactly the S_2 theorem and nothing more.

COROLLARY 5.16. *(1) If $K_S = \sup_{k\in S} d(k)/d_S(k) < \infty$, the intrinsic W_2 estimate holds with constant $\sqrt{K_S}$; for $S = \{1, 2^2, 2^3, \dots\}$, $K_S = 2$.*

(2) If S is freely generated by integers with $\mathbb{Z}$-independent exponent vectors, the estimate holds with constant 1.

Write $\mathcal{N}(H)$ for the least norm of a bounded symbol of a Helson matrix H, and $C(r) = \sup\{\mathcal{N}(H)/\|H\| : \operatorname{rank} H \le r\}$.

THEOREM 5.17 (Bounded rank). $\mathcal{N}(H) \le \sqrt{r}\,\|H\|$ *whenever* $\operatorname{rank} H \le r$, *independently of the number of variables. Moreover*

$$r^{\log_2(2/\sqrt{\pi})} \lesssim C(r) \le \sqrt{r}, \qquad \log_2(2/\sqrt{\pi}) = 0.1742\ldots$$

PROOF. $\|H\|_{S_2} \le \sqrt{r}\|H\|$, and Theorem 5.15 with $S = \mathbb{N}$, $C = 1$. For the lower bound, let $P_d = z_1 + \cdots + z_d$ and $M_d = \|P_d\|_{L^1(\mathbb{T}^d)}$. The Helson matrix of the degree-one coefficients of P_d has rank 2 and norm $\sqrt{d}$, and every symbol φ satisfies $d = |\int \varphi\overline{P_d}| \le \|\varphi\|_\infty M_d$; the symbol $(d/M_d)P_d/|P_d|$ is admissible, by rotation and permutation symmetry, so $\mathcal{N} = d/M_d$ exactly. Tensoring k copies on disjoint variables gives rank 2^k, norm $d^{k/2}$ and $\mathcal{N} = (d/M_d)^k$; by conditional expectation, extra variables do not help. The central limit theorem gives $M_d/\sqrt{d} \to \sqrt{\pi}/2$. □

The tensor examples are the only lower bounds we have, and their exponent is $\log_2$ of the ratio $\|Z\|_2/\|Z\|_1$ for a complex Gaussian Z. Any improvement must use matrices that are not tensor products of rank-two blocks.

REMARK (Maximal forms). Call $H_\varphi(F,G) = \int FG\overline{\varphi}\,\mathrm{d}m$, with $\varphi \in H^2 \cap L^\infty$, maximal if $\|H_\varphi\| = \|\varphi\|_\infty$ and minimal if $\|H_\varphi\| = \|\varphi\|_2$. The Brevig–Seip argument transfers to $\mathbb{T}^\infty$: if a maximal form attains its norm at F, G, equality in $|\int FG\overline{\varphi}| \le \|\varphi\|_\infty\|FG\|_1$ forces $|\varphi| = \|\varphi\|_\infty$ on $\{FG \ne 0\}$, which has full measure because a nonzero $H^1(\mathbb{T}^\infty)$ function vanishes only on a null set [1, Corollary 2]. So compact maximal Helson matrices are minimal.

Open

(1) Characterize the measures with $s_n(M_{\nu,R}) \asymp a_n^*$ without an index change; doubling profiles (Corollary 5.11) and separated atoms (Theorem 5.9) are two sufficient conditions.
(2) The true exponent of $C(r)$, between 0.174 and 1/2 (B7).
(3) The spectral scale for R and its complement both infinite.
(4) Certify the spectral gaps of Proposition 5.5 for R_P, and describe the critical essential spectrum when ψ is not asymptotically periodic.

CHAPTER 6

Bergman and Fock spaces of Dirichlet series

Question. What changes when Haar measure on $\mathbb{T}^\infty$ is replaced by a Bergman or Gaussian product measure?

Less than one might expect. Separation along curves is the Hellinger–Bessel inequality, which holds in all three geometries (chapter 2). Symmetry under sign changes needs a substitute for inner-outer factorization, and a contractive embedding into a space with heavier weight supplies it: Carleman's inequality for Hardy, Kulikov's theorem [31] for Bergman, and its $\beta \to \infty$ limit, proved below, for Gauss. Each embedding gives a Helson-type inequality, a two-point lemma with constant $1 + O(|x|^4)$, and the approximation step of a Hartman theorem. With these, the positive Nehari theorem and its compactness counterpart hold in every geometry, and the Bergman multiplicative Hilbert form has a bounded symbol.

New here, to our knowledge, are the Gauss limit of Kulikov's theorem and the Gauss Helson inequality, the two-point constants outside the Hardy case, the compactness criterion for every $\beta \geq 1$, the sign-averaged endpoint in the Bergman and Gauss geometries, and the bounded symbol for the Bergman multiplicative Hilbert form.

6.1. Spaces of Dirichlet series and their symbols

We keep the three geometries of section 1.1 and identify a Dirichlet series $f(s) = \sum a_n n^{-s}$ with its Bohr lift $F(z) = \sum a_n z^{\kappa(n)}$, so that $f(s) = F(\gamma(s))$ with $\gamma(s) = (p^{-s})_p$. Write $\mathscr{H}^p$, $\mathscr{A}^p_\beta$ and $\mathscr{F}^p$ for the closures of the Dirichlet polynomials in L^p of Haar measure, of m_β^∞ and of γ^∞. The monomials are orthogonal in each L^2 space, with

$$\|n^{-s}\|^2_{\mathscr{H}^2} = 1, \qquad \|n^{-s}\|^2_{\mathscr{A}^2_\beta} = \frac{1}{d_\beta(n)}, \qquad \|n^{-s}\|^2_{\mathscr{F}^2} = \kappa(n)! := \prod_p \kappa_p(n)!,$$

where $\zeta(s)^\beta = \sum d_\beta(n) n^{-s}$. The Bergman spaces are those of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [10]. For $\beta = 2$ they are the spaces of Bailleul and Lefèvre [2]. The Gauss spaces are the Fock-type spaces of Dirichlet series; along the curve γ, point evaluation has norm $\exp(\sum_p p^{-2\Re s})$.

Let X be one of these families and α a sequence. The *weighted Helson form* with moments α is

$$H_\alpha(f, g) = \sum_k (fg)_k \, \alpha(k) = \langle fg, \varphi \rangle_{X^2}, \qquad \varphi = \sum_n \frac{\alpha(n)}{\|n^{-s}\|^2_{X^2}} \, n^{-s}.$$

For the Bergman spaces these are the forms of [10] and [33], with $\varrho(n) = d_\beta(n)\alpha(n)$. A *bounded symbol* is a function $g \in L^\infty(\lambda)$ with $\int z^{\kappa(n)} g \, d\lambda = \alpha(n)$ for every n. Its orthogonal projection onto X^2 is then φ.

LEMMA 6.1. *H_α has a bounded symbol if and only if $h \mapsto \sum_k h_k \alpha(k)$ is bounded on X^1, and then $\|H_\alpha\| \le \inf \|g\|_\infty$, which equals the norm of that functional.*

PROOF. As Lemma 4.1, using $\|FG\|_{L^1(\lambda)} \le \|F\|_{L^2(\lambda)} \|G\|_{L^2(\lambda)}$. □

6.2. Contractive embeddings

THEOREM 6.2 (Kulikov). *For every $\beta \ge 1$ and every analytic polynomial g in one variable, $\|g\|_{L^2(m_{2\beta})} \le \|g\|_{L^1(m_\beta)}$, where m_1 is arc length on $\mathbb{T}$.*

This is [31, Corollary 1.3] in the case $p = 1$, $q = 2$; our normalization of m_β agrees with Kulikov's. For $\beta = 1$ it is Carleman's inequality $\|g\|_{A^2} \le \|g\|_{H^1}$. For integer β it also follows from repeated applications of [10, Theorem 1].

Let $\gamma_a = (a/\pi) e^{-a|z|^2} \, dx \, dy$ on $\mathbb{C}$, so $\gamma = \gamma_1$.

PROPOSITION 6.3 (Gauss as the limit of Bergman). *For every analytic polynomial g in one variable, $\|g\|_{L^2(\gamma_2)} \le \|g\|_{L^1(\gamma_1)}$.*

PROOF. Apply Theorem 6.2 to $g_\beta(z) = g(\sqrt{\beta} z)$ and substitute $z = w/\sqrt{\beta}$. With dA normalized area,

$$\|g_\beta\|_{L^1(m_\beta)} = \frac{\beta - 1}{\beta} \int_{|w| < \sqrt{\beta}} |g(w)| \Big(1 - \frac{|w|^2}{\beta}\Big)^{\beta - 2} \frac{dx \, dy}{\pi} \longrightarrow \|g\|_{L^1(\gamma_1)},$$

and similarly $\|g_\beta\|^2_{L^2(m_{2\beta})} \to \|g\|^2_{L^2(\gamma_2)}$, since $(2\beta - 1)\beta^{-1}(1 - |w|^2/\beta)^{2\beta - 2} \to 2e^{-2|w|^2}$. Dominated convergence applies because $(1 - x/\beta)^{\beta - 2} \le e^{-x/2}$ for $\beta \ge 4$. □

The rescaling $z = w/\sqrt{\beta}$ is the general principle behind this: the Bergman geometry with weight β tends to the Gauss geometry as $\beta \to \infty$, coordinates measured in units of $\beta^{-1/2}$. Several statements in this chapter could be obtained for the Gauss geometry by passing to that limit, but the direct proofs are just as short.

Each embedding tensorizes. Applying it in one coordinate at a time and using Minkowski's integral inequality, as in [10], gives contractive embeddings on infinitely many coordinates. Reading off the L^2 norm in coefficients:

COROLLARY 6.4 (Helson inequalities). *For every $f = \sum a_n n^{-s}$:*

$$\sum_n \frac{|a_n|^2}{d(n)} \le \|f\|^2_{\mathscr{H}^1}, \qquad \sum_n \frac{|a_n|^2}{d_{2\beta}(n)} \le \|f\|^2_{\mathscr{A}^1_\beta}, \qquad \sum_n |a_n|^2 \frac{\kappa(n)!}{2^{\Omega(n)}} \le \|F\|^2_{L^1(\gamma^\infty)}.$$

The first is Helson's inequality [24], and the second was proved for $\beta = 2$ by Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [10]. The third uses that the monomials of $L^2(\gamma_2)$ have $\|z^k\|^2 = k!/2^k$.

Kulikov also records the full Hardy scale [31, Theorem 5.1]: for $0 < p \le 2$,

$$\sum_n \frac{|a_n|^2}{d_{2/p}(n)} \le \|f\|^2_{\mathscr{H}^p}.$$

His proof tensorizes the one-variable contraction $H^p \hookrightarrow A^2_{2/p}$. This is valid also for $p < 1$, because the Minkowski exponent in the induction is $2/p \geq 1$.

6.3. Sign averages

PROPOSITION 6.5 (Two-point lemmas). *Let $x \in \mathbb{C}$, with $|x| < 1$ in the first two cases, and let g be an analytic polynomial. Then $(|g(x)|+|g(-x)|)/2 \leq c(x)\|g\|_{L^1(\lambda)}$, with*

$$c(x) = \frac{1}{1-|x|^4}, \qquad c(x) = \Lambda_\beta(|x|^2)^{1/2}, \qquad c(x) = \cosh(2|x|^2)^{1/2}$$

in the Hardy, Bergman and Gauss cases, where $\Lambda_\beta(t) = \big((1-t)^{-2\beta}+(1+t)^{-2\beta}\big)/2 = 1 + \beta(2\beta+1)t^2 + O(t^4)$. The Hardy constant is sharp.

PROOF. The Hardy case is Lemma 3.9 with $k = 2$. In the other two, by Cauchy–Schwarz and the contractive embedding into a space Y with kernel K^Y,

$$\frac{1}{2}\big(|g(x)| + |g(-x)|\big) \leq \Big(\frac{1}{2}\big(|\langle g, K^Y_x\rangle|^2 + |\langle g, K^Y_{-x}\rangle|^2\big)\Big)^{1/2} \leq \big(\frac{1}{2}\|G\|\big)^{1/2}\|g\|_Y,$$

where G is the Gram matrix of $K^Y_{\pm x}$. Its entries are real and positive: $K^Y(x,x)$ on the diagonal and $K^Y(x,-x)$ off it, so $\|G\| = K^Y(x,x) + K^Y(x,-x)$. For $Y = A^2_{2\beta}$ the kernel is $(1-\overline{w}z)^{-2\beta}$, giving $2\Lambda_\beta(|x|^2)$. For $Y = L^2(\gamma_2)$ it is $e^{2z\overline{w}}$, giving $e^{2|x|^2} + e^{-2|x|^2}$. □

All three constants are $1 + O(|x|^4)$. That is what lets them tensorize over the primes, where $\sum_p |x_p|^4 \leq \sum_p p^{-2} < \infty$. A single point evaluation, by contrast, costs $\|K_x\|_2^2 = 1 + O(|x|^2)$. Comparing the Hardy cases is instructive: the route through Carleman's inequality gives $\Lambda_1(t)^{1/2} = 1 + 3t^2/2 + O(t^4)$ instead of the sharp $1+t^2+O(t^4)$. Passing through L^2 loses only in the coefficient of the quadratic term. In the Bergman case the whole constant can be recovered from the L^1 norm near $x = 0$, and in both cases the obvious lower bound fails once $K_x + K_{-x}$ has a zero.

PROPOSITION 6.6 (Sharp two-point constants). *In the Bergman and Gauss cases, let $c(x)$ be the least constant in Proposition 6.5, let K be the kernel of X^2 (that is, of A^2_β, or of $L^2(\gamma)$ in the Gauss case), and let $\mu = \int |z|^4 \,\mathrm{d}\lambda$, which is $2/(\beta(\beta+1))$ and 2 respectively.*

1. *$c(x) \geq L(x) \coloneqq (K(x,x) + K(x,-x))/2$, that is, $\big((1-|x|^2)^{-\beta} + (1+|x|^2)^{-\beta}\big)/2$ in the Bergman case and $\cosh(|x|^2)$ in the Gauss case.*
2. *$c(x) = 1 + |x|^4/\mu + o(|x|^4)$ as $x \to 0$; that is, $1 + \beta(\beta+1)|x|^4/2 + o(|x|^4)$ and $1 + |x|^4/2 + o(|x|^4)$.*
3. *In the Bergman case, $c(x) = L(x)$ whenever $\beta \arctan|x| \leq \pi/2$: for every x when $\beta \leq 2$, and for $|x| \leq \tan(\pi/(2\beta))$ when $\beta > 2$.*
4. *$c(x) > L(x)$ in the Bergman case when $|x| > \tan(\pi/(2\beta))$, which can happen only for $\beta > 2$, and in the Gauss case for every $x \neq 0$.*

PROOF. By rotation invariance we may take $x = r > 0$. Put $H = K_r + K_{-r}$, $A = K(r,r)$ and $B = K(r,-r)$, so that $H(\pm r) = A + B = 2L$ and $\|H\|_2^2 = 4L$.

(1) The function $g = H^2$ lies in the closure of the polynomials in L^1, and $\|g\|_1 = \|H\|_2^2 = 4L$, $g(\pm r) = 4L^2$. So its two-point quotient is L.

(2) The lower bound is (1). For the upper bound, let $g = \sum a_n z^n$ be a polynomial with $\|g\|_1 = 1$ and two-point quotient $N \ge 1$, rotated so that $a_0 \ge 0$; note $a_0 \le 1$. Let $w_n = \|z^n\|_Y^2$ in the space Y of the proof of Proposition 6.5, and $q = 1 - a_0^2$. The contractive embedding gives $\sum_{n\ge1} w_n |a_n|^2 \le q$, so by Cauchy–Schwarz

$$\sum_{n\ge k} |a_n|\rho^n \le \sqrt{q}\,\Phi_k(\rho), \qquad \Phi_k(\rho)^2 = \sum_{n\ge k} \rho^{2n}/w_n \le K^Y(\rho,\rho),$$

and $\Phi_k(r) = O(r^k)$ as $r \to 0$. Let $e = (g(r) + g(-r))/2$ and $o = (g(r) - g(-r))/2$. Then $e = a_0 + a_2 r^2 + O(\sqrt{q}\,r^4)$, $o = O(\sqrt{q}\,r)$, and

$$N \le \big(|e|^2 + |o|^2\big)^{1/2}.$$

Since $a_0 \le 1 - q/2$ and $|a_2| \le \sqrt{q/w_2}$, the condition $N \ge 1$ forces $\sqrt{q} = O(r^2)$. Hence $o = O(r^3)$ and $N \le a_0 + |a_2| r^2 + O(r^6)$.

It remains to bound $1 - a_0$ below. Let $h = g - a_0$. For complex h and $a_0 > 0$, $|a_0 + h| \ge a_0 + \Re h$, and $|a_0 + h| \ge a_0 + \Re h + (\Im h)^2/(2(a_0 + |h|))$. Since $\int \Re h \,\mathrm{d}\lambda = 0$, for any disc $D_\rho = \{|z| \le \rho\}$,

$$1 - a_0 = \int \big(|a_0 + h| - a_0 - \Re h\big)\,\mathrm{d}\lambda \ge \frac{1}{2(1+\eta)} \int_{D_\rho} (\Im h)^2 \,\mathrm{d}\lambda,$$

where $\eta = \sup_{D_\rho} |h| \le \sqrt{q}\,\Phi_1(\rho)$. On the circle of radius t the mean of $(\Im h)^2 = (|h|^2 - \Re h^2)/2$ is $\frac12 \sum_{n\ge1} |a_n|^2 t^{2n} \ge \frac12 |a_2|^2 t^4$. Hence $1 - a_0 \ge |a_2|^2 \mu(\rho)/(4(1+\eta))$ with $\mu(\rho) = \int_{D_\rho} |z|^4 \,\mathrm{d}\lambda$. Let ρ tend to the boundary slowly enough that $r^2 \Phi_1(\rho) \to 0$. Then $\eta \to 0$ and $\mu(\rho) \to \mu$, so

$$N \le 1 - \frac{\mu}{4}(1 - o(1))|a_2|^2 + |a_2| r^2 + O(r^6) \le 1 + (1 + o(1))\frac{r^4}{\mu}.$$

(3) By continuity in r for a fixed polynomial, we may assume $\beta \arctan r < \pi/2$. Here $H(z) = 2(1 - r^2 z^2)^{-\beta/2} \cosh(\beta \operatorname{artanh}(rz))$ and $|\Im \operatorname{artanh}(rz)| \le \arctan r$ on the closed disc, so H has no zeros near the closed disc. By reproduction, $\int h|H|^2 \,\mathrm{d}\lambda = 2L(h(r) + h(-r))$ for h analytic near the closed disc. So the real measure $\sigma = |H|^2 \lambda - 2L(\delta_r + \delta_{-r})$ has zero mass and zero analytic moments, and its logarithmic potential $U(w) = \int \log|w - z| \,\mathrm{d}\sigma(z)$ vanishes for $|w| \ge 1$; it is continuous off $\pm r$, since the density of λ lies in L^p for some $p > 1$, and tends to $+\infty$ at $\pm r$.

We show $U \ge 0$. Angular integration gives, with $\rho_\beta(t) = (\beta - 1)(1 - t)^{\beta - 2}$,

$$\int \frac{\overline{H(z)}}{w - z}\,\mathrm{d}\lambda(z) = \frac{1}{w} \int_0^{|w|^2} H(t/w)\rho_\beta(t)\,\mathrm{d}t = \sum_{\varepsilon = \pm 1} \frac{1 - (1 - |w|^2)^{\beta-1}(1 - \varepsilon r\overline{w})^{1-\beta}}{w - \varepsilon r},$$

and reproduction applied to $(H(z) - H(w))/(w - z)$ then gives

$$2\partial_w U(w) = -(1 - |w|^2)^{\beta-1} H(w) \Big(\frac{(1 - r\overline{w})^{1-\beta}}{w - r} + \frac{(1 + r\overline{w})^{1-\beta}}{w + r} \Big).$$

At a critical point the bracket vanishes, so $|w + r|/|w - r| = (|1 - r\overline{w}|/|1 + r\overline{w}|)^{\beta-1}$; if $\Re w \ne 0$ one side exceeds 1 and the other does not, so critical points lie on the

imaginary axis. For $w = iy$, $0 < y < 1$, put $a = \arctan(ry)$ and $\gamma = \beta - 1$. Then $H(iy) = 2(1 + r^2y^2)^{-\beta/2}\cos(\beta a) > 0$ and

$$\partial_y U(iy) = \frac{2(1-y^2)^\gamma H(iy)}{(r^2+y^2)(1+r^2y^2)^{\gamma/2}}\big(r\sin(\gamma a) - y\cos(\gamma a)\big).$$

The ratio $\arctan(y/r)/\arctan(ry)$ decreases in y, because the ratio $(1+r^2y^2)/(r^2+y^2)$ of the derivatives does, so it exceeds its value $(\pi/2 - \arctan r)/\arctan r > \gamma$ at $y = 1$. Hence $\gamma a < \arctan(y/r)$, the bracket is negative, and $U(iy) > U(i) = 0$; by symmetry the same holds for $y < 0$. If U were negative somewhere, it would have a negative minimum in the disc away from $\pm r$, which is a critical point; there are none off the imaginary axis, and $U > 0$ on it. So $U \geq 0$.

If u is smooth and subharmonic near the closed disc, then

$$\int u\,\mathrm{d}\sigma = \frac{1}{2\pi}\int U\Delta u\,\mathrm{d}A \geq 0.$$

For a polynomial g put $f = g/H^2$ and apply this to $u = (|f|^2 + \varepsilon^2)^{1/2}$. Letting $\varepsilon \to 0$,

$$\|g\|_1 = \int |f||H|^2\,\mathrm{d}\lambda \geq 2L\big(|f(r)| + |f(-r)|\big) = \frac{|g(r)| + |g(-r)|}{2L}.$$

With (1), $c(r) = L$.

(4) Let $a = ic$ be a zero of H with $c > 0$ inside the domain: in the Bergman case $z = i\tau/r$, $\tau = \tan(\pi/(2\beta))$, is a zero, inside when $r > \tau$; in the Gauss case $H = 2\cosh(rz)$ vanishes at $i\pi/(2r)$. Let $F_t = H^2 + itH/(z-a)$ for real t, which lies in the closure of the polynomials. Then $F_t(\pm r) = 4L^2 + 2itL/(\pm r - a)$, and both moduli have derivative $-2Lc/(r^2+c^2)$ at $t = 0$. Since $|F_t| - |F_0| = O(t|H/(z-a)|)$, the norm has derivative $\Re(iJ)$, $J = \int \overline{H}/(z-a)\,\mathrm{d}\lambda$. Write $\mathrm{d}\lambda = w(u)\,\mathrm{d}u\;\mathrm{d}\theta/(2\pi)$ with $u = |z|^2$. On a circle $|z| < c$ expand $1/(z-a)$ in powers of z/a, and on $|z| > c$ in powers of a/z; only the inside contributes, and since H is even with real coefficients,

$$\Re(iJ) = -\frac{1}{c}\int_0^{c^2} H(iu/c)\,w(u)\,\mathrm{d}u.$$

Over the whole range of u this integral is known. With $\omega = r/c$,

$$\frac{1}{2}\int H(iu/c)w(u)\,\mathrm{d}u = \Re\int (1 - i\omega u)^{-\beta}w(u)\,\mathrm{d}u = \Re\frac{1}{1-i\omega} = \frac{1}{1+\omega^2},$$

by Euler's integral for ${}_2F_1(\beta, 1; \beta; i\omega)$, and in the Gauss case by $\int_0^\infty e^{i\omega u - u}\,\mathrm{d}u = (1 - i\omega)^{-1}$. The quotient $Q_t = N_t/\|F_t\|_1$ equals L at $t = 0$, so

$$Q_0' = -\frac{c}{2(r^2+c^2)} + \frac{1}{4c}\int_0^{c^2} H(iu/c)w(u)\,\mathrm{d}u = -\frac{1}{4c}\int_{u>c^2} H(iu/c)w(u)\,\mathrm{d}u.$$

In the Gauss case $H(iu/c) = 2\cos(\omega u)$ and $rc = \pi/2$, so $Q_0' = re^{-c^2}/(2(r^2+c^2)) > 0$. In the Bergman case $\frac{1}{2}H(iu/c) = \Re(1 + i\omega u)^{-\beta}$. Substitute $u = \tan\phi/\omega$ and put

$\tan\phi_1 = \omega$; then $1-u = \sin(\phi_1-\phi)/(\sin\phi_1\cos\phi)$, the lower limit becomes $\phi = \pi/(2\beta)$ because $\omega c^2 = \tau$, and

$$Q_0' = \frac{\beta-1}{2c\,\omega\sin^{\beta-2}\phi_1}\int_0^{\alpha}\sin(\beta(\alpha-\psi))\sin^{\beta-2}\psi\,\mathrm{d}\psi = \frac{\sin^{\beta}\alpha}{2c\,\omega\sin^{\beta-2}\phi_1},$$

where $\alpha = \phi_1 - \pi/(2\beta)$. The integral equals $\sin^{\beta}\alpha/(\beta-1)$: both sides vanish to first order at $\alpha=0$ and solve $y''+\beta^2 y = \beta\sin^{\beta-2}\alpha$. Since $\tau<\omega<1/\tau$, $0<\alpha<\pi/2$, so $Q_0'>0$. In both cases the quotient of F_t exceeds L for small $t>0$. □

For $\beta=3$, $Q_0' = c(3|x|^2-1)^3/(16|x|^2(3|x|^4+1))$ with $c^2 = 1/(3|x|^2)$.

Remark (A dual certificate). Near $x=0$ there is a shorter proof of (3), which also covers the Hardy space. Here $K(z,w) = (1-z\overline{w})^{-\beta}$. Put $O = K_r - K_{-r}$ and $\varphi = O/H = \tanh(\beta\,\mathrm{artanh}(rz))$. The Taylor coefficients of artanh are nonnegative, and those of tanh have the moduli of those of tan, which are nonnegative; so on the disc

$$|\varphi| \le \tan(\beta\,\mathrm{artanh}\,r) \le 1/2.$$

In particular H has no zeros on the closed disc. For a polynomial g let $G = g/H$, which is analytic on a neighbourhood of the closed disc, and let e, o be as in the proof of (2). Since $\langle G, H\rangle = G(r)+G(-r)$ and $H(\pm r) = 2L$,

$$e = L\int g\,\frac{\overline{H}}{H}\,\mathrm{d}\lambda, \qquad o = L\big(G(r)-G(-r)\big).$$

Let $\rho = (2L/A)\Re\varphi$, a real function with $|\rho| \le L/A \le 1$. Then

$$\begin{aligned}\int G\overline{H}\rho\,\mathrm{d}\lambda &= \frac{L}{A}\Big(\langle G\varphi, H\rangle + \langle G, \varphi H\rangle\Big)\\ &= \frac{L}{A}\big(1+\varphi(r)\big)\big(G(r)-G(-r)\big) = G(r)-G(-r),\end{aligned}$$

because φ is odd, $\varphi H = O$ and $\varphi(r) = (A-B)/(A+B)$. So $o = L\int g(\overline{H}/H)\rho\,\mathrm{d}\lambda$. Now $(|g(r)|+|g(-r)|)/2 = \max_\theta |e\cos\theta - io\sin\theta|$, and

$$|e\cos\theta - io\sin\theta| = L\left|\int g\,\frac{\overline{H}}{H}(\cos\theta - i\rho\sin\theta)\,\mathrm{d}\lambda\right| \le L\|g\|_1,$$

since $|\cos\theta - i\rho\sin\theta| \le 1$. This needs $\tan(\beta\,\mathrm{artanh}|x|) \le 1/2$. With the Hardy measure and $\beta=1$ it gives $|\rho| \le 2|x|/(1+|x|^2) \le 1$ for every x, a second proof of the Hardy constant.

In (3) and (4) together, the square bound is sharp exactly up to the first zero of $K_x + K_{-x}$: beyond it the squares H^2 stop being extremal because a zero of H can be moved in one of the two factors; in the Gauss case the gain $e^{-\pi^2/(4|x|^2)}$ is invisible to every order at $x=0$, consistent with (2).

Remark (Neumann ovals). For $\beta=2$, where λ is normalized area measure, (3) holds on the whole disc and has a geometric meaning. With $\phi(z) = z/(1-x^2z^2)$, so that $2\phi' = H$, the inequality $U \ge 0$ says that the Neumann oval $\phi(\mathbb{D})$ is a quadrature domain for subharmonic functions, with two equal nodes at $\pm x/(1-x^4)$: $\pi^{-1}\int_{\phi(\mathbb{D})} u\,\mathrm{d}A \ge (L/2)(u(t)+u(-t))$. This classical fact (see Nishry–Wennman [35]) can also be proved by shrinking the oval to two tangent discs.

PROPOSITION 6.7. *Let F be an analytic polynomial, let Q be a set of coordinates with points x_p, $p \in Q$, and fix the other coordinates at points w_p. With independent signs ε_p,*

$$\mathbb{E}_\varepsilon \big|F\big((\varepsilon_p x_p)_{p\in Q}, w\big)\big| \le \prod_{p\in Q} c(x_p) \int \big|F(\zeta, w)\big| \, \mathrm{d}\lambda^Q(\zeta).$$

For $x = \gamma_Q(s)$ with $\Re s \ge 1/2$ the constant is at most

$$C_{\mathrm{sign}} = \zeta(2), \qquad C_\beta = \prod_p \Lambda_\beta(1/p)^{1/2}, \qquad \prod_p \cosh(2/p)^{1/2} \le e^{\sum_p p^{-2}} < 1.58.$$

PROOF. As Proposition 3.10, by induction on the coordinates of Q with Fubini. For the Gauss bound use $\cosh y \le e^{y^2/2}$. □

For comparison, the Carleman route at $\beta = 1$ gives $C_1 = \zeta(2)^{3/2}\zeta(4)^{-1/2} \approx 2.03$ instead of $\zeta(2)$, and $C_2 \approx 7.26$.

The constant in Proposition 6.7 is $\prod_{p\in Q} c(x_p)$ exactly, and on the Bergman side it can now be computed.

COROLLARY 6.8 (Sharp sign averages). *In the Bergman case with $1 < \beta \le \pi/(2\arctan(2^{-1/2}))$, the best constant in Proposition 6.7, uniformly over $\Re s \ge 1/2$, over Q and over F, is*

$$C_\beta^{\mathrm{sharp}} = \prod_p L(p^{-1/2}) = \prod_p \frac{(1-p^{-1})^{-\beta} + (1+p^{-1})^{-\beta}}{2},$$

and it is attained in the closure of the polynomials. For $\beta = 2$, $C_2^{\mathrm{sharp}} = \zeta(2)^3/\zeta(4) = 5\pi^2/12 \approx 4.11$. In the Hardy case the constant $\zeta(2)$ is sharp in the same sense.

PROOF. For product polynomials $F = \prod_{p\in Q} g_p(z_p)$, with no other coordinates, both sides of Proposition 6.7 factor, so the best constant for a finite Q is $\prod_{p\in Q} c(x_p)$. Since $c \le \Lambda_\beta^{1/2}$, the factors are $1+O(p^{-2})$, and the product over all primes converges. The condition on β gives $\beta \arctan(p^{-1/2}) \le \pi/2$ for every prime, so $c(p^{-\sigma}) = L(p^{-\sigma})$ by Proposition 6.6(3), and L increases in $|x|$, so $\sigma = 1/2$ is the worst case. At $\beta = 2$, $L(p^{-1/2}) = (1-p^{-4})/(1-p^{-2})^3$. For attainment, let $h_p = (K_{r_p} + K_{-r_p})/(2L(r_p)^{1/2})$ with $r_p = p^{-1/2}$, so that $\|h_p\|_2 = 1$, $\int h_p \, \mathrm{d}\lambda = L(r_p)^{-1/2}$ and $h_p(\pm r_p)^2 = L(r_p)$. The products $H_N = \prod_{j\le N} h_{p_j}$ satisfy $\|H_N - H_M\|_2^2 = 2 - 2\prod_{M<j\le N} L(r_{p_j})^{-1/2} \to 0$, so H_N^2 converges in L^1 to some F with $\|F\|_1 = 1$, and the sign average of H_N^2 is the constant $\prod_{j\le N} L(r_{p_j})$. In the Hardy case $c(x) = (1-|x|^4)^{-1}$ for every x, and the same argument gives $\prod_p (1-p^{-2})^{-1} = \zeta(2)$. □

For larger β the finitely many primes $p < \cot^2(\pi/(2\beta))$ fall outside Proposition 6.6(3), and their factors exceed $L(p^{-1/2})$ by item (4).

6.4. Dirichlet kernels

The contractive embedding also measures cancellation. Let $D_N = \sum_{n\le N} n^{-s}$ and $d_\beta(n)$ the coefficients of ζ^β, so that $\|D_N\|_{\mathscr{A}^2_\beta}^2 = \sum_{n\le N} 1/d_\beta(n)$.

PROPOSITION 6.9. *Fix $\beta \geq 1$ and put $G_\beta(1) = \prod_p (1-1/p)^{1/\beta} \sum_{k\geq 0} k!\, p^{-k}/(\beta)_k$. Then*

$$\|D_N\|^2_{\mathscr{A}^2_\beta} \sim \frac{G_\beta(1)}{\Gamma(1/\beta)}\, N(\log N)^{1/\beta - 1},$$

and

$$(c_\beta + o(1))(\log N)^{-1/(4\beta)} \leq \frac{\|D_N\|_{\mathscr{A}^1_\beta}}{\|D_N\|_{\mathscr{A}^2_\beta}} \leq 1, \qquad c_\beta^2 = \frac{G_{2\beta}(1)\Gamma(1/\beta)}{G_\beta(1)\Gamma(1/(2\beta))}.$$

PROOF. The Dirichlet series of $1/d_\beta$ is $\zeta(s)^{1/\beta} G_\beta(s)$ with G_β analytic and nonzero near 1, and Selberg–Delange gives the asymptotic. The lower bound is $\|D_N\|_{\mathscr{A}^1_\beta} \geq \|D_N\|_{\mathscr{A}^2_{2\beta}}$, by tensorized Kulikov, with the asymptotic at 2β. □

For $\beta = 2$ this is [10, Theorem 10]. At $\beta = 1$ Harper's theorem [22] gives the true ratio $(\log\log N)^{-1/4}$, so the lower bound is far from sharp there. For $\beta > 1$ we do not know whether the ratio tends to 0; the heuristic chaos exponent $2/\sqrt{\beta}$ is subcritical, which suggests it does not. It cannot tend to 0 uniformly in β: the ratio tends to 1 when $\beta/N \to \infty$.

6.5. Positive measures in three geometries

With R, Q, ν and the measure $\mu_{\nu,R}$ as in section 4.2, let $\alpha_{\nu,R}(n) = \mathbf{1}[n_Q \in \square] \int n^{-s}\, \mathrm{d}\nu(s)$ and $[\nu]_\kappa = \sup_{0<t\leq 1} \nu((1/2, 1/2+t])t^{-\kappa}$.

THEOREM 6.10. *Let R have density $\theta \in [0,1]$ with constant B, let κ be the kernel exponent of the geometry, and let ν be a finite positive Borel measure on $(1/2,\infty)$. The following are equivalent:*

(1) $H_{\alpha_{\nu,R}}$ is bounded on $X^2 \times X^2$;
(2) $[\nu]_\kappa < \infty$;
(3) $\int |F|\, \mathrm{d}\mu_{\nu,R} \lesssim \|F\|_{X^1}$;
(4) $H_{\alpha_{\nu,R}}$ has a bounded symbol.

The ratio between the smallest symbol norm and $\|H_{\alpha_{\nu,R}}\|$ is bounded by a constant depending only on θ, B and the geometry. If $\theta = 0$, all four hold for every finite ν. Moreover $H_{\alpha_{\nu,R}}$ is compact if and only if $\nu((1/2,1/2+t]) = o(t^\kappa)$, which is automatic when $\theta = 0$, and then its bounded symbols lie in the $L^\infty(\lambda)$-closure of $C + N_\lambda$, where N_λ is the annihilator of the analytic polynomials.

PROOF. The structure is that of Theorems 4.2 and 4.6. We record the three geometry-dependent inputs.

Kernel test. Let K be the reproducing kernel of X^2 at $\gamma_R(1/2+\delta)$, restricted to the coordinates of R. Its coefficients are positive and supported on integers with $n_Q = 1$, so Tonelli's theorem gives $H_{\alpha_{\nu,R}}(K,K) = \int K(\gamma(s))^2\, \mathrm{d}\nu(s)$. Also $K(\gamma(s)) \geq \|K\|_2^2$ for $s \leq 1/2+\delta$. Hence

$$\nu((1/2,1/2+\delta]) \leq \|H_{\alpha_{\nu,R}}\|\, \|K\|_2^{-2} \leq e^{\kappa' B}(2\delta)^\kappa\, \|H_{\alpha_{\nu,R}}\|,$$

with $\kappa' = \beta$ in the Bergman case and $\kappa' = 1$ otherwise, by the lower bound $\|K\|_2^2 \geq e^{-\kappa' B}(2\delta)^{-\kappa}$ that follows from (2.1). The same computation with normalized kernels gives $\nu = o(t^\kappa)$ when $H_{\alpha_{\nu,R}}$ is compact.

Sufficiency. Average the Q-coordinates with Proposition 6.7. Then treat the remaining R-coordinates with Theorem 2.9 (2), with $p = 1$, if $\theta > 0$, or with the uniform bound $\|K_{\gamma_R(s)}\|_2^2 \le e^{\kappa'(1+B)}$ if $\theta = 0$. Treat $\Re s > 3/2$ with the point estimate. This gives $\int |F| \, d\mu_{\nu,R} \le C_{\mathrm{sign}}\big(C[\nu]_\kappa + C'\nu((1/2,\infty))\big)\|F\|_{X^1}$. Symbols then come from Lemma 6.1.

Compactness. For the part of ν above $1/2+\varepsilon$, the moments satisfy $\alpha(n) \le \nu((1/2,\infty))n^{-1/2-\varepsilon}$, and Corollary 6.4 bounds the tail of the functional beyond $n > K$ by the square root of

$$\sum_{n>K} d(n)\, n^{-1-2\varepsilon}, \qquad \sum_{n>K} d_{2\beta}(n)\, n^{-1-2\varepsilon}, \qquad \sum_{n>K} \frac{2^{\Omega(n)}}{\kappa(n)!}\, n^{-1-2\varepsilon}.$$

These are tails of convergent series, the last being $\prod_p \exp(2p^{-1-2\varepsilon})$. The part below $1/2+\varepsilon$ is small by the sufficiency estimate. The rest of the argument is that of Theorem 4.6. □

In the Bergman case with $R = \emptyset$, every $H_{\alpha_{\nu,\emptyset}}$ is bounded, compact, and has a bounded symbol of norm at most $C_\beta\, \nu((1/2,\infty))$. Without the sign average, Ma, Lu and Zu show that boundedness requires $\nu \lesssim t^\beta$ and compactness requires $\nu = o(t^\beta)$ [33].

6.6. The Bergman multiplicative Hilbert form

Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [10] introduced the form on $\mathscr{A}^2 \times \mathscr{A}^2$, $\mathscr{A} = \mathscr{A}_2$, with analytic symbol

$$\varphi(s) = \sum_{n\ge 2} \frac{d(n)}{\sqrt{n}\log^2 n}\, n^{-s},$$

the Bergman analogue of the multiplicative Hilbert matrix. They proved that it is bounded and not compact, and that φ defines a bounded functional on $\mathscr{A}^p$ for $1 < p < \infty$. Ma, Lu and Zu [33, Theorem 1.1] have since determined its spectrum on the subspace $\mathscr{A}^2_0$ of series with $a_1 = 0$: the essential spectrum is $[0,1]$, absolutely continuous of multiplicity one, and above it there are finitely many eigenvalues, at least one, all simple. Its moments are $\alpha(n) = 1/(\sqrt{n}\log^2 n) = \int_0^\infty n^{-1/2-u} u\, du$, so it is the form of the measure $u\, du$ on the curve: the Bergman counterpart of Lebesgue measure.

The Bergman analogue of Problem 4 asks whether this form has a bounded symbol. It does.

THEOREM 6.11. *There is C such that $|\sum_{n\ge2} a_n/(\sqrt{n}\log^2 n)| \le C\|f\|_{\mathscr{A}^1}$ for every $f \in \mathscr{A}^1$, where the series converges in its natural order. Consequently, the Bergman multiplicative Hilbert form has a bounded symbol in $L^\infty(m_2^\infty)$.*

PROOF. For polynomials, the Gamma integral gives $\sum_{n\ge2} a_n/(\sqrt{n}\log^2 n) = \int_0^\infty (f(1/2+u) - a_1)\, u\, du$. On $(0,1)$ apply Theorem 2.9 (2) in the Bergman geometry $\beta = 2$ with all primes, where $\kappa = 2$ and $\nu = u\, du$ has $[\nu]_2 = 1/2$. For $u \ge 1$ use $|a_n| \le d(n)\|f\|_{\mathscr{A}^1}$, read off from the reproducing property. The bounded symbol then comes from Lemma 6.1.

For natural-order convergence, compare the partial sum up to N with the Abel sum at $\varepsilon = 1/\log N$. The Bergman Helson inequality with d_4 (Corollary 6.4) and the dyadic estimate $\sum_{e^u \le n < e^{2u}} d_4(n)/(n \log^4 n) \le 16e^2$ make the difference tend to 0. Translations $f \mapsto f(\cdot + \varepsilon)$ are contractions of $\mathscr{A}^1$ [10, Theorem 11], so the Abel sums converge to the value of the bounded functional. □

The same subtraction as in Corollary 4.11 also handles the finite measure $u\mathbf{1}_{(0,1]}(u)\,\mathrm{d}u$, with correction terms controlled by the d_4 Helson inequality. Combined with Theorem 6.10, it gives bounded symbols for the sign-averaged variants: for instance, the Bergman Hilbert form restricted to pairs with $(mn)_Q$ a square.

PROPOSITION 6.12 (The Hardy pairing is unbounded). *The functional $L(f) = \sum_{n\ge 2} a_n d(n)/(\sqrt{n}\log^2 n)$, the pairing of f with φ in $\mathscr{H}^2$, is unbounded on $\mathscr{H}^1$. With $f_\delta = \zeta(s+1/2+\delta)^2/\zeta(1+2\delta)$, one has $\|f_\delta\|_{\mathscr{H}^1} = 1$ and $L(f_\delta) \sim 1/(3\zeta(2)\delta)$.*

PROOF. $\|f_\delta\|_{\mathscr{H}^1} = \|\zeta(s+1/2+\delta)\|^2_{\mathscr{H}^2}/\zeta(1+2\delta) = 1$. By $1/\log^2 n = \int_0^\infty t n^{-t}\,\mathrm{d}t$ and $\sum d(n)^2 n^{-s} = \zeta(s)^4/\zeta(2s) \sim \zeta(2)^{-1}(s-1)^{-4}$,

$$\sum_{n\ge 2} \frac{d(n)^2}{n^{1+\delta}\log^2 n} \sim \frac{1}{\zeta(2)} \int_0^\infty \frac{t\,\mathrm{d}t}{(\delta+t)^4} = \frac{1}{6\zeta(2)\delta^2},$$

and $\zeta(1+2\delta) \sim 1/(2\delta)$. Truncated Euler products give polynomial witnesses. □

So the Bergman symbol is bounded against the Bergman pairing, where the weight $d(n)$ cancels, and the natural Hardy analogue fails by a power of δ; the pointwise count u^{-3} against $u\,\mathrm{d}u$ was already the right size.

6.7. Comparison

	Hardy	Bergman (β)	Gauss
transition maps	Möbius	Möbius	translations
energy $E(r,s)$	$\sum \rho_p^2$	$\sum \rho_p^2$	$\sum \lvert r_p - s_p\rvert^2$
Hellinger affinity	$\le e^{-E/4}$	$\le e^{-\beta E/4}$	$= e^{-E/4}$
kernel exponent κ	θ	$\beta\theta$	θ
contractive embedding	Carleman	Kulikov	Proposition 6.3
Helson weight	$1/d(n)$	$1/d_{2\beta}(n)$	$\kappa(n)!/2^{\Omega(n)}$
two-point constant	$(1-\lvert x\rvert^4)^{-1}$ (sharp)	$\Lambda_\beta(\lvert x\rvert^2)^{1/2}$	$\cosh(2\lvert x\rvert^2)^{1/2}$
sign-average constant	$\zeta(2)$	C_β	< 1.58

The Gauss column is the cleanest: its Hellinger affinity is exact, and its sign-average constant is the smallest. The Hardy column is the only one with a sharp two-point constant for every x. In the Bergman column the sharp constant is the square bound exactly up to the first zero of $K_x + K_{-x}$, and not beyond it; in the Gauss column only the leading term is known (Proposition 6.6).

Open questions

QUESTION 5. What are the sharp two-point constants in the Gauss geometry, where the square bound always fails (Proposition 6.6(4)), and in the Bergman geometry beyond the first zero of $K_x + K_{-x}$?

Part 3

Where Nehari's theorem fails: Schatten classes

CHAPTER 7

Schatten-class forms without bounded symbols

The question of this chapter was posed by Brevig and Perfekt.

PROBLEM 6 (Brevig and Perfekt [13, Question 3]). *Is there a multiplicative Hankel form in S_p, $2 < p < \infty$, without a bounded symbol, and if so, for which p? Brevig and Perfekt conjectured that such forms exist for every $p > 2$.*

Let p_* be the infimum of such p. Helson's theorem gives $p_* \geq 2$ in the sense that S_2 forms always have bounded symbols, and Brevig and Perfekt [13] proved $p_* \leq (1 - \log \pi / \log 4)^{-1} = 5.7388\ldots$. This chapter proves $p_* \leq 4$, which settles Problem 6 for every $p > 4$, by a construction in the Gaussian model, and shows that the threshold 4 is built into the method: for the natural blocks it is the identity $\|\varphi\|_{L^4} = \|A(\varphi)\|_{S_4}$ of Janson, Peetre and Rochberg. Bampouras [4] found the bound $p_* \leq 4$ independently; the two proofs are compared after Theorem 7.9. The moment criterion in the generality of admissible blocks, the threshold corollary and the inequality $\|\varphi\|_{L^4} \leq \|H_\varphi\|_{S_4}$ on finite tori are, to our knowledge, new. Chapter 8 shows that nothing nearby goes below 4.

7.1. Blocks and the Gaussian model

For an analytic polynomial φ on $\mathbb{T}^d$ write $H_\varphi(f, g) = \langle fg, \varphi \rangle$, with matrix $(\hat{\varphi}(\alpha + \beta))_{\alpha,\beta}$.

LEMMA 7.1 (Brevig–Perfekt [13]). *If $p \geq 1$ and polynomials f, φ in finitely many variables satisfy $|\langle f, \varphi \rangle| > \|H_\varphi\|_{S_p} \|f\|_1$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. Otherwise the closed graph theorem gives $|\langle f, \varphi \rangle| \leq C_p \|H_\varphi\|_{S_p} \|f\|_1$ for all polynomials. Pairing, L^1 norm and Schatten norm are all multiplicative over products in disjoint variables, so m independent copies raise the ratio to the mth power. □

Let μ_d be the law of d independent standard complex Gaussians on $\mathbb{C}^d$. For $\varphi = \sum c_\alpha z^\alpha$ the Gaussian Hankel form $(f, g) \mapsto \langle fg, \varphi \rangle_{L^2(\mu_d)}$ has, in the basis $z^\alpha / \sqrt{\alpha!}$, the matrix $A(\varphi)_{\alpha\beta} = c_{\alpha+\beta}(\alpha + \beta)!/\sqrt{\alpha!\beta!}$. Let $D(\varphi)$ be the norm of $f \mapsto \langle f, \varphi \rangle_{L^2(\mu_d)}$ on analytic polynomials with the $L^1(\mu_d)$ norm.

PROPOSITION 7.2 (Transfer). *Let $p \geq 2$. If $D(\varphi) > \|A(\varphi)\|_{S_p}$ for a polynomial φ on $\mathbb{C}^d$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. Replace z_j by $L_{N,j} = N^{-1/2} \sum_{\ell \leq N} z_{j,\ell}$ on disjoint blocks of Steinhaus variables. For fixed k, $\|L_N^k\|_2^2 \to k!$, $\|H_{L_N^k}\|_{S_2}^2 \to 2^k k!$ (a coefficient of index

ν occurs $\prod(\nu_\ell+1)$ times, and only squarefree indices survive), and $\mathbb{E}|L_N|^{2k} \le k!$. These multiply over blocks, so $\|H_{\varphi_N}\|_{S_2} \to \|A(\varphi)\|_{S_2}$. The compression of H_{φ_N} to the normalized monomials of degree at most K converges entrywise to $A(\varphi)$. Since compression is an orthogonal projection in Hilbert–Schmidt norm, the remainder tends to 0 in S_2, hence in S_p. The central limit theorem and uniformly bounded second moments give $\|f(L_N)\|_1 \to \|f\|_{L^1(\mu_d)}$, and the pairings converge by orthogonality. Apply Lemma 7.1. □

7.2. Every $p > 4$

Let $Z_1, Z_2, \dots$ be independent standard complex Gaussians and $P_n = Z_1 \cdots Z_n$, so $\mathbb{E}|P_n|^2 = 1$ and $\mathbb{E}|P_n|^4 = 2^n$.

LEMMA 7.3 (Spectrum). *For $c > 0$ let $\lambda = (1+\sqrt{1+4c^2})/2$. The nonzero singular values of $A(1+cP_n)$ are λ, $\lambda - 1$, and c with multiplicity $2^n - 2$. Hence $\|A(1+cP_n)\|_{S_p} - \lambda \le 2^n c^p/p$ for $p \ge 2$.*

PROOF. Only squarefree monomials contribute, each paired with its complement. The pair $\{1, P_n\}$ carries $\left(\begin{smallmatrix} 1 & c \\ c & 0 \end{smallmatrix}\right)$, and the other $2^{n-1}-1$ pairs carry $\left(\begin{smallmatrix} 0 & c \\ c & 0 \end{smallmatrix}\right)$. Since $\lambda - 1 \le c$ and $\lambda \ge 1$, concavity of $x^{1/p}$ gives the estimate. □

LEMMA 7.4 (First variation of L^1). *Let W be rotation invariant with $\mathbb{E}|W|^2 = 1$, $M = \mathbb{E}|W|^4$, and let $u = W/r$, $G = \mathbb{E}(|u|^2-1)_+$, $T = \mathbb{E}(|u|^2; |u| > 1/2)$. For $t \ge 0$,*

$$\mathbb{E}\big|(1+u)^2 - tu^2\big| \le 1 + r^{-2} - tG + 16Mt^2r^{-4} + 8t^{5/4}T.$$

PROOF. For $w \ne 0$,

$$|w - tv| = |w| - t\Re(v\overline{w})/|w| + R, \qquad 0 \le R \le \min(2t|v|, 4t^2|v|^2/|w|).$$

Take $w = (1+u)^2$, $v = u^2$. The main term has mean $\mathbb{E}|1+u|^2 = 1 + r^{-2}$. Averaging the linear term over the angle,

$$\int_{\mathbb{T}} \rho^2\zeta^2 \frac{1+\rho\overline{\zeta}}{1+\rho\zeta}\, \mathrm{d}m(\zeta) = (\rho^2-1)_+,$$

by expanding in powers of $\rho\zeta$ for $\rho < 1$ and of $1/(\rho\zeta)$ for $\rho > 1$; so it has mean $-tG$. On $|u| \le 1/2$, $R \le 16t^2|u|^4$. On $|u| > 1/2$, $R \le 2^{5/4}t^{5/4}|u|^{5/2}|1+u|^{-1/2}$, and the conditional mean of $\rho^{1/2}|1+u|^{-1/2}$ given $|u| = \rho \ge 1/2$ is at most $2^{5/4}$. □

LEMMA 7.5 (Large deviations). *Let $\Psi(s) = \log \mathbb{E}|Z_1|^s = \log\Gamma(1+s/2)$. For $s > 2$ put $\beta = \Psi'(s)$, $I = s\beta - \Psi(s)$, $r_n = e^{\beta n}$, and let G_n, T_n be as in Lemma 7.4 for $W = P_n$, $r = r_n$. Then $\beta > 0$, $T_n \le 2^{s-2}e^{-nI}$ and $n^{-1}\log G_n \to -I$.*

PROOF. Ψ is strictly convex with $\Psi(0) = \Psi(2) = 0$, so $\beta > 0$. With $x = |P_n|/r_n$, $\mathbb{E}x^s = e^{-nI}$, and $(x^2-1)_+ \le x^s$, $x^2\mathbf{1}_{x>1/2} \le 2^{s-2}x^s$ give the upper bounds. For the lower bound, tilt by $e^{u\log|P_n| - n\Psi(u)}$ with $u > s$: under the tilted law $\log|P_n|/n$ concentrates at $\Psi'(u) > \beta$, and converting back costs $e^{n(\Psi(u)-u\Psi'(u)+2(\Psi'(u)-\beta))+o(n)}$. Let $u \downarrow s$. □

THEOREM 7.6. *For every $p > 4$ and all large n there are $c_n, r_n, t_n > 0$ such that $\varphi_n = 1 + c_nP_n$ and $f_n = (1+P_n/r_n)^2 - t_n(P_n/r_n)^2$ satisfy $|\langle f_n, \varphi_n\rangle| > \|A(\varphi_n)\|_{S_p}\|f_n\|_1$. Consequently $p_* \le 4$.*

PROOF. Take $s > 4$ and the data of Lemma 7.5, and put $B = 4\beta - \log 2$, so $2^n/r_n^4 = e^{-nB}$. Strict convexity and $\Psi(4) = \log 2$ give $I > B$, and $(p\beta - \log 2) - (2I - B) \to (p-4)\Psi'(4) > 0$ as $s \downarrow 4$; fix s and $\varepsilon > 0$ with $2I - B + \varepsilon < p\beta - \log 2$. Let $c_n = r_n/(r_n^2 - 1)$, so that $\lambda_n = r_n^2/(r_n^2 - 1)$, and $t_n = e^{-n(I-B+\varepsilon)}$.

By Lemma 7.3, the Schatten excess is $\|A(\varphi_n)\|_{S_p} - \lambda_n \le (2^p/p)e^{-n(p\beta - \log 2)}$. The pairing is $\langle f_n, \varphi_n\rangle = 1 + 2c_n/r_n = \lambda_n(1 + r_n^{-2})$, because $P_n^2 \perp \varphi_n$. By Lemmas 7.4 and 7.5, $(1 + r_n^{-2}) - \|f_n\|_1 \ge t_n G_n(1 - o(1))$, the two error terms being $e^{-n(\varepsilon + o(1))}$ and $e^{-n((I-B+\varepsilon)/4 + o(1))}$ relative to $t_n G_n$. Hence $\langle f_n, \varphi_n\rangle/\|f_n\|_1 - \lambda_n \ge t_n G_n/4 = e^{-n(2I - B + \varepsilon + o(1))}$, which beats the excess. Apply Proposition 7.2. □

The exponent 4 enters once: the second-order cost of the perturbation is $t^2\mathbb{E}|P_n|^4/r_n^4$, and $\mathbb{E}|Z|^4 = 2$ makes it $t^2 2^n/r_n^4$, which balances the 2^n residual singular values exactly at $p = 4$.

COROLLARY 7.7 (Dimension growth). *For $p > 4$ let $C_p(d)$ be the best constant in $|\langle f, \varphi\rangle| \le C\|H_\varphi\|_{S_p}\|f\|_{L^1(\mathbb{T}^d)}$, which is the minimal-symbol constant on $\mathbb{T}^d$. Then $C_p(d) \ge A_p e^{c_p d}$ with $c_p > 0$.*

PROOF. A finite pair with ratio $R_p > 1$ in d_p variables exists by Theorem 7.6 and Proposition 7.2; k disjoint copies give $C_p(kd_p) \ge R_p^k$, and $C_p(d)$ is nondecreasing in d. □

The rate degenerates as $p \downarrow 4$, and whether $C_p(d)$ stays bounded for some $p \le 4$ is open. In a fixed number of variables the constant may well be finite for every $p < \infty$. Bampouras and Perfekt [8] prove this for $d = 2$: on the product Hardy space of two half-planes, every Hankel operator in some S_p, $p < \infty$, has a symbol in product BMO, that is, a bounded symbol. By the Cayley transform in each variable the same holds on $\mathbb{T}^2$, so $C_p(2) < \infty$ for every p. They expect the same in every dimension, while for $p = \infty$ the question is open already for $d = 2$, and Ortega-Cerdà and Seip [38] show that the constant there, where finite, grows exponentially in d. So the counterexamples of this chapter live in the limit $d \to \infty$, and for $p > 4$ the corollary gives the exponential rate.

7.3. The moment criterion and why it stops at 4

Bampouras's proof integrates over the scale instead of fixing it. Its structure is clearest for a general block. Call a polynomial W on $\mathbb{T}^N$ *admissible* if $\|W\|_2 = 1$, $W(0) = 0$, $\int W^2\overline{W}\,\mathrm{d}m = 0$, $\int g|W|^2\,\mathrm{d}m = \hat{g}(0)$ for analytic g, and the law of W is rotation invariant. Every normalized homogeneous polynomial is admissible, and so are products of admissible blocks in disjoint variables.

LEMMA 7.8. *For admissible F and $\varphi_t = 1 - t^2 + tF$, the span of 1 and F reduces H_{φ_t}, with compression $\left(\begin{smallmatrix} 1-t^2 & t \\ t & 0 \end{smallmatrix}\right)$, and $\|H_{\varphi_t}\|_{S_p}^p = 1 + t^{2p} + t^p(\|H_F\|_{S_p}^p - 2)$.*

PROOF. $H_F(F, g) = \int g|F|^2 = \hat{g}(0)$ and $H_F(F, F) = \int F^2\overline{F} = 0$, so H_F swaps 1 and F and maps their orthogonal complement into itself. □

THEOREM 7.9 (Moment criterion). *Let W be admissible and $p > 2$. If $\|W\|_{L^4} > \|H_W\|_{S_p}$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. Let

$$D_\eta(u) = 1 + u^2 - \int_{\mathbb{T}} |(1+u\zeta)^2 - \eta u^2 \zeta^2| \, \mathrm{d}m(\zeta), \qquad I_\kappa(\eta) = \int_0^\infty D_\eta(u) u^{-\kappa-1} \, \mathrm{d}u;$$

by [4], $I_\kappa(\eta) > 0$ for $\kappa \in (2,4)$ and small η. By continuity choose $\kappa \in (2, \min(p,4))$ with $\|W\|_\kappa > \|H_W\|_{S_p}$. Suppose the conclusion fails. For $F = \prod_{j \le m} W(z^{(j)})$, with $A = \|H_W\|_{S_p}^p$, the test $F_t = (1+tF)^2 - \eta t^2 F^2$ and Lemma 7.8 give

$$\|F_t\|_1 \ge (1+t^2)(1 + t^{2p} + (A^m - 2)t^p)^{-1/p},$$

while rotation invariance gives $\|F_t\|_1 = 1 + t^2 - \int D_\eta(t|F|) \, \mathrm{d}m$. Integrate against $t^{-\kappa-1} \, \mathrm{d}t$ and substitute $u = t|F|$: $I_\kappa(\eta)(\mathbb{E}|W|^\kappa)^m \le C_{p,\kappa} A^{m\kappa/p}$. Taking mth roots gives $\|W\|_\kappa \le \|H_W\|_{S_p}$, a contradiction. □

Both proofs of $p_* \le 4$ test a constant-plus-product symbol against a perturbed square, and both get 4 from a Gaussian fourth moment. Bampouras works on finite tori with linear blocks and integrates the scale out against $t^{-\kappa-1} \, \mathrm{d}t$, as in the proof above, which reduces everything to the moments $e^{\Psi(\kappa)}$. The proof of Theorem 7.6 fixes the scale and optimizes it by large deviations, with rates $I = s\Psi'(s) - \Psi(s)$, the Legendre duals of those moments. His route is shorter; this one gives explicit parameters for every $p > 4$, and with them certified finite witnesses, for instance at $p = 9/2$.

PROPOSITION 7.10 (L^4 is dominated by S_4). *For every polynomial φ on $\mathbb{T}^d$, $\|\varphi\|_{L^4} \le \|H_\varphi\|_{S_4}$. In the Gaussian model $\|\varphi\|_{L^4(\mu)} = \|A(\varphi)\|_{S_4}$ [27].*

PROOF. $\|H_\varphi\|_{S_4}^4 = \sum_{\beta,\beta'} |\sum_\alpha \overline{c_{\alpha+\beta}} c_{\alpha+\beta'}|^2$ and $\|\varphi\|_4^4 = \sum_\delta |R(\delta)|^2$ with $R(\delta) = \sum_{\nu-\nu'=\delta} c_\nu \overline{c_{\nu'}}$. Writing $\delta = \delta^+ - \delta^-$ with disjoint supports, the pairs with $\nu - \nu' = \delta$ are exactly $\nu = \alpha + \delta^+$, $\nu' = \alpha + \delta^-$, so $|R(\delta)|^2$ is the term $(\beta,\beta') = (\delta^-, \delta^+)$ of the first sum, and distinct δ give distinct terms. □

COROLLARY 7.11 (The method's threshold). *(1) For $p \le 4$ the criterion fails for every admissible block on every finite torus, since $\|W\|_{L^\kappa} \le \|W\|_{L^4} \le \|H_W\|_{S_4} \le \|H_W\|_{S_p}$.*

(2) For a normalized linear form $L = \sum_{i \le N} c_i z_i$ it succeeds exactly for $p > 4\log 2 / \log(2 - \sum |c_i|^4)$; with equal coefficients this is 6.84, 5.43, 4.95, 4.32 for $N = 2, 3, 4, 10$, and $4 + 2/(N \log 2) + O(N^{-2})$ in general.

(3) If H_W has rank two, then W is linear: a monomial of top degree D and a nested chain of divisors give a triangular $(D+1) \times (D+1)$ submatrix. So the rank-two version of the criterion is exactly the linear case.

PROOF. (2) $\|H_L\|_{S_p}^p = 2$ and $\|L\|_4^4 = 2 - \sum |c_i|^4$. □

So three variables already beat the Brevig–Perfekt exponent, and for Gaussian homogeneous blocks the crossing $\|W\|_{L^p} = \|A(W)\|_{S_p}$ always happens exactly at $p = 4$. Going below 4 with Lemma 7.1 requires a block outside the admissible class, so that the $\{1, F\}$ splitting fails, or a test family not of the form $(1+tF)^2 - \eta t^2 F^2$.

Open

(1) Is $p_* = 4$? Equivalently for these methods: the L^1–L^4 question of chapter 8.
(2) Does $C_p(d)$ grow with d for some $p \leq 4$? Is $C_p(d)$ finite for every $p < \infty$ when $d \geq 3$, as it is for $d = 2$?

CHAPTER 8

Barriers: why 4, and why not 2

Question. Can the Gaussian-block method of chapter 7 reach any $p \le 4$? And what keeps every construction away from $p = 2$?

Throughout, μ is Gaussian probability measure, $A(\varphi)$ the Gaussian Hankel matrix and $D(\varphi)$ the norm of $f \mapsto \langle f, \varphi\rangle$ on analytic polynomials in $L^1(\mu)$, as in section 7.1. A symbol gives an S_p counterexample through Proposition 7.2 when $D(\varphi) > \|A(\varphi)\|_{S_p}$; we call this a *strict block* at p. Gaussian hypercontractivity gives $D(\varphi) \le \|A(\varphi)\|_{S_2}$, and $\|A(\varphi)\|_{S_4} = \|\varphi\|_{L^4(\mu)}$ [27].

The answers: the constant-plus-product family that reaches every $p > 4$ has no strict block at $p \le 4$, for every n and every test, so 4 is exactly its threshold; affine and even quadratic symbols, and all zero-free Hilbert–Schmidt symbols, are equally stuck; and every family of bounded degree in a bounded number of variables is excluded on an interval $(2, 2+\varepsilon)$. Whether *any* Gaussian symbol beats S_4 is the L^1–L^4 question of section 8.4. The Paley–Wiener threshold 4 of Bampouras has a different origin (section 8.5). The sharpness of the product family at 4, the barriers for affine, even quadratic and zero-free symbols, the interval $(2, 2+\varepsilon)$, the disc blocks and the polydisc growth are, to our knowledge, new.

8.1. The product family is sharp at 4

Let $P_n = Z_1 \cdots Z_n$. By Lemma 7.3, $\|A(c_0 + c_1 P_n)\|_{S_4}^4 = |c_0|^4 + 4|c_0 c_1|^2 + 2^n |c_1|^4$.

THEOREM 8.1. *For every $n \ge 1$ and all $c_0, c_1 \in \mathbb{C}$, $D(c_0 + c_1 P_n) \le \|A(c_0 + c_1 P_n)\|_{S_4}$. Consequently the infimum of exponents with a strict block in this family is exactly 4.*

The cases $n = 1$ and $n \ge 2$ are proved differently.

The affine case

For $k > 0$ and $R = |Z|$ put $A = \mathbb{E}(1 - k^2 R^2)_+$ and $K = \mathbb{E}[R \min(kR, 1)]$.

LEMMA 8.2. *$Q(k) = A^4 + 4A^2K^2 + 2K^4$ increases strictly on $(0, \infty)$ from the limit 1 at $k = 0$.*

PROOF. With $B = \mathbb{E}[R^2 \mathbf{1}_{kR<1}]$, $A' = -2kB$ and $K' = B$, so $Q' = 8B\Phi$ with $\Phi = K^3 + A^2K - kA^3 - 2kAK^2$. Put $t = 1/k$, $x = t^2$, $e = e^{-x}$. Then $A = 1-(1-e)/x$, and the Mills bound $\int_t^\infty e^{-s^2}\,\mathrm{d}s \ge te^{-t^2}/(2t^2+1)$ gives $K \ge K_e = t^{-1}(1 - e(x+1)/(2x+1))$. For $x \le 1$, $K_e > 2kA$ reduces to $2 + 3x - 2x^2 > e(x^2 + 5x + 2)$, which follows from the fourth-order Taylor bound for e^{-x} and the identity $(2+3x-2x^2) - (x^2+5x+2)(1 - x + x^2/2 - x^3/6 + x^4/24) = x^2(1-x)(x^3 + 2x^2 - 4x + 24)/24$; then

$\Phi = K^2(K-2kA) + A^2(K-kA) > 0$. For $x \ge 1$, on the region $K \ge kA$ one has $\partial_K \Phi \ge 0$ and $\partial_A \Phi \le 0$, so replacing e by the larger $1/T(x)$, $T(x) = \sum_{j \le 4} x^j/j!$, only lowers Φ; the resulting expression is $x^{3/2}(x+2)^3 P(x)/(13824(2x+1)^3 T(x)^3)$ with $P(x) = x^3[8x^2(x-9/4)^2 + 75x^2/2 - 163x + 1326] + 2268x^2 + 6408x + 6912 > 0$. □

Proof of Theorem 8.1 for $n = 1$. For $b = \min(kR, 1)$ the unimodular function

$$\psi_k(R\zeta) = \frac{1 + b\zeta}{1 + b\overline{\zeta}}$$

has analytic coefficients $1 - b^2, b, 0, \dots$ on each circle, so it represents the functional of $A + Kz$, and $D(1 + (K/A)z) \le 1/A < (1 + 4(K/A)^2 + 2(K/A)^4)^{1/4}$ by Lemma 8.2. The ratio K/A increases continuously from 0 to ∞, so scaling and rotation cover every symbol with $c_0 c_1 \ne 0$; for $c_0 = 0$, $D(c_1 z) = 2|c_1|/\sqrt{\pi} < 2^{1/4}|c_1|$. □

Products of at least two factors

For $M \ge 4$ put $S_M(c) = (1 + 4c^2 + Mc^4)^{1/4}$ and $L_M(b) = \sup_{c \ge 0}(1 + bc)/S_M(c)$.

Lemma 8.3 (Reduction). *The theorem holds for $n \ge 2$ if every polynomial $q(w) = a_0 + a_1 w + \cdots$ with $a_0 \ne 0$ satisfies $\mathbb{E}|q(P_n)| \ge |a_0| L_{2^n}(|a_1/a_0|)$.*

Proof. Averaging a test over the rotations with $\zeta_1 \cdots \zeta_n = 1$ keeps exactly the monomials with equal exponents, preserves the pairing and contracts L^1; so tests may be taken of the form $q(P_n)$. Then $|a_0| + c|a_1| \le S_M(c)\mathbb{E}|q(P_n)|$ is the claim. If $a_0 = 0$, coefficient extraction gives $\mathbb{E}|q(P_n)| \ge |a_1|(\sqrt{\pi}/2)^n \ge |a_1| M^{-1/4}$. □

Lemma 8.4 (Circle estimate). *If G is analytic on a neighbourhood of $\overline{\mathbb{D}}$ with $G(0) = 1$ and $|G'(0)| = s$, then $\int_{\mathbb{T}} |G| \, \mathrm{d}m \ge h(s)$, where $h(s) = 1 + s^2/4$ for $s \le 2$ and $h(s) = s$ for $s \ge 2$.*

Proof. Factor $G = uv$ with $\|u\|_2 \|v\|_2 = \|G\|_1$ and pair with $1 + \lambda z$: $1 + \lambda s \le \|G\|_1 (1 + \sqrt{1 + 4\lambda^2})/2$. Take $\lambda = 2s/(4 - s^2)$ for $s < 2$; for $s \ge 2$ extract the coefficient. □

For $n = 2$: with $\beta = |a_1/a_0|$, the circle estimate and rotation invariance give $\mathbb{E}|q(Z_1 Z_2)| \ge |a_0| \mathbb{E} h(\beta |Z_1 Z_2|)$. Since $x \mapsto h(\beta\sqrt{x})$ is concave, Jensen gives $\mathbb{E} h(\beta|W|) \ge \mathbb{E} h(\beta\sqrt{2}|\Re W|)$ for $W = Z_1 Z_2$, and $\sqrt{2}|\Re W|$ is distributed as $X/\sqrt{2}$ with X exponential. Then $\mathbb{E} h(bX) = 1 + (b^2/2)(1 - e^{-2/b}) \ge \sqrt{1 + b^2}$, because $\sinh(1/b) \ge 1/b$. This is $L_4(b) = \sqrt{1 + b^2/2}$ at $b = \beta/\sqrt{2}$.

The induction from n to $n+1$ conditions on Z_{n+1}: the polynomial $w \mapsto q(wZ_{n+1})$ has coefficients a_0 and $a_1 Z_{n+1}$, so it suffices to prove the envelope inequality

$$\mathbb{E} L_M(b|Z|) \ge L_{2M}(b) \qquad (M \ge 4,\ b \ge 0). \tag{8.1}$$

Lemma 8.5 (The log-Gamma semigroup). *There are positive random variables X_a, $a \ge 0$, with $X_0 = 1$, $X_1 \overset{d}{=} |Z|^2$, $\mathbb{E} X_a^s = \Gamma(1 + s)^a$ and $X_{a+a'} \overset{d}{=} X_a X'_{a'}$. The semigroup $P_a g(t) = \mathbb{E} g(t X_a)$ has generator*

$$\mathcal{A} g(t) = \int_0^1 \frac{g(tu) - g(t) + (1 - u) t g'(t)}{(-\log u)(1 - u)} \, \mathrm{d}u,$$

with positive kernel and $\mathcal{A}(t^s) = \log\Gamma(1+s)\,t^s$.

PROOF. $\log|Z|^2$ is infinitely divisible [34]: the Lévy measure $e^y\,\mathrm{d}y/(|y|(1-e^y))$ on $(-\infty,0)$ has Laplace exponent $\int_0^1 (u^s-1+s(1-u))/((-\log u)(1-u))\,\mathrm{d}u$, whose second derivative is $\sum_{k\ge1}(k+s)^{-2}$ and which vanishes at $s=0,1$, so it equals $\log\Gamma(1+s)$. Put $X_a = e^{\xi_a}$. □

Write $f_M(t) = L_M(\sqrt{t})$; then (8.1) is $P_1 f_M \ge f_{2M}$. The family $h_a = f_{2^a M}$ moves with speed $(\log 2)N\partial_N f_N|_{N=2^a M}$, and a comparison principle for $\mathcal{A}$ on functions of square-root growth (Dynkin's formula, then differentiating $s\mapsto P_s h_{1-s}$) shows $h_1 \le P_1 h_0$ as soon as h_a is a subsolution:

THEOREM 8.6. *For* $M \ge 4$ *and* $t>0$, $\mathcal{A}f_M(t) > (\log 2) M\partial_M f_M(t)$.

PROOF. Let c be the maximizer in $L_M(\sqrt{t})$, so $\sqrt{t} = (2c+Mc^3)/(1+2c^2)$, and put $y = Mc^2$. The envelope theorem gives

$$tf_M' = \frac{c^2(2+y)}{2(1+2c^2)S_M(c)}, \qquad M\partial_M f_M = -\frac{y}{2(2+y)}\,tf_M',$$

and a direct computation in the variable c shows that f_M is increasing and concave, $f_M^{(3)}>0$, and $-tf_M''/f_M' \le y/(2(1+y))$.

If $y \le 4/3$: since $f_M^{(3)}>0$, Taylor's formula shows that $(f_M(tu)-f_M(t)+(1-u)tf_M'(t))/(1-u)^2$ increases in u, so $\mathcal{A}f_M \ge (\log 2)(1-f_M+tf_M')$, using $\int_0^1 (1-u)/(-\log u)\,\mathrm{d}u = \log 2$. The inequality $1-f_M+tf_M' > M\partial_M f_M$ becomes $(1+2c^2)S_M(c) > 1+3c^2+Mc^4/4$, whose fourth-power difference times M^5 is a polynomial in M and y with nonnegative brackets on $y\le 4/3$.

If $y \ge 4/3$: let $\alpha = (2+y)/(2+2y) \in (1/2, 5/7]$. The curvature bound integrates to the minorant $f_M(tu) \ge f_M(t) + tf_M'(t)(u^\alpha-1)/\alpha$, which touches at $u=1$, so positivity of the kernel gives $\mathcal{A}f_M(t) \ge tf_M'(t)\log\Gamma(1+\alpha)/\alpha$. Finally $\log\Gamma(1+\alpha) > -(\log 2)(1-\alpha)/2$ on $[1/2,5/7]$, by Taylor expansion at 2 with $\gamma > 47/24 - 2\log 2$, $\log 2 < 25/36$ and $\zeta(2) > 13/8$, and $(1-\alpha)/\alpha = y/(2+y)$. □

This completes the proof of Theorem 8.1. The exponent 4 is the point where the 2^n residual singular values of size $|c_1|$ and the fourth moment $\mathbb{E}|P_n|^4 = 2^n$ balance; at S_4 the Hankel norm is $\|c_0 + c_1 P_n\|_{L^4}$, and the theorem says that the L^1 dual norm never exceeds it.

8.2. Other symbol families

PROPOSITION 8.7 (Even quadratic symbols). *Every analytic polynomial* $f = \sum a_n z^n$ *in one Gaussian variable has* $\|f\|_1^2 \ge |a_0|^2 + |a_2|^2$. *Hence* $D(c_0+c_2z^2) \le (|c_0|^2+4|c_2|^2)^{1/2} < \|A(c_0+c_2z^2)\|_{S_4}$ *for* $c_2 \ne 0$.

PROOF. Pass to the even part $g(z^2)$. With $\beta = |a_2/a_0|$, Lemma 8.4 on each circle and $x=r^2$ give $\|f\|_1/|a_0| \ge \int_0^\infty h(\beta x)e^{-x}\,\mathrm{d}x = 1+(\beta^2/2)(1-e^{-2/\beta}) \ge \sqrt{1+\beta^2}$, as in the case $n=2$ above. Then $\langle f,\varphi\rangle = a_0\overline{c_0} + 2a_2\overline{c_2}$ and Cauchy–Schwarz. The Hankel block has $\|A\|_{S_4}^4 = |c_0|^4 + 8|c_0c_2|^2 + 24|c_2|^4$, which exceeds $(|c_0|^2+4|c_2|^2)^2$ by $8|c_2|^4$. □

PROPOSITION 8.8 (Zero-free symbols). *Suppose that φ is entire without zeros and that $\|A(\varphi)\|_{S_2} < \infty$. Then $D(\varphi)$ equals the operator norm of $A(\varphi)$, so φ has no strict block at any p.*

PROOF. The growth bound $|\varphi(z)| \le \|A\|_{S_2} e^{|z|^2/4}$ and Borel–Carathéodory make $\log\varphi$ a quadratic polynomial, and the Hilbert–Schmidt condition forces $\varphi = e^{az^2+bz+c}$ with $|a| < 1/4$. A Weyl translation, unitary on analytic L^2, isometric on analytic L^1 and compatible with products, removes b, and a rotation makes $a \ge 0$. Write $a = q/(2(1+q^2))$ with $0 \le q < 1$. The bounded function $\sqrt{1+q^2}\exp(iq\Im z^2)$ represents e^{az^2}, so $D \le \sqrt{1+q^2}$; and $f = g^2$ with $g = e^{qz^2/2}$ gives $\langle f, \varphi\rangle/\|g\|_2^2 = \sqrt{1+q^2}$, so $\|A\| \ge \sqrt{1+q^2}$. □

For monomials the computation is explicit: $D(z^m) = m!/\Gamma(1+m/2)$ by circular averaging, and $A(z^m)$ is anti-diagonal. For even m this equals $\|A\|_{S_\infty}$; for $m = 2k+1$ a strict block needs $p > \log 2/\log(\sqrt{k!(k+1)!}/\Gamma(k+3/2)) \ge (1 - \log\pi/\log 4)^{-1}$. So no one-variable homogeneous symbol improves on Brevig and Perfekt.

8.3. Near $p = 2$

THEOREM 8.9 (Bounded degree and dimension). *For all $d, K \ge 1$ there is $\varepsilon_{d,K} > 0$ such that $D(\varphi) \le \|A(\varphi)\|_{S_p}$ for $2 \le p \le 2 + \varepsilon_{d,K}$ and every polynomial φ on $\mathbb{C}^d$ of total degree at most K, against tests of any degree.*

PROOF. *Near constants.* Normalize $c_0 = 1$ and let $\tau = \max_{|\alpha|\ge 1}|c_\alpha|$. A test with quotient above 1 and $f(0) = 1$ has, by the Gaussian Carleman inequality $\sum \alpha! 2^{-|\alpha|}|a_\alpha|^2 \le \|f\|_1^2$, all coefficients of degree at most K of size $O(\tau)$. Let g be the square root of f truncated at degree K. On tori of radius r with $r_j \le R = c\tau^{-1/K}$, $|g - 1| \le 1/2$, and the unimodular function $g/\overline{g}$ has analytic projection of degree at most K, so $\int_{\mathbb{T}^d}|f(r\zeta)|\,\mathrm{d}m \ge \int_{\mathbb{T}^d}|g(r\zeta)|^2\,\mathrm{d}m$. Integrating radially, the region outside the polydisc of radius R costs $O(e^{-R^2})$, so $D(\varphi) \le \|A(\varphi)\|_{S_\infty} + Ce^{-b\tau^{-2/K}}$. On the other hand, with $m_\alpha = \alpha! c_\alpha$, the 2×2 minors in rows $0, e_j$ and columns $0, \alpha$ show $s_2(A) \gtrsim \tau^{K+1}$, so $\|A\|_{S_p} - \|A\|_{S_\infty} \gtrsim \tau^{p(K+1)}$, which wins for small τ.

Away from constants. If $D(\varphi) = \|A(\varphi)\|_{S_2}$ for a nonconstant φ, a maximizing sequence would converge to $\varphi(2z)$ up to normalization and force equality in Gaussian Carleman, which holds only for exponentials [46]. So $D < \|A\|_{S_2}$, and by compactness of the normalized symbol sphere outside the constant neighbourhood, $D \le (1-\eta_0)\|A\|_{S_2}$ there. Since $\operatorname{rank} A \le \binom{d+K}{K} = r$, $\|A\|_{S_p} \ge r^{1/p-1/2}\|A\|_{S_2}$, which exceeds $(1-\eta_0)\|A\|_{S_2}$ for p close to 2. □

PROPOSITION 8.10 (Quadratic compression). *For $\varphi = c + b^T Z + Z^T C Z$ on $\mathbb{C}^d$ with $C = C^T$, the nonzero singular values of $A(\varphi)$ are those of*

$$\begin{pmatrix} c & b^T & \sqrt{2}\|C\|_F \\ b & 2C & 0 \\ \sqrt{2}\|C\|_F & 0 & 0 \end{pmatrix}.$$

If $\operatorname{rank} C = r$, a unitary change of variables reduces φ to $r+1$ Gaussians, preserving D and all singular values. So quadratic symbols approaching $p = 2$ need $\operatorname{rank} C \to \infty$.

PROOF. The quadratic row of the full matrix is a vector q with $\|q\|^2 = 2\|C\|_F^2$; a unitary congruence in the quadratic block rotates it to one coordinate. Takagi factorization diagonalizes C, and Theorem 8.9 applies with $d = r + 1$, $K = 2$. □

REMARK (Entropy form of a strict block). For a finite-rank block, put $q_j = s_j^2/\|A\|_{S_2}^2$ and $\Delta = \log(\|A\|_{S_2}/D) \geq 0$. Then $D > \|A\|_{S_p}$ exactly when $\Delta < (1/2 - 1/p)H_{p/2}(q)$, with H_α the Rényi entropy; in particular rank r forces $\Delta < (1/2 - 1/p)\log r$. Both Δ and H_α add under tensor products. So a sequence of blocks succeeding at exponents $p \downarrow 2$ needs either Helson deficits $\Delta \to 0$ at bounded rank, which Theorem 8.9 rules out at bounded degree and dimension, or entropy growing faster than the deficit. Local expansions confirm the picture: for $\varphi = 1 + \varepsilon b$, $f = 1 + \varepsilon h$ the log quotient is $-\varepsilon^2\|h - 2b\|_2^2/4 + o(\varepsilon^2)$, optimal but not positive at $h = 2b$; and squeezed symbols $e^{tz^2}P(z)$ lose as $t \uparrow 1/4$.

8.4. The L^1–L^4 question

PROBLEM 7 (The L^1–L^4 question). *Is $|\int f\overline{\varphi}\,\mathrm{d}\mu_d| \leq \|f\|_{L^1(\mu_d)}\|\varphi\|_{L^4(\mu_d)}$ for all analytic polynomials f, φ on $\mathbb{C}^d$ and all d? Equivalently, does every polynomial φ have a bounded representative ψ with $\|\psi\|_\infty \leq \|\varphi\|_{L^4}$?*

Since $\|\varphi\|_{L^4} = \|A(\varphi)\|_{S_4}$, a counterexample gives a strict block at every p slightly below 4, hence $p_* < 4$. A positive answer shows that 4 is the limit of the Gaussian method, and with Proposition 7.10 it gives $D(\varphi) \leq \|H_\varphi\|_{S_4}$ for torus blocks too, closing every route through Lemma 7.1 that passes through either model. It would not decide p_*: blocks on a fixed finite torus that are far from Gaussian, like the Ortega-Cerdà–Seip examples for bounded forms, are not covered. Every family in this chapter satisfies the inequality. The natural remaining candidates are cumulative products $c_0 + \sum_j c_j Z_1 \cdots Z_j$ and quadratic symbols of large rank; local searches have found no violation.

8.5. The Paley–Wiener threshold

Bampouras [6] proved that Nehari's theorem fails in S_p, $p > 4$, for Hankel operators on Paley–Wiener spaces $PW(\Omega)$ of convex $\Omega \subset \mathbb{R}^n$ with a curved boundary piece. The cap counting of Proposition 8.12 reorganizes his proof around a scale-uniform L^1 bound, which shows that the coincidence with the $\mathbb{T}^\infty$ threshold is not structural.

LEMMA 8.11 (Lossless cap sums). *Let $\hat{b}$ be supported in the unit ball, $\rho > 0$, and $|\xi_i - \xi_j| \geq 2\rho$ for $i \neq j$. Then for every integer $k > n/4$,*

$$\Big\|\sum_j c_j e^{2\pi i\langle \xi_j, x\rangle}\rho^n b(\rho x)\Big\|_{L^1(\mathbb{R}^n)} \leq \|(1 + |x|^2)^k b\|_2\,\|(1 + |x|^2)^{-k}\|_2\,\|c\|_{\ell^2}.$$

PROOF. Cauchy–Schwarz with weight $w = (1 + |\rho x|^2)^k$, and Plancherel for wf, whose terms have disjoint Fourier supports $B(\xi_j, \rho)$ because $(1 - \Delta/4\pi^2)^k$ preserves supports. □

PROPOSITION 8.12. *Let N caps $\hat{\varphi}_i = \beta((\cdot - z_i)/\rho)$ in 2Ω satisfy the hypotheses of the lemma and give orthogonal Hankel blocks, and let a bound $\omega_\Omega = \chi_\Omega * \chi_\Omega$ on*

them. For $\psi = \sum \varphi_i$ *and* $p \geq 2$,

$$\frac{\langle \psi, \psi \rangle}{\|\psi\|_1 \|H_\psi\|_{S_p}} \geq c\, N^{1/2-1/p} (\rho^n/a)^{1/p}.$$

PROOF. The pairing is $N\rho^n \|\beta\|_2^2$; Lemma 8.11 gives $\|\psi\|_1 \lesssim \sqrt{N}$; and Bampouras's bound

$$\|H_{\varphi_i}\|_{S_p} \leq \|\hat{\varphi}_i\, \omega^{1/p}\|_{L^{p'}},$$

with orthogonality, gives $\|H_\psi\|_{S_p} \lesssim N^{1/p} \rho^{n/p'} a^{1/p}$. □

With $\rho^n/a \approx N^{-\gamma}$, the threshold is $p = 2(1+\gamma)$. The factor 2 is square-root cancellation in L^1; $\gamma = (n-1)/k$ is geometric, for $\approx \varepsilon^{-k}$ caps of radius ε^2 on the boundary. Full curvature gives $k = n-1$ and the threshold 4 in every dimension; partial curvature gives larger thresholds, for instance 6 for $n = 3$, $k = 1$. The mechanism has no analogue on $\mathbb{T}^\infty$: there $\|H_\psi\| \geq \|H_\psi 1\| = \|\psi\|_2$ for every symbol, so a sum of symbols cannot split into orthogonal blocks, and decoupling must come from tensor products, where 4 is $\mathbb{E}|Z|^4 = 2$. Accordingly the $\mathbb{T}^\infty$ barriers do not transfer verbatim. The Paley–Wiener analogue of Problem 7 asks whether $|\langle h, \varphi \rangle| \leq C \|h\|_1 \|H_\varphi\|_{S_4}$; at $p = 4$ the counting bound above stops growing, and numerically the quotients stay bounded.

At $p = 2$ the Paley–Wiener question goes back to Brevig and Perfekt.

PROBLEM 8 (Brevig and Perfekt [14]). *Does every Hankel operator on* $PW(D)$*,* D *the unit disc, that belongs to* S_2 *have a bounded symbol?*

The answer is yes. Bampouras and Perfekt [7] proved it for balls in every dimension, and Bampouras [5] then for every convex set free of affine lines. Both go through a duality that turns the problem into a Paley–Wiener form of Helson's inequality. Put $\omega(\zeta) = |D \cap (\zeta - D)|$, the density of $\xi + \eta$ for ξ, η uniform in D. Then $\|H_\varphi\|_{S_2}^2 = \int_{2D} |\hat{\varphi}|^2 \omega$, and Cauchy–Schwarz, with equality for $\hat{\varphi} = \hat{h}/\omega$, shows that the best constant C in $|\langle h, \varphi \rangle| \leq C \|h\|_1 \|H_\varphi\|_{S_2}$ is given by

$$C^2 = \sup_h \frac{1}{\|h\|_1^2} \int_{2D} \frac{|\hat{h}|^2}{\omega}, \tag{8.2}$$

the supremum taken over nonzero $h \in L^1(\mathbb{R}^2)$ with $\hat{h}$ supported in $2\overline{D}$. By the Hahn–Banach theorem, $C < \infty$ is exactly a positive answer to Problem 8. Near $\partial(2D)$ the weight ω vanishes like $\operatorname{dist}(\zeta, \partial(2D))^{3/2}$, so $1/\omega$ is not integrable and (8.2) is not trivial. Bampouras and Perfekt construct $\psi \in L^\infty(\mathbb{R}^2)$ whose Fourier transform equals $1/\omega$ on $2D$, as a distribution, from the boundary values of $(1 - |x|^2 \pm i0)^{-3/2}$; then, for $\hat{h} \in C_c^\infty(2D)$,

$$\int_{2D} \frac{|\hat{h}|^2}{\omega} = \int |\hat{h}|^2 \hat{\psi} = \langle h * \psi, h \rangle \leq \|\psi\|_\infty \|h\|_1^2,$$

so $C^2 \leq \|\psi\|_\infty$. Bampouras's theorem for general convex $\Omega \subset \mathbb{R}^n$ comes with the explicit constant $2^n (2n)!\, \Gamma(n/2+1)^2/(n!)^3 \leq 8^n$ in Helson's inequality, for the weight $|\Omega \cap (2x - \Omega)|$. The constant cannot be 1, already at $p = 2$; this is the first assertion below, and with the symbol ψ it pins down the growth of the constant on

products of discs. On $\mathbb{T}^\infty$, by contrast, Helson's inequality gives the constant 1 in S_2 (Theorem 5.15).

PROPOSITION 8.13 (Disc blocks). *Let $D \subset \mathbb{R}^2$ be the unit disc. For every $p \geq 2$ there are Schwartz functions h, φ with Fourier supports in $2\overline{D}$ and*

$$|\langle h, \varphi \rangle| > \|h\|_1 \|H_\varphi\|_{S_p}.$$

Consequently, for every $p \geq 2$, the best constant in $|\langle h, \varphi \rangle| \leq C\|h\|_1\|H_\varphi\|_{S_p}$ on $PW(D^m)$ grows exponentially in m.

PROOF. Let $v = |D|$ and $k = \check{\chi}_D$, and put $h_0 = k^2$. Then $\widehat{h_0} = \chi_D * \chi_D$ is supported in $2\overline{D}$, $h_0 \geq 0$, $h_0(0) = v^2$ and $\|h_0\|_1 = \|\chi_D\|_2^2 = v$; moreover $h_0 > 0$ almost everywhere, since k is real analytic and not identically zero. Let $\beta \geq 0$ be smooth, even and nonzero, with support in $2D$ away from the origin, and $h_1 = \check{\beta}$. Then h_1 is real, $\int h_1 = \beta(0) = 0$ and $h_1(0) = \int \beta > 0$. By dominated convergence, $\|h_0 + \varepsilon h_1\|_1 = v + \varepsilon \int h_1 + o(\varepsilon) = v + o(\varepsilon)$, while $(h_0 + \varepsilon h_1)(0) = v^2 + \varepsilon h_1(0)$. So $h = h_0 + \varepsilon h_1$ satisfies $h(0) > v\|h\|_1$ for small $\varepsilon > 0$. Replacing h by $h(rx)\eta(\delta x)$, with $\hat{\eta}$ supported in D, $\eta(0) = 1$, $r \uparrow 1$ and $\delta < 2(1-r)$, makes it Schwartz with the same Fourier support condition and preserves the strict inequality, since these functions converge to h in L^1 and have the same value at 0.

For the symbol take $\hat{\varphi}_\ell = a_\ell \in C_c^\infty(2D)$ with $0 \leq a_\ell \leq 1$ and $a_\ell \to 1$ on $2D$. The kernels $a_\ell(\xi + \zeta)$ converge in $L^2(D \times D)$ to the constant 1, whose operator has rank one and norm v; hence $\|H_{\varphi_\ell}\|_{S_p} \to v$ for every $p \geq 2$, while $\langle h, \varphi_\ell \rangle = \int \hat{h}\, a_\ell \to h(0)$. For large ℓ the quotient exceeds 1. As in Ortega-Cerdà and Seip [38], on D^m the tensor powers $h^{\otimes m}$ and $\varphi^{\otimes m}$ have pairing, L^1 norm and S_p norm equal to the mth powers of those on D. □

COROLLARY 8.14 (Polydiscs). *Let ψ be the symbol of Bampouras and Perfekt [7], and let C_m be the best constant in $|\langle h, \varphi \rangle| \leq C_m\|h\|_1\|H_\varphi\|_{S_2}$ on $PW(D^m)$. Then*

$$C_1^m \leq C_m \leq \|\psi\|_\infty^{m/2}, \qquad C_1 > 1.$$

PROOF. The lower bound is the tensor argument of Proposition 8.13 applied to near extremals for C_1, and $C_1 > 1$ is that proposition at $p = 2$. For the upper bound, the density of $\xi + \eta$ for ξ, η uniform in D^m is $\omega^{\otimes m}$, so (8.2) holds on D^m with this weight. The Fourier transform of $\psi^{\otimes m}$ is $\hat{\psi}^{\otimes m}$, which equals $(1/\omega)^{\otimes m}$ on $(2D)^m$, since tensor products of test functions are dense there, and $\|\psi^{\otimes m}\|_\infty = \|\psi\|_\infty^m$. The computation above, with h ranging over the dense set of $h \in L^1$ with $\hat{h} \in C_c^\infty((2D)^m)$ and Fatou's lemma for the rest, gives $C_m^2 \leq \|\psi\|_\infty^m$. Bampouras's constant, with $n = 2m$, gives an upper bound of the same exponential form. □

So there is no Paley–Wiener analogue of Helson's dimension-free inequality: on D^m the constant grows exactly exponentially, and on D^∞ it is infinite. For balls the tensor argument is not available, and the argument of Proposition 8.13 gives only a constant larger than 1; whether the constant for the ball B^n grows with n is open.

For polygons there is no threshold at all. Bampouras and Perfekt [8, Corollary 1.2] prove that every Hankel operator on $PW(P)$, P a polygon, that lies in

some S_p, $p < \infty$, has a bounded symbol, by lifting a theorem on the product Hardy space of two half-planes. This is the limit $k = 0$ of the counting above, where the threshold $2(1 + (n-1)/k)$ is infinite: the obstruction above S_2 is curvature.

For $2 < p \le 4$ the disc remains open. Bampouras and Perfekt show that for the disc and $2 \le p < 6$, where the Besov description of S_p is available, Nehari's theorem for S_p implies $\int_{2D} |\hat{h}|^{p'}/\omega \lesssim \|h\|_1^{p'}$. A function h violating this inequality for some $p' \in [4/3, 2)$ would therefore show that Nehari's theorem fails for $S_p(PW(D))$ at the dual exponent p. Their proof of the case $p' = 2$ uses L^2 through the identity $\int |\hat{h}|^2 \hat{\psi} = \langle h * \psi, h \rangle$, which has no analogue for $p' < 2$.

Open

(1) Problem 7.
(2) A dimension- and degree-free interval $(2, 2 + \varepsilon)$ with no strict blocks, or a family of blocks with exponents tending to 2.
(3) Nehari's theorem for $S_p(PW(D))$, $2 < p \le 4$; by Bampouras and Perfekt it would give $\int_{2D} |\hat{h}|^{p'}/\omega \lesssim \|h\|_1^{p'}$.
(4) Does the S_2 constant for the ball B^n grow with n?

Part 4

The multiplicative Hilbert matrix

CHAPTER 9

Factorization and π

For a Dirichlet series $f = \sum a_n n^{-s}$ write $c_n = 1/(\sqrt{n}\log n)$ and

$$\ell(f) = \sum_{n\geq 2} \frac{a_n}{\sqrt{n}\log n} = \int_0^\infty \big(f(1/2+\sigma) - a_1\big)\,\mathrm{d}\sigma.$$

Let C be the norm of ℓ on $\mathscr{H}^1_0$, the series with $a_1 = 0$. Write $r(\sigma) = (p^{-1/2-\sigma})_p$ for the prime curve, F for the Bohr lift of f, and $J(F) = \int_0^\infty |F(r(\sigma))|\,\mathrm{d}\sigma$. Let C_J be the least constant in $J(F) \leq C_J \|F\|_1$ on H^1_0. Since $|\ell(f)| \leq J(F)$, we have $C \leq C_J$.

Part IV is about four inequalities that share the constant π in one variable (Proposition 9.1).

	One variable	On $\mathbb{T}^\infty$
Hilbert form on $H^2 \times H^2$	π	π [15]; also with the row $m = 1$ adjoined (Theorem 9.2)
Signed functional ℓ on H^1_0	π	finite [39]; $\pi \leq C \leq 5.515$ (Proposition 10.4)
Fejér–Riesz functional J on H^1_0	π	finite [39]; $C \leq C_J \leq 5.515$
Hardy: $\sum \lvert a_n\rvert c_n$ on H^1	π	open (Problem 10)

PROBLEM 9 (The sharp constant). *Is $C = \pi$, the norm of the multiplicative Hilbert matrix on $\mathscr{H}^2$?*

PROBLEM 10 (Perfekt [39]). *Is $|a_1| + \sum_{n\geq 2} |a_n|/(\sqrt{n}\log n) \lesssim \|f\|_{\mathscr{H}^1}$ for every Dirichlet polynomial f?*

In one variable the four constants coincide for one reason: every H^1 function is a product of two H^2 functions with the same norm, and Hilbert's inequality bounds the product. On $\mathbb{T}^\infty$ the first row survives with the same constant, while factorization fails, even in its weak form, because Nehari's theorem does [38]. Part IV measures what that failure costs, row by row. This chapter treats the H^2 side: the multiplicative Hilbert matrix and its near-extremal vectors (section 9.2), and the equivalent forms of $C = \pi$, each with the reason it is not automatic (section 9.3). Chapter 10 goes above π: explicit upper bounds, a Gaussian scaling limit in which π is very probably exceeded, the absolute inequality on pieces of $\mathscr{H}^1$, and partial sums.

Brevig, Perfekt, Seip, Siskakis and Vukotić introduced the multiplicative Hilbert matrix, showed that its norm on $\mathscr{H}^2$ is π, and observed that it has a bounded symbol exactly when $C < \infty$ [15]; the question became Problem 3.2 of Saksman and Seip [43]. Partial answers replaced $\mathscr{H}^1$ by $\mathscr{H}^p$, $p > 1$, by Bayart and Brevig [9], and

$\log n$ by $(\log n)^\beta$, $\beta > 1$, by Bondarenko, Brevig, Saksman and Seip [11]. Perfekt [39, Theorem 1.1 and Corollary 1.2] proved $C_J < \infty$, with a constant of order $3 \cdot 10^5$; so $C < \infty$ and the multiplicative Hilbert matrix has a bounded symbol. He also asked Problem 10. Perfekt and Pushnitski proved that the multiplicative Hilbert matrix has purely absolutely continuous spectrum $[0, \pi]$ [41]. Results below that carry no other attribution are, to our knowledge, new.

The *sharp Fejér–Riesz conjecture* $C_J = \pi$ implies $C = \pi$. Both are very probably false (section 10.2).

9.1. One variable

PROPOSITION 9.1. *Let* $f = \sum_{k\ge0} a_k z^k \in H^1(\mathbb{T})$, *and let* $g = \sum_j b_j z^j$ *and* $h = \sum_k d_k z^k$ *belong to* $H^2(\mathbb{T})$.

(1) (Hilbert) $|\sum_{j,k} b_j d_k/(j+k+1)| \le \pi \|g\|_2 \|h\|_2$.

(2) (Fejér–Riesz) $\int_0^1 |f(x)|\,\mathrm{d}x \le \pi \|f\|_1$.

(3) (Hardy) $\sum_k |a_k|/(k+1) \le \pi \|f\|_1$.

(4) $|\sum_k a_k/(k+1)| \le \pi \|f\|_1$.

In each, π *is best possible.*

PROOF. (1) The entries are positive, so we may take $b, d \ge 0$. Put $A = j + 1/2$. The function $y \mapsto 1/((A+y)\sqrt{y})$ is convex on $(0,\infty)$, so its value at $k+1/2$ is at most its integral over $(k, k+1)$, and

$$\sum_{k\ge0} \frac{(k+1/2)^{-1/2}}{j+k+1} \le \int_0^\infty \frac{\mathrm{d}y}{(A+y)\sqrt{y}} = \pi (j+1/2)^{-1/2}.$$

Schur's test with the weight $(k+1/2)^{-1/2}$ on rows and columns gives the bound. The same test applies to the smaller entries $\rho^{j+k+1}/(j+k+1)$, $0 < \rho < 1$.

(2) Write $f = IO$ with I inner and O outer, and put $g = IO^{1/2}$, $h = O^{1/2}$. Then $f = gh$ and $\|g\|_2^2 = \|h\|_2^2 = \|f\|_1$. For $0 < \rho < 1$, $\int_0^\rho |g(x)|^2\,\mathrm{d}x = \sum_{j,k} b_j \overline{b_k} \rho^{j+k+1}/(j+k+1) \le \pi \|g\|_2^2$ by (1), and similarly for h. Cauchy–Schwarz and $\rho \to 1$ give (2).

(3) With $f = gh$ as in (2), $|a_k| \le \sum_{i+j=k} |b_i||d_j|$, so (1) applied to $(|b_i|)$ and $(|d_j|)$ gives (3).

(4) For polynomials $\sum_k a_k/(k+1) = \int_0^1 f(x)\,\mathrm{d}x$; both sides are continuous on H^1 by (2) and (3), so (4) follows from either.

Sharpness. Let $b_k = (k+1)^{-1/2-\varepsilon}$, $g = \sum b_k z^k$ and $f = g^2$. The coefficients of f are positive, $\|f\|_1 = \|g\|_2^2$, and each left side in (1)–(4), with $h = g$, equals $Q = \sum_{j,k} b_j b_k/(j+k+1)$. Since $j+k+1 \le (j+1)+(k+1)$ and $y \mapsto y^{-1/2-\varepsilon}/(J+y)$ decreases, with $J = j+1$,

$$\sum_k \frac{b_k}{j+k+1} \ge \int_1^\infty \frac{y^{-1/2-\varepsilon}\,\mathrm{d}y}{J+y} \ge J^{-1/2-\varepsilon}\Big(I_\varepsilon - \frac{J^{-1/2+\varepsilon}}{1/2-\varepsilon}\Big),$$

where $I_\varepsilon = \int_0^\infty u^{-1/2-\varepsilon}(1+u)^{-1}\,\mathrm{d}u$. Summing against $b_j = J^{-1/2-\varepsilon}$ gives $Q \ge I_\varepsilon \|g\|_2^2 - O(1)$. As $\varepsilon \to 0$, $\|g\|_2^2 = \sum_J J^{-1-2\varepsilon} \to \infty$ and $I_\varepsilon = \pi/\cos(\pi\varepsilon) \to \pi$, so $Q/\|f\|_1 \to \pi$. □

The dictionary to Dirichlet series is $z^k \leftrightarrow n^{-s}$, $\int_0^1 x^k\,\mathrm{d}x = 1/(k+1) \leftrightarrow \int_0^\infty n^{-1/2-\sigma}\,\mathrm{d}\sigma = c_n$, and $1/(j+k+1) \leftrightarrow c_{mn}$. Only the first step of the proof, the factorization $f = gh$, has no counterpart on $\mathbb{T}^\infty$.

9.2. The multiplicative Hilbert matrix

Let $\mathscr{M} = (c_{mn})_{m,n\ge2}$, of norm π [15], and let $\mathscr{R} = (c_{mn})_{m\ge1,n\ge2}$ be $\mathscr{M}$ with the row $m=1$ adjoined. For $g \in H^2$, $h \in H^2_0$ one has $\ell(gh) = \langle \mathscr{R}\hat{h}, \bar{\hat{g}}\rangle$, so $C \ge \|\mathscr{R}\|$.

THEOREM 9.2. *$\|\mathscr{R}\| = \pi$. If π_d^* is the norm of the restriction to the integers generated by the first d primes, then $\pi_d^* \uparrow \pi$, and*

$$\frac{C}{\pi} = \lim_{d\to\infty} \frac{C_d}{\pi_d^*},$$

where C_d is the norm of ℓ on the same restricted $\mathscr{H}^1_0$.

PROOF. We apply Schur's test with weights $u_n = (n\log n)^{-1/2}$ on columns and $v_1 = 1$, $v_m = (m\log m)^{-1/2}$ on rows; it suffices that $\sum_n c_{mn}u_n \le \pi v_m$ and $\sum_m c_{mn}v_m \le \pi u_n$.

Rows $m \ge 2$. With $x = \log m$, the summand $1/(t\sqrt{\log t}(x+\log t))$ decreases in t, so $v_m^{-1}\sum_n c_{mn}u_n \le \sqrt{x}\int_1^\infty dt/(t\sqrt{\log t}(x+\log t)) = 2\sqrt{x}\int_0^\infty dr/(x+r^2) = \pi$.

Row $m = 1$. With $g(t) = 1/(t\log^{3/2} t)$ decreasing, $\sum_n g(n) \le g(2) + g(3) + 2/\sqrt{\log 3} < 3.091 < \pi$, using $\log 2 > 0.69$ and $\log 3 > 1.09$.

Columns. With $y = \log n$ and $f_y(t) = 1/(t\sqrt{\log t}(\log t + y))$,

$$u_n^{-1}\sum_m c_{mn}v_m = \frac{1}{\sqrt{y}} + \sqrt{y}\sum_{m\ge2} f_y(m) \le \frac{1}{\sqrt{y}} + \sqrt{y}\Big(\frac{\pi}{\sqrt{y}} - \int_1^2 f_y + f_y(2)\Big),$$

since $\int_1^\infty f_y = \pi/\sqrt{y}$. So it suffices that $\int_1^2 f_y - f_y(2) \ge 1/y$ for $y \ge \log 2$. With $a = \log 2$, $A = \sqrt{a}$ and $u = \sqrt{a/y} \in (0,1]$, this reads

$$\Psi(u) = 2A\frac{\arctan u}{u} - \frac{1}{2A(1+u^2)} - 1 \ge 0.$$

Put $h(u) = \arctan u - u/(1+u^2)$, $D = u^3 - 2a(1+u^2)^2h$ and $J = u(3-4a) - 8a(1+u^2)h$. Then $\operatorname{sgn}\Psi' = \operatorname{sgn} D$, $D' = uJ$, and $J'' = -16a(h + 2u/(1+u^2)) < 0$. Since $J(0) = 0 < J'(0)$, $J(1) < 0$, $D(0) = 0$ and $D(1) = 1 - 2a(\pi-2) < 0$, the function Ψ increases and then decreases, so its minimum is at $u = 0$ or $u = 1$, where $\Psi(0) = 2A - 1/(2A) - 1 > 0$ and $\Psi(1) = \pi A/2 - 1/(4A) - 1 > 0$ because $A > 0.83$.

Limits. Monotonicity in d is clear, and every finitely supported test vector lives in some $\mathcal{S}_d$. The same density argument gives $C_d \uparrow C$. □

Equivalently $\mathscr{R}^*\mathscr{R} = \mathscr{M}^2 + c\otimes c \le \pi^2$: the rank-one term is absorbed although it is not zero. Since $\pi_d^* \le 1.71$ for $d \le 7$, finite-prime numerics are far from the regime in which C/π is decided.

9.2.1. Near-extremal vectors

Let $N:\ell^2(\{2,3,\dots\})\to L^2(0,\infty)$, $Nb(\sigma)=\sum b_n n^{-1/2-\sigma}$, so $\mathscr{M}=N^*N$, and let S_m be the isometry $e_n\mapsto e_{mn}$. Then $NS_m=w_mN$ with $w_m(\sigma)=m^{-1/2-\sigma}$.

THEOREM 9.3 (Slow variation). *If b_k are unit vectors with $\langle\mathscr{M}b_k,b_k\rangle\to\pi$, then for all positive integers m,m'*

$$\sum_n b_k(mn)\overline{b_k(m'n)}\to(mm')^{-1/2}.$$

PROOF. Put $g_k=Nb_k$. From $\|(\pi-\mathscr{M})b\|^2\le\pi\langle(\pi-\mathscr{M})b,b\rangle$, $b_k=\pi^{-1}N^*g_k+o(1)$. Weak limits are eigenvectors, and $\mathscr{M}$ has none [41], so $b_k\rightharpoonup 0$. As N restricted to (δ,∞) is Hilbert–Schmidt, $\int_\delta^\infty|g_k|^2\to 0$, hence $\|w_mg_k-m^{-1/2}g_k\|\to 0$. Therefore $S_m^*b_k=\pi^{-1}N^*w_mg_k+o(1)=m^{-1/2}\pi^{-1}\mathscr{M}b_k+o(1)$, and $\|\mathscr{M}b_k\|\to\pi$. □

Since $\mathscr{M}$ is real and positive, $|\ell(h^2)|=|\langle\mathscr{M}b,\bar b\rangle|\le\langle\mathscr{M}b,b\rangle$ for $h=\sum b_nn^{-s}$, so the theorem applies to every sequence with $\|h_k\|_2=1$ and $|\ell(h_k^2)|\to\pi$.

COROLLARY 9.4 (Marginals and fixed factors). *Let S be a finite set of primes and $r_0=(p^{-1/2})_{p\in S}$. For such h_k:*

1. *the marginal of $|h_k|^2dm$ on $\mathbb{T}^S$ tends weakly to the Poisson measure $P_{r_0}\,\mathrm{d}m_S$;*
2. *for every polynomial $u\neq 0$ in the variables of S, overlapping those of h_k or not,*

$$\limsup_k\frac{|\ell(uh_k^2)|}{\|uh_k^2\|_1}\le\pi\,\frac{|u(r_0)|}{\int|u|P_{r_0}\,\mathrm{d}m_S}\le\pi.$$

PROOF. (1) For coprime S-integers m,m', the Fourier coefficient at $\kappa(m)-\kappa(m')$ is $\sum_t b_k(mt)\overline{b_k(m't)}$, and every character of $\mathbb{T}^S$ arises from exactly one such reduced pair. (Without coprimality the identity is false: $m=m'=2$ gives the constant character.) (2) $\ell(uh_k^2)=\int_0^\infty u(r(\sigma))g_k(\sigma)^2\,\mathrm{d}\sigma$ and $|g_k|^2$ concentrates at $\sigma=0$, so this is $u(r_0)\ell(h_k^2)+o(1)$; the denominator is $\int|u||h_k|^2\,\mathrm{d}m\to\int|u|P_{r_0}\,\mathrm{d}m_S$ by (1). □

For a single low prime no extremality is needed: if $g\in H^1$ depends on one prime and $h\in H_0^2$ on the others, then $|\ell(gh^2)|\le\pi\|gh^2\|_1$, by inner-outer factorization of g and Theorem 9.2.

9.3. Equivalent forms and obstructions

Let $G=\widehat{\mathbb{Q}_{>0}}\cong\mathbb{T}^\infty$ and $P=\{q\in\mathbb{Q}_{>0}:q\ge 1\}$, a total positive cone containing $\mathbb{N}$. For $b:P\to\mathbb{C}$ let $\mathcal{H}_b(u,v)=\sum_{q,r\in P}\hat u(q)\hat v(r)b(qr)$ on $H^2_P(G)$.

THEOREM 9.5 (Completion principle).

$$C=\min\bigl\{\|\mathcal{H}_b\|:b(n)=c_n\ \textit{for all integers } n\ge 2\bigr\},$$

with $b(1)$ free. So $C=\pi$ if and only if the multiplicative Hilbert coefficients extend to the ordered rationals with Hankel norm π.

Proof. A Hahn–Banach extension Λ of ℓ to $L^1(G)$ gives $b(q) = \Lambda(\chi_q)$ with $\|\mathcal{H}_b\| \le C$. Conversely, the Nehari theorem for ordered groups [47] gives a symbol of norm $\|\mathcal{H}_b\|$ that represents ℓ on $\mathscr{H}^1_0$. □

Remark (What the reformulation does and does not give). By compactness, $C > \pi$ would be witnessed by one finite set $F \subset P$ and one trace-class matrix X on F with $\sum_{qr=t} X(q,r) = 0$ for $t \notin \mathbb{N}_{\ge 2}$; rank-one X never works, since then the form is a compression of $\mathscr{R}$. By Helson–Lowdenslager factorization [25], $C = \pi$ is equivalent to $|\sum_n c_n \sum_{qr=n} \hat{u}(q)\hat{v}(r)| \le \pi\|u\|_2\|v\|_2$ for $u, v \in H^2_P(G)$ whose product is an integer Dirichlet polynomial; the factors need not have spectrum in $\mathbb{N}$, already for $z_1 + z_2$. The natural completions all fail: continuous interpolation $b(q) = F(\log q)$, the integrated flat representatives, and $b = 0$ off $\mathbb{N}$ are unbounded, and no completely positive Schur damping fixes the integer entries non-trivially. A bounded completion must cancel the non-square-summable entries $\sum_m c_{Am}c_{Bm}$, $2 \le A < B$ coprime, through its non-integer layers.

9.3.1. Curve factorization

Proposition 9.6 (Curve factorization). *For $F \in H^1_0$ let $\Phi(F)$ be the infimum of $\|G\|_2\|H\|_2/\|F\|_1$ over $G \in H^2_0$, $H \in H^2$ with $|F(r(\sigma))| \le |G(r(\sigma))H(r(\sigma))|$ for all $\sigma > 0$. Then $J(F) \le \pi\Phi(F)\|F\|_1$.*

Proof. $J(F) \le \sum_{m\ge1,n\ge2} c_{mn}|\hat{H}(m)||\hat{G}(n)| \le \pi\|G\|_2\|H\|_2$ by Theorem 9.2, which applies to absolute values since the matrix is positive. □

In one variable inner-outer factorization gives $\Phi \le 1$. On $\mathbb{T}^\infty$ global weak factorization fails [38], but Φ asks only for a majorant on the curve, which is far weaker.

Proposition 9.7 (Curve factorization fails). *For every $F \in H^1_0$ and $\sigma > 0$, with $\zeta_\sigma = \zeta(1 + 2\sigma)$,*

$$\Phi(F)\|F\|_1 \ge \frac{|F(r(\sigma))|}{\sqrt{\zeta_\sigma(\zeta_\sigma - 1)}}.$$

Consequently $\Phi(K_{r(\sigma)}(K_{r(\sigma)} - 1)) > 1$ for every $\sigma > 0$, and $\Phi(F) > 1.0151$ for $f = 2^{-s} + (2/5)3^{-s}$.

Proof. For admissible $G \in H^2_0$, $H \in H^2$, $|G(r)H(r)| \le \|G\|_2\sqrt{\zeta_\sigma - 1} \cdot \|H\|_2\sqrt{\zeta_\sigma}$. For $F = K(K-1)$ with $K = K_{r(\sigma)}$, $F(r(\sigma)) = \zeta_\sigma(\zeta_\sigma - 1)$ while $\|F\|_1 < \|K\|_2\|K - 1\|_2 = \sqrt{\zeta_\sigma(\zeta_\sigma - 1)}$, strictly, because $|1 - 1/K|$ is not constant. For the binomial, $\|F\|_1 = \int |1 + 2z/5|\,\mathrm{d}m \le 1.04048$, and at $\sigma = 3$, $|f(7/2)| = 0.0969\ldots$ against $\sqrt{\zeta(7)(\zeta(7) - 1)} \le 0.09179$. □

The mechanism is again Ortega-Cerdà–Seip: for $2^{-s} + \varepsilon 3^{-s}$ the evaluation gains to first order in ε at a suitable point while the L^1 norm pays only to second order. Near extremizers the picture is different: for the two-prime near-extremizer of Proposition 10.4, the convex program for Φ and its dual agree at $\Phi \approx 0.994$. So the reduction survives in the weaker form that matters: $C = \pi$ if $\Phi(F_k) \to 1$ along some extremizing sequence.

REMARK (A warning from one variable). Building a symbol from the phases of extremal squares, $\psi = \pi F/|F|$, does not work directly. In one variable, with $F = (1-rz)^{-1}$ and $r \to 1$, the limit symbol has analytic coefficients $1/(k+1/2)$, so it represents $\sum a_k/(k+1/2)$ rather than $\sum a_k/(k+1)$. The multiplicative analogue is the family $\ell_c(f) = \sum a_n/(\sqrt{n}(\log n + c)) = \int_0^\infty e^{-c\sigma} f(1/2+\sigma)\,\mathrm{d}\sigma$, all of H^2 norm π. A phase-limit symbol would settle one member ℓ_{c_*}; passing to $c = 0$ is the completion problem of Theorem 9.5.

9.3.2. Small primes: a linear-symbol effect

Numerically $C_d/\pi_d^* \approx 1.06$ for $d = 2, 3, 4$, which once suggested $C > \pi$. For primes $p < q$ let $C_{p,q}$ and $\pi_{p,q}^*$ be the two-prime analogues.

PROPOSITION 9.8. *As $p, q \to \infty$ with $q/p \to 1$, $C_{p,q}/\pi_{p,q}^* \to \pi/(2\sqrt{2}) = 1.1107\ldots$.*

PROOF. With $a = c_p$, $b = c_q$ the functional is $a\hat{f}(p)+b\hat{f}(q)+R$, where R has square-summable coefficients of size $o(a)$ and so dual norm $o(a)$. Hence $C_{p,q} = D(az_1 + bz_2) + o(a)$, where D is the dual norm, and similarly $\pi_{p,q}^* = \sqrt{a^2+b^2} + o(a)$. For $a = b$, $D(z_1+z_2) = 2/\|z_1+z_2\|_1 = \pi/2$, with extremal symbol $(z_1+z_2)/|z_1+z_2|$. □

(p,q)	$(2,3)$	$(5,7)$	$(11,13)$	$(101,103)$	$(1009,1013)$
$C_{p,q}/\pi_{p,q}^* \geq$	1.0612	1.0905	1.1038	1.1101	1.1107

The same ratio appears exactly as a factorization gap: for $q_d = \prod_{j\leq d}(z_{2j-1} + z_{2j})$,

$$\|q_d\|_1 = (4/\pi)^d, \qquad \|q_d\|_{H^2\odot H_0^2} = 2^{d/2}, \qquad \frac{\|q_d\|_{H^2\odot H_0^2}}{\|q_d\|_1} = \Big(\frac{\pi}{2\sqrt{2}}\Big)^d.$$

Since $|\ell(f)| \leq \pi\|f\|_{H^2\odot H_0^2}$ by Theorem 9.2, this shows that the rectangular norm alone can never prove $C = \pi$: the projective norm and the H^1 norm are not equivalent on H_0^1.

Open

(1) Is there a bounded completion of (c_n) to the ordered rationals with norm π (Theorem 9.5)?
(2) If $C > \pi$, then $\Phi(F_k) \not\to 1$ along extremizing sequences. Is there a direct proof? ($\Phi \leq 1$ fails in general, Proposition 9.7.)

CHAPTER 10

Beyond π

This chapter is about the constants of chapter 9 themselves. Perfekt's theorem makes them finite, and section 10.1 gives explicit upper bounds; together with the rectangular norm they bracket C between π and 5.515, and they show that no counterexample to $C = \pi$ uses only primes below 10^{216}. The centre of the chapter is section 10.2: for quadratic combinations of Euler kernels at many scales near the critical line, the H^1 norm has an exact Gaussian limit. This gives a lower bound for C by a constant of a log-correlated Gaussian field, and we conjecture that this is where C is decided. A probabilistic certificate puts the Gaussian constant above π. Section 10.4 proves the absolute inequality on every piece of $\mathscr{H}^1$ cut out by a degree constraint the Euler product can see, and section 10.5 treats partial sums.

10.1. Perfekt's theorem and explicit bounds

Perfekt [39, Theorem 1.1] proved $\int_0^1 |f(1/2+\sigma)|\,\mathrm{d}\sigma \lesssim \|f\|_{\mathscr{H}^1}$, by localizing the flat representative of chapter 1 and controlling the overlaps through separation and concentration estimates. Theorem 2.5 reproves this for every $0 < p < \infty$ through the Hellinger–Bessel inequality, with an explicit constant. For the multiplicative Hilbert functional this gives the following.

COROLLARY 10.1 (Perfekt's inequality [39]). *For every $f \in \mathscr{H}^1$,*

$$\Big|a_1 + \sum_{n\ge 2} \frac{a_n}{\sqrt{n}\log n}\Big| \le \Big(5 + 2\sqrt{2} + \sum_{n\ge 2} \frac{1}{n^{3/2}\log n}\Big)\|f\|_{\mathscr{H}^1} < 9\,\|f\|_{\mathscr{H}^1}.$$

In particular, the norm C of the multiplicative Hilbert functional on $\mathscr{H}^1_0$ satisfies $\pi \le C < 9$, the lower bound coming from the multiplicative Hilbert matrix on $\mathscr{H}^2$ [15]. Proposition 10.4 lowers 9 to 5.515.

PROOF. The left side is $|\int_0^1 f(1/2+\sigma)\,\mathrm{d}\sigma + \int_1^\infty (f(1/2+\sigma) - a_1)\,\mathrm{d}\sigma|$. Use Theorem 2.5 with $p = 1$ on the first integral and $|a_n| \le \|f\|_1$ on the second. The series is about 1.113. □

REMARK (The limit of $|F|$-only methods). For Lebesgue measure on $(0,\delta)$ along the curve, the Gram kernel of Theorem 2.1 is asymptotically $2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$, homogeneous of degree -1, whose Mellin multiplier is $\Gamma(1/4)^2/\sqrt{2\pi} = 5.2441\ldots$. The H^2 analogue gives the Carleman kernel $1/(\sigma+\tau)$, of norm π. So, up to uniformity of the asymptotics, the unrestricted Gram estimate of Theorem 2.1 stops at 5.244. This does not rule out arguments that use the special structure of $u = |F|^{1/2}$. Since J is subadditive and $\|\sum F_i\|_1 \ge$

$\sum\|F_i\|_1 - 2\sum_{i<j}(\|F_i\|_1\|F_j\|_1)^{1/2}H(\nu_i,\nu_j)$, profiles at well-separated scales cannot create an excess either; extremizing sequences for $\mathscr{M}$ spread over a long range of $\log(1/\sigma)$, so the sharp constant is a per-unit-scale quantity.

The local constant of Theorem 2.7 handles $\sigma \le 2$. Beyond that, a second refinement uses Helson's inequality in the odd primes together with an exact factorization in z_2.

LEMMA 10.2 (Vector-valued outer factorization). *If V is a nonzero $\mathbb{C}^N$-valued analytic polynomial on $\mathbb{T}$, there are an outer $g \in H^2(\mathbb{T})$ and $W \in H^2(\mathbb{T};\mathbb{C}^N)$ with $V = gW$, $\|g\|_2^2 = \int\|V\|\,\mathrm{d}m$ and $\|W\|_2^2 \le \int\|V\|\,\mathrm{d}m$.*

PROOF. Let g be outer with $|g|^2 = \|V\|$ on $\mathbb{T}$. Since $\log\|V\|$ is subharmonic, $\|V\| \le |g|^2$ in $\mathbb{D}$, so $W = V/g$ is holomorphic with $\|W\| \le |g|$. □

For a positive measure ν on $(0,\infty)$ put $c_N(\nu) = \int N^{-1/2-\sigma}\,\mathrm{d}\nu$ and let A_ν be the nonnegative matrix with rows $k \ge 0$, columns (j,m) with $j \ge 0$, m odd and $(j,m) \ne (0,1)$, and entries $\sqrt{d(m)}\,c_{2^{k+j}m}(\nu)$.

THEOREM 10.3 (Tail bound). *For $F \in H^1_0(\mathbb{T}^\infty)$, $\int|F(r(\sigma))|\,\mathrm{d}\nu(\sigma) \le \|A_\nu\|\,\|F\|_1$, and $\|A_\nu\|^2 \le \sum_{N\ge2} d(N)c_N(\nu)^2 - \sum_{k\ge1} c_{2^k}(\nu)^2$. The same holds on $\mathcal{P}$-smooth functions with the sums restricted accordingly.*

PROOF. Write $F = \sum_{m\ \mathrm{odd}} F_m(z_2)w^{\kappa(m)}$. For each z_2, Helson's inequality in w gives $(\sum_m|F_m(z_2)|^2/d(m))^{1/2} \le \int|F(z_2,w)|\,\mathrm{d}m(w)$. Apply Lemma 10.2 to $V = (F_m/\sqrt{d(m)})_m$: $F_m = \sqrt{d(m)}gW_m$ with $\|g\|_2\|W\|_2 \le \|F\|_1$ and $\hat W_1(0) = 0$. Expand $F(r(\sigma))$ in powers of a_σ and integrate in ν. For the Hilbert–Schmidt bound, the entries with $k+j = e$ for $N = 2^e m$ number $e+1$, and $(e+1)d(m) = d(N)$. □

PROPOSITION 10.4 (Computer-assisted bounds). *The following bounds hold.*

1. *For every $F \in H^1_0$, $J(F) \le 5.5150\,\|F\|_1$. Hence $\pi \le C \le C_J \le 5.5150$.*
2. *If $F \in H^1_0$ uses only primes $p \le P$ with $\prod_{p\le P}(1-1/p)^{-1} \le 890$, in particular $P \le 10^{216}$, then $J(F) \le 3.1407\|F\|_1 < \pi\|F\|_1$.*
3. *For the first d primes,*

d	1	2	3	4	5	6	7
$C_d \le$	1.19914	1.58582	1.72142	1.80494	1.84758	1.88301	1.90873

Moreover $C_1 = \pi_1^*$, $1.54792 \le C_2 \le 1.58582$ and $1.0617 \le C_2/\pi_2^* \le 1.0878$.

PROOF. (1) On $(0,2)$ use Theorem 2.7. On $(2,\infty)$ use Theorem 10.3 with Lebesgue measure: the finite part of A_ν with $N \le 2000$ has norm at most 0.27086 by Schur's test with weights verified in ball arithmetic, and the rest has Hilbert–Schmidt norm below 10^{-6}.

(2) Theorem 10.3 with Lebesgue measure gives $J(F)^2/\|F\|_1^2 \le \int_0^\infty \sigma(\Pi_P(\sigma)^2 - 1)\,\mathrm{d}\sigma - \lambda_0$, where $\Pi_P(\sigma) = \prod_{p\le P}(1-p^{-1-\sigma})^{-1} \le \min\{\zeta(1+\sigma), \Pi_P(0)\}$ and $\lambda_0 = \mathrm{Li}_2(1/2)/\log^2 2$. With $\Pi_P(0) \le 890$ the integral is at most 11.07528 (ball arithmetic), so the bound is $\sqrt{9.86343} < 3.1407$. The bound $\prod_{p\le x}(1-1/p)^{-1} < e^\gamma \log x\,(1 + 1/(2\log^2 x))$ of Rosser and Schoenfeld [42] gives 885.9 at $x = 10^{216}$.

(3) Upper bounds: Theorem 10.3 with Schur weights verified in ball arithmetic and a Hilbert–Schmidt tail. $d = 1$: Nehari's theorem in one variable identifies C_1 with the Hankel matrix that defines π_1^*. Lower bound for C_2: an explicit polynomial in z_2, z_3 of degree at most 11 in each variable, with $\ell(q) = 1.54870\ldots$ exactly and $\|q\|_1 \le 1.00051$ by a midpoint rule with certified Taylor errors on an 8192^2 grid. □

The tail bound is nearly tight for its part: $F = z_2$ has $\int_2^\infty |F(r)|\,\mathrm{d}\sigma = 2^{-5/2}/\log 2 \approx 0.255$. No refinement of this scheme reaches $\Gamma(1/4)^2/\sqrt{2\pi}$ itself, and making finitely many primes exact does not lower the boundary constant: the modified kernel is still asymptotic to L_0 near 0.

By (2), a counterexample to $C = \pi$ must involve a prime above 10^{216}. The finite-prime constants themselves are much smaller, and by (3) the two-prime excess of section 9.3.2 is real and capped, between six and nine per cent.

10.2. The Gaussian scaling limit

The results so far say where a counterexample would have to live. It must use primes beyond 10^{216} (Proposition 10.4), mix degrees (Theorem 10.13), and cannot gain from a fixed finite set of primes (Corollary 9.4). Products of normalized Euler kernels at many scales near the critical line do all three. Their H^1 norms have an exact Gaussian limit, and in that limit π is very probably exceeded.

Fix scales $a_1, \ldots, a_m > 0$ and write $a_i = e^{t_i}$. For $\varepsilon > 0$ let

$$k_i^\varepsilon(z) = \zeta(1 + 2\varepsilon a_i)^{-1/2} \prod_p \bigl(1 - p^{-1/2-\varepsilon a_i} z_p\bigr)^{-1},$$

the normalized reproducing kernel at the point $r(\varepsilon a_i)$. For a real symmetric $m \times m$ matrix A put

$$F_\varepsilon = \sum_{i,j} A_{ij} k_i^\varepsilon k_j^\varepsilon, \qquad F_\varepsilon^0 = F_\varepsilon - F_\varepsilon(0) \in H_0^1.$$

PROPOSITION 10.5. *As $\varepsilon \to 0$, $\langle k_i^\varepsilon, k_j^\varepsilon \rangle \to G_{ij}$ and $\ell(F_\varepsilon^0) \to \sum_{i,j} A_{ij} H_{ij}$, where*

$$G_{ij} = \operatorname{sech} \frac{t_i - t_j}{2}, \qquad H_{ij} = \frac{t_i - t_j}{\sinh((t_i - t_j)/2)}, \quad H_{ii} = 2.$$

Moreover $H \preceq \pi G$ for every choice of scales.

PROOF. The Gram entry is $\zeta(1 + \varepsilon(a_i + a_j))/\sqrt{\zeta(1 + 2\varepsilon a_i)\zeta(1 + 2\varepsilon a_j)}$, and $\zeta(1 + t) = t^{-1} + O(1)$. The functional is $\ell(F) = \int_0^\infty (F(r(\sigma)) - F(0))\,\mathrm{d}\sigma$, and the entry for (i, j) equals $\int_0^\infty (\zeta(1+\varepsilon a_i+\sigma)\zeta(1+\varepsilon a_j+\sigma) - 1)\,\mathrm{d}\sigma$ divided by the same normalization. Replacing both zeta factors by their poles costs $O(\log 1/\varepsilon)$, which is $o(1)$ after normalization. The substitution $\sigma = \varepsilon x$ then gives $2\sqrt{a_i a_j} \int_0^\infty dx/((a_i+x)(a_j+x)) = H_{ij}$.

For the last claim, $\pi \operatorname{sech}(x/2) - x/\sinh(x/2)$ is a positive definite function on $\mathbb{R}$: the Fourier transforms of $\operatorname{sech}(x/2)$ and $x/\sinh(x/2)$ are $2\pi \operatorname{sech}(\pi\xi)$ and $2\pi^2 \operatorname{sech}^2(\pi\xi)$, and $2\pi^2 \operatorname{sech}(\pi\xi)(1 - \operatorname{sech}(\pi\xi)) \ge 0$. □

The last statement explains why squares never beat π here. If $A = uu^T$, then $F_\varepsilon = (\sum u_i k_i^\varepsilon)^2$, its norm tends to $u^* G u$, and $|u^T H u| \le u^* H u \le \pi u^* G u$. Any

excess must come from indefinite quadratics. Their norms are not determined by G, and the next theorem computes them.

Fix a reference scale a_0 among the a_i. Let $D = (D_i)$ be a centred circular complex Gaussian vector with $\mathbb{E} D_i \overline{D_k} = C_{ik}$, where

$$C_{ik} = \log \frac{(a_i + a_0)(a_k + a_0)}{2a_0(a_i + a_k)} = \int_0^\infty (e^{-a_i x} - e^{-a_0 x})(e^{-a_k x} - e^{-a_0 x}) \, \frac{\mathrm{d}x}{x},$$

and put $R_i^{(j)} = \sqrt{a_i / a_0}\,(a_0 + a_j)(a_i + a_j)^{-1} e^{D_i}$.

THEOREM 10.6.

$$\lim_{\varepsilon \to 0} \|F_\varepsilon^0\|_1 = \mathcal{N}(A) \coloneqq \frac{1}{m} \sum_{j=1}^{m} \mathbb{E}\, \frac{m\,|R^{(j)T} A R^{(j)}|}{|R^{(j)}|^2}.$$

Consequently $C \geq |\sum_{i,j} A_{ij} H_{ij}| / \mathcal{N}(A)$ *whenever* $\mathcal{N}(A) > 0$.

PROOF. With $q_\varepsilon = m^{-1} \sum_j |k_j^\varepsilon|^2$, a probability density, one has exactly

$$\|F_\varepsilon\|_1 = \frac{1}{m} \sum_j \int \frac{m\,|\sum A_{ik} r_i r_k|}{\sum_i |r_i|^2}\, |k_j^\varepsilon|^2 \, \mathrm{d}m, \qquad r_i = k_i^\varepsilon / k_0^\varepsilon.$$

Under $|k_j^\varepsilon|^2 dm$ the coordinates z_p are independent and Poisson-kernel distributed, with $\mathbb{E} z_p^n = p^{-n(1/2 + \varepsilon a_j)}$. The logarithm of r_i is $\log(a_i/a_0)/2 + o(1)$ plus the linear sum

$$\sum_p p^{-1/2} \big(p^{-\varepsilon a_i} - p^{-\varepsilon a_0} \big) z_p$$

plus a remainder of degree at least two. That remainder tends to 0 in mean and in L^2, by domination with p^{-2}. The linear sum has mean

$$\sum_p p^{-1} \big(p^{-\varepsilon(a_i + a_j)} - p^{-\varepsilon(a_0 + a_j)} \big) \to \log \frac{a_0 + a_j}{a_i + a_j},$$

by $\sum_p p^{-1-t} = \log(1/t) + c + o(1)$. In the same way its covariances tend to C_{ik}, and its pseudo-covariances vanish because $\mathbb{E}(z_p - \mathbb{E} z_p)^2 = 0$. By the Lindeberg theorem, the vector (r_i) converges in law to $R^{(j)}$. The integrand is a bounded continuous function of $\log r$, bounded by $m\|A\|$, so the integrals converge. Finally $F_\varepsilon(0) = O(\varepsilon)$. □

So proving $C > \pi$ is reduced to bounding one finite-dimensional Gaussian expectation from above. The anchored form of D makes this practical: with $L(t) = \log\cosh(t/2)$, the increments $D(t_j + s) - D(t_j)$ form a stationary field with covariance $L(s) + L(u) - L(s - u)$. Squares give exactly $u^* G u$, which provides an exact control variate.

Let C_{Gauss} be the supremum of $|\sum_{i,j} A_{ij} H_{ij}| / \mathcal{N}(A)$ over all m, all scales $a_1, \ldots, a_m > 0$, and all real symmetric A with $\mathcal{N}(A) > 0$. It is a constant of the log-correlated Gaussian field D, and involves no arithmetic.

COROLLARY 10.7. $C \geq C_{\mathrm{Gauss}}$.

PROOF. This is the last statement of Theorem 10.6. □

CONJECTURE 1. $C = C_{\mathrm{Gauss}} > \pi$.

The evidence for the inequality is section 10.2 below. The evidence for the equality is structural. Near-extremal vectors for the multiplicative Hilbert matrix tend weakly to zero, put asymptotically no mass at $\sigma > \delta$, and vary slowly under multiplication (Theorem 9.3). For ℓ on $\mathscr{H}_0^1$, section 10.3 proves that every extremizing sequence either concentrates at $\sigma = 0$ or produces an extremizer, and that the constant at $\sigma = 0$, suitably normalized, does not see any finite set of primes (Theorem 10.9 and Proposition 10.11). So the conjecture splits into two statements: that ℓ does not attain its norm, and that the concentrated problem is governed by the Gaussian limit of Theorem 10.6. If the second fails for quadratics, the natural replacement allows kernel polynomials of higher degree, whose norms we have not computed.

REMARK (The excess, numerically). Take the scales $t_i = i$, $|i| \le T/2$, and let A range over quadratics in the top few generalized eigenvectors of (H, G). Then minimize $\mathcal{N}(A)$ for fixed $\sum A_{ij}H_{ij}$ on one sample and evaluate on an independent one. The best ratios are then $\sum A_{ij}H_{ij}/\mathcal{N}(A) \approx 3.1386$, 3.1422, 3.1442, 3.1450 for $T = 120$, 160, 200, 240, against 3.1326, 3.1363, 3.1381, 3.1392 for squares. Indefinite quadratics thus gain about 0.005 over squares at every span. The squares' deficit decays with the span, and the two cross between $T = 120$ and $T = 160$. A refinement of the grid changes nothing.

REMARK (A probabilistic certificate). For $T = 240$ and one explicit quadratic in six modes, normalized so that $\sum A_{ij}H_{ij} = 1$ (to 10^{-15}, in interval arithmetic), 10^8 samples drawn from a seed fixed in advance give $\mathcal{N}(A) \le 0.3179567$. The tool is a one-sided Bernstein inequality for the control-variate residual, with explicit Lipschitz, coupling and rounding budgets. The threshold is $1/\pi = 0.3183099$, so

$$C \ge 3.1450 > \pi.$$

This is not a proof. It holds except on an event of probability at most $2 \cdot 10^{-30}$, assuming that the pseudo-random generator behaves like independent draws and that floating-point arithmetic obeys its rounding model. A deterministic certificate needs a bound for one bounded Gaussian integral in 480 real dimensions, to about $4 \cdot 10^{-4}$. Grids are hopeless at that dimension, and quadratic majorants provably fail (their best bound is the nuclear norm of A).

The certificate is consistent with everything above. The kernels involve all primes, and the excess appears only as $\varepsilon \to 0$, so only primes far beyond 10^{216} matter. The quadratic is not a square, and it mixes all degrees. If it is right, then $C > \pi$, the sharp Fejér–Riesz conjecture fails, and $\Phi(F_k) \not\to 1$ along extremizing sequences.

REMARK (Fewer scales). Splitting the primes into K interleaved classes and summing the squares of the top eigenvectors on each class gives ratios 1.29, 0.61, 0.47 for $K = 1, 2, 3$: the Ortega-Cerdà–Seip gain from independent blocks is real, about $\sqrt{K}$, but sparse classes give trace-class matrices (Corollary 5.3) whose top eigenvalues fall like $1/K$. None of these constructions exceeds the square bound, and the Euler-product family $\prod_{p \le y}(1 + 2z_p/\sqrt{p}) - 1$ stays below $e^\gamma < \pi$. All of

them use a bounded number of scales. The excess of section 10.2 needs about one hundred and fifty.

10.3. Extremizing sequences

For the multiplicative Hilbert matrix, near-extremal vectors tend weakly to zero because $\mathscr{M}$ has no eigenvectors (Theorem 9.3). In H^1 weak-* limits do not split the norm, since L^1 norms are not additive under weak convergence. The substitute is an averaging argument and the following lemma, in which analyticity is what matters. Write E_d for the conditional expectation onto the first d coordinates, and P_x for the Poisson kernel of $x \in \mathbb{D}^d$, viewed as a function on $\mathbb{T}^\infty$.

LEMMA 10.8. *Let $F_N \in H^1(\mathbb{T}^\infty)$ with $\sup_N \|F_N\|_1 < \infty$ and $F_N \to u$ almost everywhere. Then $u \in H^1(\mathbb{T}^\infty)$, and the Fourier coefficients of F_N converge to those of u.*

PROOF. Let $K = \sup_N \|F_N\|_1$. By Fatou, $u \in L^1$. Let μ be a weak-* cluster point of $F_N\,\mathrm{d}m$; passing to a subsequence, $F_N\,\mathrm{d}m \to \mu$ weak-*. Since $|G|^{1/2}$ is plurisubharmonic for G analytic, and both sides are continuous on H^1,

$$|E_d G(x)|^{1/2} \le \int |G|^{1/2} P_x\,\mathrm{d}m \qquad (G \in H^1,\ x \in \mathbb{D}^d).$$

Fix M and put $g = |u - F_M|^{1/2}$. Apply this to $G = F_N - F_M$ and let $N \to \infty$. On the left, $E_d F_N(x) \to \int P_x\,\mathrm{d}\mu$. On the right, $|F_N - F_M|^{1/2} \to g$ almost everywhere with $\||F_N - F_M|^{1/2}\|_2^2 \le 2K$, hence weakly in L^2. So the Poisson extension h_d of the analytic measure $\nu = E_d\mu - E_d F_M$ on $\mathbb{T}^d$ satisfies $|h_d(x)| \le (P_x[E_d g])^2$. As $\rho \to 1$ the functions $(P_{\rho\zeta}[E_d g])^2$ converge in $L^1(\mathbb{T}^d)$, so $\{h_d(\rho\,\cdot)\}$ is uniformly integrable; therefore ν is absolutely continuous, and letting $\rho \to 1$ along a suitable sequence gives $|E_d\mu - E_d F_M| \le (E_d g)^2$ almost everywhere on $\mathbb{T}^d$.

By Doob's maximal inequality $\sup_d (E_d g)^2 \in L^1$, and $\{E_d F_M\}_d$ is uniformly integrable, so the martingale $(E_d\mu)_d$ is uniformly integrable. Hence $\mu = F\,\mathrm{d}m$ with $F \in L^1$, and martingale convergence gives $|F - F_M| \le |u - F_M|$ almost everywhere, for every M. Letting $M \to \infty$ gives $F = u$. So $\mu = u\,\mathrm{d}m$, which has analytic spectrum, and the whole sequence converges. □

THEOREM 10.9 (Concentration or an extremizer). *Let (f_k) be an extremizing sequence: $f_k \in \mathscr{H}^1_0$, $\|f_k\|_{\mathscr{H}^1} = 1$ and $\ell(f_k) \to C$. Then either*

$$\int_\delta^\infty |f_k(1/2 + \sigma)|\,\mathrm{d}\sigma \to 0 \qquad \textit{for every } \delta > 0,$$

or ℓ attains its norm: there is $u \in \mathscr{H}^1_0$, $u \ne 0$, with $|\ell(u)| = C\|u\|_{\mathscr{H}^1}$.

PROOF. For $\|f\|_1 \le 1$ and $a_1 = 0$ we have $|f(1/2+\sigma)| \le \zeta(1+2\sigma)$ by Proposition 1.2 and $|f(1/2+\sigma)| \le \zeta(1/2+\sigma) - 1$ for $\sigma \ge 1$, an integrable majorant on $[\delta, \infty)$. Helson's inequality $\sum |a_n|^2/d(n) \le \|f\|_1^2$ [24] makes the tails $\sum_{n>N} |a_n| n^{-1/2-\sigma}$ small uniformly on the unit ball. Hence, if the coefficients of f_k converge to those of u, then $f_k(1/2+\sigma) \to u(1/2+\sigma)$ for every $\sigma > 0$, and $\int_\delta^\infty |f_k| \to \int_\delta^\infty |u|$ by dominated convergence.

Suppose that $\int_\delta^\infty |f_k| \ge \eta > 0$ along a subsequence. Pass to a further subsequence whose coefficients converge, to those of a measure μ, and then, by Komlós's theorem [30] applied to the real and imaginary parts, to one whose Cesàro means F_N converge almost everywhere. Then $F_N \in \mathscr{H}^1_0$, $\|F_N\|_1 \le 1$ and $\ell(F_N) \to C$, so $\|F_N\|_1 \to 1$; the coefficients of F_N still converge to those of μ. By Lemma 10.8, $\mu = u\,dm$ with $u \in \mathscr{H}^1_0$ and $F_N \to u$ almost everywhere, and $\int_\delta^\infty |u| \ge \eta$, so $u \ne 0$. The Brezis–Lieb lemma [16] gives $\|F_N - u\|_1 \to 1 - \|u\|_1$. Since $F_N - u \in \mathscr{H}^1_0$,

$$C = \lim|\ell(F_N)| \le |\ell(u)| + C\lim\|F_N - u\|_1 = |\ell(u)| + C(1 - \|u\|_1),$$

so $|\ell(u)| \ge C\|u\|_1$. □

An extremizer, if there is one, is rigid.

PROPOSITION 10.10 (Extremizers). *Let $F \in H^1_0$ with $\|F\|_1 = 1$ and $\ell(F) = C$.*

(1) $\ell(QF) = C\int Q|F|\,dm$ for every analytic polynomial Q.
(2) F depends on infinitely many variables.
(3) F is not of the form h^2 with $h \in H^2$.

PROOF. (1) The inequality $|1+w| \le 1 + \Re w + |w|^2/2$ holds for every complex w, because the square of the right side exceeds $|1+w|^2$ by $(\Re w + |w|^2/2)^2$ and the right side equals $(1+|1+w|^2)/2$. So, for complex t,

$$\begin{aligned} C + \Re\big(t\,\ell(QF)\big) &\le |\ell(F(1+tQ))| \le C\|F(1+tQ)\|_1 \\ &\le C + C\Re\Big(t\int Q|F|\,dm\Big) + \frac{C|t|^2}{2}\|Q\|_\infty^2, \end{aligned}$$

and letting $t \to 0$ in every direction gives (1).

(2) Suppose F depends only on the variables of a finite set S of primes. For $p \notin S$, $\int z_p|F|\,dm = 0$, so by (1) $\int_0^\infty p^{-\sigma} g(\sigma)\,d\sigma = 0$, where $g(\sigma) = F(r(\sigma))$. The function g is real analytic on $(-1/2, \infty)$, bounded on $[0,\infty)$, and not identically zero since $\ell(F) \ne 0$; if $g(\sigma) \sim c\sigma^m$ at 0, then Watson's lemma gives $\int_0^\infty e^{-x\sigma} g(\sigma)\,d\sigma \sim c\,m!\,x^{-m-1} \ne 0$ as $x \to \infty$, which contradicts its vanishing at $x = \log p$ for every large prime p.

(3) If $F = h^2$ with $h = \sum b_n n^{-s}$, then $b_1 = 0$ and $C = |\langle \mathscr{M} b, \bar b\rangle| \le \langle \mathscr{M}|b|, |b|\rangle \le \pi\|b\|^2 = \pi$. Since $C \ge \pi$, $|b|$ would be an eigenvector of $\mathscr{M}$ with eigenvalue π, and there are none [41]. □

In the first alternative of Theorem 10.9 the constant is local. For $0 < \delta \le 1$ let $C^{(\delta)}$ be the norm of $f \mapsto \int_0^\delta f(1/2+\sigma)\,d\sigma$ on $\mathscr{H}^1$, and $C^{(\delta)}_{>P}$ its norm on the series supported on integers with no prime factor $p \le P$. Put $c_P = \prod_{p\le P}(1 - 1/p)$.

PROPOSITION 10.11 (Small primes). *For every P there is K_P with $|C^{(\delta)} - C^{(\delta)}_{>P}/c_P| \le K_P\delta$ for $0 < \delta \le 1$. If ℓ does not attain its norm, then $C \le C^{(\delta)}$ for every δ, so*

$$C \le \liminf_{\delta\to 0} \frac{C^{(\delta)}_{>P}}{c_P} \qquad \text{for every } P.$$

PROOF. Let S be the primes up to P, write $z = (\zeta, w)$ and $r(\sigma) = (r_S(\sigma), r'(\sigma))$, and put $r_0 = r_S(0)$. For a polynomial F, $F(r(\sigma)) = \int_{\mathbb{T}^S} F(\zeta, r'(\sigma)) P_{r_S(\sigma)}(\zeta)\,dm(\zeta)$,

and $|P_{r_S(\sigma)} - P_{r_0}| \le K_P\sigma$ for $\sigma \le 1$ because r_0 lies inside the polydisc. The function $H(w) = F(r_0, w)$ has $\|H\|_1 \le \|F\|_1/c_P$: apply $|g(a)| \le \|g\|_{H^1(\mathbb{T})}/(1-|a|^2)$ in each variable of S and integrate. Splitting $P_{r_S(\sigma)} = P_{r_0} + (P_{r_S(\sigma)} - P_{r_0})$,

$$\Big|\int_0^\delta F(r(\sigma))\,\mathrm{d}\sigma\Big| \le \frac{C^{(\delta)}_{>P}}{c_P}\|F\|_1 + K_P\delta\int_{\mathbb{T}^S}\int_0^1 |F(\zeta, r'(\sigma))|\,\mathrm{d}\sigma\,\mathrm{d}m(\zeta),$$

and the last integral is at most $(5+2\sqrt{2})\|F\|_1$ by Theorem 2.5, applied for each ζ to $w \mapsto F(\zeta, w)$. Conversely, for g supported on integers free of primes $p \le P$, put $f = ug$ with $u = \prod_{p\le P}(1-1/p)(1-p^{-1/2}z_p)^{-2}$. The variables are disjoint, so $\|f\|_1 = \|u\|_1\|g\|_1 = \|g\|_1$, and $u(r_S(\sigma)) = c_P\prod_{p\le P}(1-p^{-1-\sigma})^{-2} = c_P^{-1} + O_P(\sigma)$. Hence $\int_0^\delta f(r)\,\mathrm{d}\sigma = c_P^{-1}\int_0^\delta g(r')\,\mathrm{d}\sigma + O_P(\delta)\|g\|_1$. For the last statement, apply Theorem 10.9: along an extremizing sequence $\int_0^\delta f_k(1/2+\sigma)\,\mathrm{d}\sigma \to C$. □

The factor $c_P = \|K_{r_0}\|_2^{-2}$ is the price of removing the small primes, and nothing else is. What remains for Conjecture 1 is thus whether ℓ attains its norm, and whether the local constant $\lim_P \liminf_{\delta\to 0} C^{(\delta)}_{>P}/c_P$, in which every coordinate is small, is decided by the Gaussian limit; for that, kernels at many scales would have to be shown to be asymptotically optimal, which needs a central limit theorem for $\log|f|$ under $|f|\,\mathrm{d}m$ along concentrating sequences.

10.4. The absolute inequality

In one variable the absolute inequality is part (3) of Proposition 9.1; the same argument works for Paley–Wiener spaces whenever Nehari's theorem, equivalently weak factorization, holds [7, Lemma 7.5]. If $f = \sum_j g_j h_j$ with $\sum_j \|g_j\|_{\mathscr{H}^2}\|h_j\|_{\mathscr{H}^2} \lesssim \|f\|_{\mathscr{H}^1}$, then $|a_n|$ is dominated by the Dirichlet convolutions of the coefficient moduli, and the multiplicative Hilbert matrix, of norm π, bounds the weighted sum. On $\mathbb{T}^\infty$ this weak factorization fails, since Nehari's theorem does [38], and the S_2 substitute of chapter 5 controls only $\sum_j(\|g_j\|_{\mathscr{H}^2}\|h_j\|_{\mathscr{H}^2})^2$.

Problem 10 is open. This section proves the inequality on every piece of $\mathscr{H}^1$ cut out by a degree constraint that the Euler product can see: one homogeneous degree, one degree in a subset of the primes, integers composed of a thin set of primes, and the balanced four-class diagonals together with summable unions of them. It also proves it for products across two complementary prime sets, and for sums of r such products at the sharp cost $\sqrt{r}$. All of these rest on Helson's inequality. It then identifies the signed critical exponent on prime sets of density θ, which is $(\log n)^{-\theta}$, against the absolute exponent $(\log n)^{-1}$. It ends with the homogeneous dual constants.

10.4.1. An arithmetic criterion

Lemma 10.12. *Let $S \subseteq \mathbb{N}$ and $B(S) = \sum_{n\in S,\, n\ge 2} d(n)/(n\log^2 n)$. Every $f \in \mathscr{H}^1$ supported on S satisfies*

$$|a_1| + \sum_{n\ge 2}\frac{|a_n|}{\sqrt{n}\log n} \le \big(1 + B(S)^{1/2}\big)\|f\|_{\mathscr{H}^1}.$$

Moreover $B(S) = \int_0^\infty \sigma \sum_{n\in S,\, n\ge 2} d(n) n^{-1-\sigma}\, \mathrm{d}\sigma$.

PROOF. Use Cauchy–Schwarz with weights $d(n)$ and Helson's inequality [24], which states that $\sum |a_n|^2/d(n) \le \|f\|_{\mathscr{H}^1}^2$. For the formula, $\int_0^\infty \sigma n^{-\sigma}\, \mathrm{d}\sigma = 1/\log^2 n$. □

So every result in the next subsection is a bound on a Dirichlet series of $d(n)$ restricted to S, near $\sigma = 0$. The weight $\sigma\, \mathrm{d}\sigma$ is exactly what makes $\sigma^{-2+\varepsilon}$ integrable and σ^{-2} not: the full set $S = \mathbb{N}$ gives $\zeta(1+\sigma)^2 \asymp \sigma^{-2}$ and $B = \infty$, which is why the method cannot settle the full inequality.

10.4.2. Pieces

For a set $\mathcal{P}$ of primes let $\Omega_{\mathcal{P}}(n) = \sum_{p\in\mathcal{P}} v_p(n)$. Enumerate the primes $p_1 < p_2 < \cdots$ and let $\mathcal{P}_j$, $1 \le j \le 4$, be the residue classes of indices modulo 4.

THEOREM 10.13. *The absolute inequality* $|a_1| + \sum_{n\ge 2} |a_n|/(\sqrt{n}\log n) \le C\|f\|_{\mathscr{H}^1}$ *holds for* f *supported on each of the following sets, with* C *as stated.*

(1) $\{\Omega(n) = m\}$*, with* C *absolute and independent of* m*. For* $m \ge 1$*, and also for* $\{\Omega(n) \in \{1,2\}\}$*, the sum alone satisfies* $\sum_{n\ge 2} |a_n|/(\sqrt{n}\log n) \le (2389/256)^{1/2}\|f\|_{\mathscr{H}^1} < \pi\|f\|_{\mathscr{H}^1}$*, uniformly in* m *and in the number of primes.*

(2) $\{\Omega_{\mathcal{P}}(n) = k\}$*, where* $\sum_{p\in\mathcal{P}} p^{-1-\sigma} \le \delta\log(1/\sigma) + C_0$ *and* $\sum_{p\notin\mathcal{P}} p^{-1-\sigma} \le (1-\delta)\log(1/\sigma) + C_0$ *on* $(0,1]$*,* $0 < \delta \le 1$*;* C *depends on* δ, C_0 *only, not on* k*. The degrees in the primes outside* $\mathcal{P}$ *are unrestricted.*

(3) The integers composed of primes in $\mathcal{P}$*, where* $\sum_{p\in\mathcal{P}} p^{-1-\sigma} \le \delta\log(1/\sigma) + C_0$ *with* $\delta < 1$*; for instance either alternating half of the primes.*

(4) The union over $a \in \mathcal{A} \subseteq \mathbb{Z}^3$ *of the diagonals* $\{\Omega_{\mathcal{P}_j}(n) - \Omega_{\mathcal{P}_4}(n) = a_j,\ j \le 3\}$*, with* $C \lesssim (\sum_{a\in\mathcal{A}} (1+|a|_\infty)^{-1})^{1/2}$*; for a single diagonal* C *is absolute.*

PROOF. In each case we bound $B(S)$ through Lemma 10.12, by marking degrees in the Euler product $\prod_p (1 - u_p p^{-1-\sigma})^{-2}$ and comparing coefficientwise. For $0 < \sigma \le 1$, expanding the logarithm gives the majorant $\exp(2\sum_p u_p p^{-1-\sigma})E(u)$, where E collects the terms $u^\ell/(\ell p^\ell)$, $\ell \ge 2$, and has nonnegative coefficients and $E(1) < \infty$; for $\sigma \ge 1$, every coefficient is at most $\zeta(1+\sigma)^2 - 1 \le C2^{-\sigma}$.

(1) With one marker u, $\sum_p p^{-1-\sigma} \le \log(2/\sigma)$ gives $[u^m] \le [u^m](2/\sigma)^{2u}E(u)$. Integrating σ^{1-2u} over $(0,1)$ gives $4^u/(2(1-u))$, and the coefficient of u^m in $4^u E(u)/(1-u)$ is a partial sum of nonnegative coefficients, hence at most $4E(1)$. For the explicit constant, $a_1 = 0$ when $m \ge 1$, and the proof of Lemma 10.12 bounds the sum by $B_m^{1/2}\|f\|_{\mathscr{H}^1}$, where $B_m = \sum_{\Omega(n)=m} d(n)/(n\log^2 n) < 2389/256 < \pi^2$ is an arithmetic estimate uniform in m; for two layers $B_1 + B_2 < 141/16$.

(2) Mark only the primes of $\mathcal{P}$. The majorant is $C\sigma^{-2(1-\delta)}\sigma^{-2\delta u}e^{2C_0 u}E(u)$, and $\int_0^1 \sigma^{-1+2\delta(1-u)}\, \mathrm{d}\sigma = 1/(2\delta(1-u))$; conclude as in (1).

(3) $\prod_{p\in\mathcal{P}}(1-p^{-1-\sigma})^{-2} \le C\sigma^{-2\delta}$ and $\int_0^1 \sigma^{1-2\delta}\, \mathrm{d}\sigma < \infty$. For the alternating halves, $0 \le \sum_j (-1)^{j-1} p_j^{-s} \le 2^{-s}$ gives $\delta = 1/2$.

(4) Each class $\mathcal{P}_j$ satisfies $\sum_{p\in\mathcal{P}_j} p^{-1-\sigma} \le \log(1/\sigma)/4 + C_0$, by the same alternating comparison. With $T = \log(1/\sigma)$ and $\mu = T/2$, the coefficients of $\exp(\mu\sum_j u_j)$ are $e^{4\mu}$ times the joint law of four independent Poisson variables of mean μ, and the factor $e^{4\mu} = \sigma^{-2}$ cancels against $\sigma\,\mathrm{d}\sigma = e^{-2T}\,\mathrm{d}T$. So the contribution of a diagonal is $\int_0^\infty \mathbb{P}(N_j - N_4 = a_j,\ j \le 3)\,\mathrm{d}\mu$, convolved with the coefficients of E. The integral is at most $C(1+|a|_\infty)^{-1}$: conditioning on N_j, N_4 for the largest $|a_j|$ gives the factor $(1+\mu)^{-1}$ from the other two classes, and a tilted Fourier bound gives $\mathbb{P}(N_j - N_4 = A) \lesssim (1+\mu)^{-1/2} e^{-\min(A, A^2/\mu)/8}$. The shifts from E are absorbed using $E(3/2,\dots,3/2) < \infty$, and for $\sigma \ge 1$ every selected n has $\Omega(n) \ge |a|_\infty$, which costs $(3/2)^{-|a|_\infty}$. Different diagonals are disjoint; sum over $\mathcal{A}$. □

Since $J(F) \le \sum_n |a_n| c_n$, part (1) proves the sharp Fejér–Riesz conjecture on every homogeneous layer: any counterexample mixes degrees in an essential way.

Part (4) needs four classes: with three, the Poisson concentration factor is $(1+T)^{-1}$, whose integral diverges. That is a limit of the method, not a counterexample.

PROPOSITION 10.14 (Products across complementary prime sets). *Let $\mathcal{P}_0, \mathcal{P}_1$ be the primes of odd and even index.*

1. *If $g, h \in \mathscr{H}^1$ are supported on integers composed of primes in $\mathcal{P}_0$ and $\mathcal{P}_1$ respectively, then $f = gh$ satisfies the absolute inequality with an absolute constant, and $\|f\|_{\mathscr{H}^1} = \|g\|_{\mathscr{H}^1}\|h\|_{\mathscr{H}^1}$.*
2. *If $f = \sum_{j\le r} g_j h_j$ with such factors, then the absolute inequality holds with constant $C\sqrt{r}$.*
3. *The cost $\sqrt{r}$ of separated representations is sharp. For*

$$f_N = \sum_{j\le N} (p_{2j-1}p_{2j})^{-s},$$

every representation of f_N as a sum of products $u_k v_k$ with separated supports costs at least $N/2$, while $\|f_N\|_{\mathscr{H}^1} \le \sqrt{N}$.

PROOF. (1) The lifts use disjoint variables. Group m, n into blocks $2^j \le \log n < 2^{j+1}$. By (3) of Theorem 10.13 with $\sigma = 2^{-j}$, each block has $\sum d(n)/(n\log n) \le C$, and $\sqrt{\log m \log n}/(\log m + \log n) \le \sqrt{2}\,2^{-|j-k|/2}$ across blocks. Cauchy–Schwarz per pair of blocks, a sum over diagonals $j-k$, and Helson's inequality for g and h finish.

(2) The span E of the lifts of the g_j in L^1 has dimension at most r, so the Kadets–Snobar theorem [28] gives a projection onto E of norm at most $\sqrt{2r}$, represented by a kernel $K(x) \in E$ with $\operatorname{ess\,sup}\|K(x)\|_1 \le \sqrt{2r}$. Since every w-section of the lift of f lies in E, $f(z,w) = \int K(x)(z) f(x,w)\,\mathrm{d}m(x)$, which is a representation with projective cost at most $\sqrt{2r}\|f\|_{\mathscr{H}^1}$. Apply (1) to each term.

(3) Projecting each u_k, v_k to degree one and Helson's inequality with $d(p) = 2$ give $\sum_j |\hat{u}_k(p_{2j-1})|^2 \le 2\|u_k\|^2$ and similarly for v_k; summing the coefficients at $p_{2j-1}p_{2j}$ gives $N \le 2\sum_k \|u_k\|\|v_k\|$, while $\|f_N\|_{\mathscr{H}^1} \le \|f_N\|_{\mathscr{H}^2} = \sqrt{N}$. □

Remark (Product Hardy inequality). The tensorized Hardy inequality

$$\sum_{\alpha} \frac{|a_\alpha|}{\prod_j (\alpha_j+1)} \le \pi^d \|f\|_{H^1(\mathbb{T}^d)},$$

sharp by tensoring near-extremal squares for the Hilbert matrix. On the integers generated by d primes it reads $\sum_n |a_n|/d(n) \le \pi^d \|f\|_{\mathscr{H}^1}$. So no dimension-free inequality holds for the weight $1/d(n)$; the question is only about smaller weights such as $1/(\sqrt{n}\log n)$.

10.4.3. Signed functionals on prime sets

Corollary 10.15. *Let R have density $\theta \in (0,1]$. For $f \in \mathscr{H}^1$ supported on integers composed of primes in R,*

$$\Big| a_1 + \sum_{n\ge 2} \frac{a_n}{\sqrt{n}(\log n)^\theta} \Big| \le C\|f\|_{\mathscr{H}^1},$$

and the series converges in its natural order.

Proof. For polynomials, $\sum_{n\ge2} a_n/(\sqrt{n}(\log n)^\theta) = \Gamma(\theta)^{-1}\int_0^\infty (f(1/2+\sigma) - a_1)\sigma^{\theta-1}\,\mathrm{d}\sigma$. On $(0,1)$ use Theorem 3.2 with $p=1$; on $(1,\infty)$ use $|f(1/2+\sigma)-a_1| \le C2^{-\sigma}\|f\|_{\mathscr{H}^1}$. For natural-order convergence we follow Perfekt's Tauberian argument for $\theta = 1$ [39], comparing partial sums with Abel sums at $1/\log N$: the upper density bound gives $\sum_{e^u \le n < e^{2u}} d(n)/(n(\log n)^{2\theta}) \le C$ on R-integers, which with Helson's inequality makes the difference tend to 0, and horizontal translation is contractive on $\mathscr{H}^1$. □

Compare part (3) of Theorem 10.13: on a prime set of density $\theta < 1$ the *absolute* sum with weight $(\log n)^{-1}$ is bounded, while the *signed* sum is bounded down to the weight $(\log n)^{-\theta}$, which is critical by Corollary 5.3. Between the two exponents, cancellation is what matters. For $\theta = 1$ the two exponents meet, and whether the absolute inequality holds there is exactly Perfekt's question.

10.4.4. Fractional primitives of the zeta function

Bondarenko, Brevig, Saksman and Seip [11] considered, for $\beta > 0$, the functionals

$$\varphi_\beta(f) = a_1 + \sum_{n\ge2} \frac{a_n}{\sqrt{n}(\log n)^\beta} = a_1 + \frac{1}{\Gamma(\beta)}\int_0^\infty \big(f(1/2+\sigma) - a_1\big)\sigma^{\beta-1}\,\mathrm{d}\sigma.$$

In one variable the analogue is $f \mapsto \Gamma(\beta)^{-1}\int_0^1 f(x)(1-x)^{\beta-1}\,\mathrm{d}x$, with coefficients $\Gamma(j+1)/\Gamma(j+1+\beta)$. It is bounded on $H^p(\mathbb{D})$ for $p>1$ exactly when $\beta > 1/p$, and for $p \le 1$ exactly when $\beta \ge 1/p$ [11, Theorem 3.3].

Problem 11 (Bondarenko, Brevig, Saksman and Seip [11, Section 3]). *Let $0 < p \le 2$. Is $\varphi_{1/p}$ bounded on $\mathscr{H}^p$ if and only if $p \le 1$?*

The case $p = 1$ is Perfekt's theorem, and $p = 2$ is trivial. The embedding of Theorem 2.5, valid for every p, settles the rest.

Theorem 10.16. *Let $0 < p \le 2$. The functional $\varphi_{1/p}$ is bounded on $\mathscr{H}^p$ if and only if $p \le 1$.*

PROOF. Put $\beta = 1/p$. By Theorem 1.7 and Cauchy–Schwarz, $|f(1/2+\sigma)|^p \le \zeta(1+2\sigma)\|f\|_{\mathscr{H}^p}^p$; letting $\sigma \to \infty$ gives $|a_1| \le \|f\|_{\mathscr{H}^p}$.

Boundedness for $p \le 1$. Write $\|\cdot\|$ for the $\mathscr{H}^p$ norm. On $(0,1)$, since $1-p \ge 0$ and $\zeta(1+2\sigma) \le 3/(2\sigma)$,

$$|f(1/2+\sigma)|\sigma^{\beta-1} = |f|^p|f|^{1-p}\sigma^{\beta-1} \le (3/2)^{\beta-1}|f(1/2+\sigma)|^p\|f\|^{1-p},$$

so $\int_0^1 |f(1/2+\sigma)|\sigma^{\beta-1}\,\mathrm{d}\sigma \le (3/2)^{\beta-1}(5+2\sqrt{2})\|f\|$ by Theorem 2.5. For $\sigma \ge 1$ let F be the Bohr lift of a Dirichlet polynomial f, put $\rho = 2^{\sigma-1}r(\sigma)$, so that $\rho_q \le q^{-3/2}$, and $G(\lambda) = F(\lambda\rho)$. For $|\lambda| = 1$, plurisubharmonicity of $|F|^p$ gives $|G(\lambda)|^p \le \|f\|^p \sup P_{\lambda\rho} \le \|f\|^p\zeta(3/2)^2/\zeta(3) < 6\|f\|^p$. Since $G(0) = a_1$, the Schwarz lemma applied to $G - a_1$ gives $|f(1/2+\sigma) - a_1| = |G(2^{1-\sigma}) - a_1| \le 2\cdot 6^\beta 2^{1-\sigma}\|f\|$. Together with $|a_1| \le \|f\|$ this bounds every part of the integral by a constant times $\|f\|$. No triangle inequality was used, so the argument holds for $p < 1$, and the functional extends from Dirichlet polynomials.

Unboundedness for $1 < p \le 2$. Let P_k be the Poisson kernel of $r(\sigma_k)$, $\sigma_k = 2^{-k}$, and $F_k = K_{r(\sigma_k)}^{2/p}\zeta(1+2\sigma_k)^{-1/p}$, so that $|F_k|^p = P_k$ and F_k has positive coefficients. Since $\zeta(1+\sigma_k+\sigma) \ge \zeta(1+2\sigma_k) \ge (2\sigma_k)^{-1}$ for $\sigma \le \sigma_k$,

$$\varphi_\beta(F_k) \ge \frac{p\,\sigma_k^\beta}{\Gamma(\beta)}\big(\zeta(1+2\sigma_k)^\beta - \zeta(1+2\sigma_k)^{-\beta}\big) \ge \frac{p}{\Gamma(\beta)}\big(2^{-\beta} - 4^{-\beta}\big) =: c_p > 0$$

for $k \ge 2$. Let $g = \sum_{k=2}^{M+1} F_k$. Hölder's inequality with weights $w_k = P_k^{1/(2(p-1))}$, followed by $(\sum_j w_j)^{p-1} \le \sum_j w_j^{p-1}$ (valid since $0 < p-1 \le 1$), gives pointwise

$$|g|^p \le \Big(\sum_k P_k w_k^{1-p}\Big)\Big(\sum_j w_j\Big)^{p-1} \le \sum_{j,k}(P_jP_k)^{1/2},$$

so $\|g\|_{\mathscr{H}^p}^p \le \sum_{j,k} H(\omega_{r(\sigma_j)}, \omega_{r(\sigma_k)})$. By Lemma 2.3 and $1-\rho(a,b)^2 = (1-a^2)(1-b^2)/(1-ab)^2$,

$$H(\omega_{r(\sigma)}, \omega_{r(\tau)}) \le \Big(\frac{\zeta(1+\sigma+\tau)^2}{\zeta(1+2\sigma)\zeta(1+2\tau)}\Big)^{1/4} \le \Big(\frac{4\sigma\tau(1+\sigma+\tau)^2}{(\sigma+\tau)^2}\Big)^{1/4},$$

which is at most $2\cdot 2^{-|j-k|/4}$ for $\sigma = \sigma_j$, $\tau = \sigma_k$. So every row of the Gram sum is below 20, and $\|g\|_{\mathscr{H}^p}^p \le 20M$, while $\varphi_\beta(g) \ge c_pM$. Since $p > 1$, the ratio $\varphi_\beta(g)/\|g\|_{\mathscr{H}^p} \ge c_p 20^{-1/p}M^{1-1/p}$ is unbounded. The functions F_k involve all primes; truncating to the primes $q \le N$, taking Taylor sections, and letting $N \to \infty$ gives the same conclusion for Dirichlet polynomials, by monotone convergence in the positive coefficients. □

The mechanism is the one-variable one: normalized kernels at dyadic scales are seen with weight of order one by $\varphi_{1/p}$, while their pth powers are almost orthogonal when $p > 1$. On $\mathbb{T}^\infty$ the almost orthogonality is the decay of the Hellinger affinity across scales.

10.4.5. Homogeneous dual constants

Let c_m be the norm of $\varphi \mapsto \Lambda_\varphi$ from homogeneous degree-m symbols in $H^2(\mathbb{T}^\infty)$ to $(H^1)^*$; equivalently, the best constant in $\|f\|_2 \le c_m \|f\|_1$ for homogeneous f of degree m.

PROPOSITION 10.17. *$\sqrt{m!}/\Gamma(1+m/2) \le c_m \le 2^{m/2}$, so $c_m^{1/m} \to \sqrt{2}$; moreover $c_{m+n} \ge c_m c_n$ and $c_m \ge (2/\sqrt{\pi})^m$. Consequently any inequality $\|\Lambda_\varphi\| \le C(\sum_m w_m \|\varphi_m\|_2^2)^{1/2}$ over homogeneous components needs $w_m \ge C^{-2} m!/\Gamma(1+m/2)^2$, so $\liminf w_m^{1/m} \ge 2$.*

PROOF. Projection onto degree m is an H^1 contraction, which gives the equivalence. The upper bound is Helson's inequality with $\prod(\nu_j+1) \le 2^m$. For the lower bound take $((z_1+\cdots+z_d)/\sqrt{d})^m$ and let $d \to \infty$: the moments converge to those of a complex Gaussian, $\mathbb{E}|Z|^q = \Gamma(1+q/2)$. Supermultiplicativity comes from disjoint variables, and $c_1 = 2/\sqrt{\pi}$ is Brevig's value [12]. □

10.5. Partial sums

Partial sums lie outside the story of π: the operators S_N are not coefficient functionals, and they share with the rest of Part IV only the tools, Helson's inequality and Harper's low moments. We include them because the $\mathscr{H}^1$ problem is open and those tools give something.

PROBLEM 12 (Saksman and Seip [43, Problem 4.5]). *What is the precise growth of $\|S_N\|_{\mathscr{H}^1 \to \mathscr{H}^1}$, where $S_N \sum a_n n^{-s} = \sum_{n \le N} a_n n^{-s}$?*

For Problem 12, the order of growth is known for every p except $p = 1$. For $1 < p < \infty$ the operators S_N are uniformly bounded, by Helson's theorem on half-spaces of ordered groups applied to $\mathbb{Q}_+$ ordered by size. On $\mathscr{H}^\infty$, Balasubramanian, Calado and Queffélec [3] proved $\|S_N\| \lesssim \log N$, and Chen and Fu [18] have shown that $\|S_N\| \ge (1/(2\pi) + o(1)) \log N$, so the order is $\log N$. For $0 < p < 1$, Bondarenko, Brevig, Saksman and Seip [11] showed that $\log \|S_N\|_{\mathscr{H}^p \to \mathscr{H}^p}$ is of order $\log N / \log\log N$. On $\mathscr{H}^1$ they proved $\log\log N \lesssim \|S_N\| \lesssim \log N / \log\log N$. The lower bound is a one-variable fact: on series in the powers 2^{-ks}, S_N is the partial sum of degree $\log_2 N$ on $H^1(\mathbb{D})$. Any larger lower bound must therefore use many primes.

PROPOSITION 10.18. *There is an absolute C such that, for $f \in \mathscr{H}^1$ and $g = S_N f = \sum_{n \le N} a_n n^{-s} \ne 0$,*

$$\frac{\|g\|_{\mathscr{H}^1}}{\|f\|_{\mathscr{H}^1}} \le C\Big(1 + \log \frac{\|g\|_{\mathscr{H}^2}}{\|g\|_{\mathscr{H}^1}}\Big).$$

PROOF. Bondarenko, Brevig, Saksman and Seip [11] prove that

$$\|S_N f\|_{\mathscr{H}^{1-\varepsilon}} \le C_0 \varepsilon^{-1} \|f\|_{\mathscr{H}^1}.$$

Hölder's inequality gives

$$\|g\|_1 \le \|g\|_{1-\varepsilon}^{(1-\varepsilon)/(1+\varepsilon)} \|g\|_2^{2\varepsilon/(1+\varepsilon)}, \qquad \text{hence} \qquad \frac{\|g\|_1}{\|f\|_1} \le \frac{C_0}{\varepsilon} \Big(\frac{\|g\|_2}{\|g\|_1}\Big)^{2\varepsilon/(1-\varepsilon)}.$$

Take $\varepsilon = 1/(2(1+\log(\|g\|_2/\|g\|_1)))$. □

COROLLARY 10.19. *Uniformly over all tails:*

(1) if $S_N f = \sum_{n \le N} n^{-s}$*, then* $\|S_N f\|_{\mathscr{H}^1} \lesssim (1 + \log\log\log N)\|f\|_{\mathscr{H}^1}$*;*
(2) if $S_N f = \sum_{n \le N} n^{-1/2-s}$*, then* $\|S_N f\|_{\mathscr{H}^1} \lesssim (1 + \log\log N)\|f\|_{\mathscr{H}^1}$*;*
(3) if every n *in the support of* $S_N f$ *has* $\Omega(n) \le m$*, then* $\|S_N f\|_{\mathscr{H}^1} \lesssim (m+1)\|f\|_{\mathscr{H}^1}$*.*

PROOF. (1) Harper's theorem [22] gives $\|\sum_{n \le N} n^{-s}\|_1 \asymp \sqrt{N}(\log\log N)^{-1/4}$. (2) Gerspach's pseudomoment bound [21] gives $\|\sum_{n \le N} n^{-1/2-s}\|_1 \asymp (\log N)^{1/4}$ against an $\mathscr{H}^2$ norm $\asymp (\log N)^{1/2}$. (3) $d(n) \le 2^m$ on the support, so Helson gives $\|g\|_2 \le 2^{m/2}\|g\|_1$. □

The known lower bound for $\|S_N\|_{\mathscr{H}^1 \to \mathscr{H}^1}$ is of order $\log\log N$. A large amplification therefore needs an output with an exponentially large ratio of $\mathscr{H}^2$ to $\mathscr{H}^1$ norms. The Dirichlet kernel is not such an output, and neither is the partial sum of $\zeta(1/2+s)$, which can at most match the known order. Vertical blow-up does not help either (section 3.5).

Open

(1) Prove $C > \pi$ deterministically, by bounding the Gaussian integral of Theorem 10.6 for one explicit quadratic.
(2) Prove Conjecture 1. By Theorem 10.9 and Proposition 10.11 this splits in two: does ℓ attain its norm on $\mathscr{H}^1_0$, and is the local constant at $\sigma = 0$, with every coordinate small, decided by kernels at many scales?
(3) A per-unit-scale formulation: the window scaling limit of J at $\sigma \asymp 1/\log P$, presumably governed by critical multiplicative chaos.
(4) Close the gap $1.0617 \le C_2/\pi_2^* \le 1.0878$, and certify lower bounds for C_3; and close $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ versus π, which needs joint phase information about infinitely many large primes.
(5) The absolute inequality $\sum |a_n|/(\sqrt{n}\log n) \lesssim \|f\|_{\mathscr{H}^1}$ (Problem 10). By Lemma 10.12, any proof must use more than Helson's inequality on the full set of integers.
(6) Is $\|S_N\|_{\mathscr{H}^1 \to \mathscr{H}^1}$ of order $\log\log N$, or larger?

APPENDIX A

Problems and their status

The book states twelve numbered problems: eight from the literature, one from the author's thesis, and three that arise here. The table records what is known about each, as of October 2026; several of the answers rest on recent preprints. Of the twelve, three are answered in this book (Problems 2, 5 and 11), three have been answered elsewhere, one of them also by a second proof here and with a new Bergman analogue (Problems 1, 4 and 8), four are answered in part, and two are open.

	Source	Status
Problem 1	Saksman–Seip [43, Problem 2.1]	Answered, modulo a preprint: yes exactly for $p \geq 2$. No for $0 < p < 2$ by Harper [22]; yes for $p > 2$ announced by Chen et al. [17]. Here: the rate $N_p(\varepsilon) \lesssim \varepsilon^{-1/2}$ for every p (Theorem 3.5).
Problem 2	Søvik [45]	Answered in the negative: $C_0 - 2\pi$ lies between $2.8 \cdot 10^{-7}$ and 10^{-3}, and C_0 is attained (Propositions 3.7 and 3.8).
Problem 3	Perfekt–Pushnitski [40]	Partial. Yes for every partially averaged curve measure, in three geometries (Theorems 4.2 and 6.10), and for energy-separated families (Theorem 3.1).
Problem 4	Saksman–Seip [43, Problem 3.2]	Answered by Perfekt [39]; second proof in Corollary 4.11; the Bergman analogue is answered in Theorem 6.11.
Problem 5	This book	Answered: yes under regular variation (Theorem 5.4; under C^2 regularity, with the finer spectral type, Ma–Lu–Zu [33]); no in general, with the band structure proved and the gaps found numerically (Proposition 5.5).
Problem 6	Brevig–Perfekt [13, Question 3]	Partial. Such forms exist for every $p > 4$ (Theorem 7.6; also Bampouras [4]); open for $2 < p \leq 4$, where the natural families are sharp at 4 (chapter 8). In two variables none exist, for any $p < \infty$ [8].
Problem 7	This book	Open. Every family treated in chapter 8 satisfies the inequality.
Problem 8	Brevig–Perfekt [14]	Answered, in preprints: for balls by Bampouras–Perfekt [7], for convex sets free of lines by Bampouras [5]. The constant exceeds 1 and grows exactly exponentially on polydiscs (Proposition 8.13 and Corollary 8.14). For S_p, $2 < p \leq 4$, open on the disc; no threshold for polygons.
Problem 9	This book	Open. $C < \infty$ by Perfekt [39]; $\pi \leq C \leq 5.515$ (Proposition 10.4); a probabilistic certificate gives $C \geq 3.145$ (section 10.2).
Problem 10	Perfekt [39]	Partial. True on every piece of Theorem 10.13 and for products across prime sets (Proposition 10.14).
Problem 11	Bondarenko–Brevig–Saksman–Seip [11]	Answered: yes (Theorem 10.16).
Problem 12	Saksman–Seip [43, Problem 4.5]	Partial. On $\mathscr{H}^1$, $\log\log N \lesssim \|S_N\| \lesssim \log N / \log\log N$ [11]; on $\mathscr{H}^\infty$ the order is $\log N$ [18]. An output-sensitive bound here (Proposition 10.18).

Further open questions

(1) *Two-point stability.* Is the factor 2 in Corollary 1.8 necessary (chapter 1)?
(2) *What Hellinger–Bessel loses.* Is there a geometric condition, between ℓ^2-bounded affinities and separation, under which the Gram condition is necessary (Proposition 2.12)?
(3) *Vertical lines.* The rate of $N_p(\varepsilon)$ for $p < 2$, and the exact value of C_0 in Proposition 3.7 (chapter 3).
(4) *Spectra.* Which measures have $s_n \asymp a_n^*$ without an index change? Certify the spectral gaps of Proposition 5.5. The exponent of the bounded-rank Nehari constant, between 0.174 and 1/2 (chapter 5).
(5) *Bergman spaces.* The sharp two-point constants: in the Bergman geometry beyond the first zero of $K_x + K_{-x}$ (up to it the square bound is sharp), and in the Gauss geometry, where the square bound always fails, the constant itself (Proposition 6.6); and whether Bergman Dirichlet kernels cancel for $\beta > 1$ (chapter 6).
(6) *Near $p = 2$.* Is there a dimension- and degree-free interval above 2 with no strict Gaussian blocks (chapter 8)?
(7) *Fixed dimension.* Is the S_p Nehari constant on $\mathbb{T}^d$ finite for every $p < \infty$ when $d \geq 3$, as it is for $d = 2$ (chapter 7)? Does the S_2 constant for the ball B^n grow with n, as it does for polydiscs (chapter 8)?
(8) *Spectral type.* For prime sets of density θ, is the critical spectrum absolutely continuous, and when are there eigenvalues above it? For all primes in the Bergman geometry this is known (chapter 5).

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